

Polycontrolled PROPs for Qudit Circuits: A Uniform Complete Equational Theory For Arbitrary Finite Dimension

Abstract

We present a finite schematic axiomatisation of quantum circuits over d -level systems (qudits), uniform in every finite dimension $d \geq 2$. For each d we define a PROP $\mathbf{CQC}_d$ equipped with a family of control functors, treating control as a primitive categorical constructor. Using a translation between qudit circuits and the LOPP calculus for linear optics based on d -ary Gray codes, we obtain for each d a finite set of local axiom schemata $\mathbf{QC}_d$ that is sound and complete for unitary d -level circuits: two circuits denote the same unitary if and only if they are inter-derivable using axioms involving at most three wires. The generators are compatible with standard universal qudit gate families, yielding a sound equational basis for circuit rewriting and optimisation-by-rewriting. Conceptually, this extends the qubit circuit completeness results of Clément et al. to arbitrary finite dimension, and instantiates the control-as-constructor approach of Delorme and Perdrix in this setting, while keeping the axiom shapes uniform in d .

CCS Concepts

• **Theory of computation** → **Quantum computation theory**; **Categorical semantics**; *Equational logic and rewriting*.

Keywords

Qudit circuits, Polycontrolled PROPs, controlled quantum circuits, equational theories, categorical quantum mechanics

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1 Introduction

Quantum circuits provide a standard low-level syntax for finite-dimensional quantum computation: wires carry d -level quantum data, boxes represent unitary gates, and a circuit denotes a unitary operator on the tensor power $(\mathbb{C}^d)^{\otimes n}$ [15, 24]. Accordingly, they serve as the basic representation both for algorithm design and for hardware-aware compilation and verification; in this setting, optimisation is typically justified *via* equational circuit rewriting. Moreover, correctness arguments in verification and compilation are routinely phrased as local diagrammatic rewrites, closed under both sequential and parallel composition. Consequently, from a

logical perspective, this raises a foundational completeness question: can equality of unitary circuits be axiomatised by a *finite*, syntax-directed equational theory in the circuit PROP?

In the PROP formalism (a strict symmetric monoidal category whose objects are wire counts) such a theory is given by a finite set of local rewrite *schemata* between circuit diagrams, closed under both sequential and parallel composition. For qubits ($d = 2$), complete circuit-level axiomatisations are already known [6, 8, 9]. A key lesson from this line of work is the central role of *control*. If control is not treated as primitive, then completeness forces axiom schemata that quantify over arbitrarily many control wires. By contrast, treating control functorially in a controlled PROP yields finite, bounded-arity axiom sets [14, 17].

Meanwhile, graphical calculi such as ZX/ZH/ZW admit complete equational theories for broad classes of qubit processes [1, 4, 5, 11, 19, 28]. However, these languages describe general linear maps rather than circuit-level unitaries; moreover, extracting circuits can be non-local—and, indeed, hard in general [12]. This motivates circuit-specific rewrite systems.

For $d > 2$, many platforms support native d -level systems (qudits); consequently, qudit-native design and compilation can reduce resource counts or exploit non-binary structure in gate libraries and higher-dimensional circuit design [20, 25, 27, 31]. From an equational perspective, several diagrammatic and circuit formalisms for qudits have been proposed. Qudit and “qufinite” variants of the ZX- and ZH-calculi provide universal graphical languages parameterised by the dimension; moreover, they admit completeness results in some settings [30, 33]. More recently, qufinite ZXW-calculi, together with finite-dimensional ZX and ZW calculi, have yielded complete equational theories for the whole of finite-dimensional Hilbert spaces, uniformly in d [13, 26, 34]. These results are foundationally powerful; nevertheless, they operate at the level of *arbitrary* linear maps and therefore do not directly yield bounded-arity PROP presentations of a circuit language for *unitary* gates with a native control constructor. At the circuit level, by contrast, completeness results are largely fragmentary. There are complete rewrite systems for specific subtheories, notably for qutrit stabiliser quantum mechanics in ZX and, more recently, for multi-qutrit Clifford circuits [20]. Beyond ZX-style diagrammatics, further work has developed graphical calculi for multi-qudit computations based on generalised Clifford algebras and related algebraic structures [21, 22]. However, these approaches either target broad classes of linear maps (rather than a circuit syntax for unitary gates), focus on particular fragments, or treat only fixed small dimensions. Consequently, they do not provide a finite, uniform-in- d PROP presentation of the *circuit* category whose objects are wire arities and whose morphisms are unitary d -level circuits over a fixed universal gate signature, with axioms given by local schemata of bounded wire arity.

There is no dimension-uniform finitary presentation of unitary qudit circuits satisfying the following combination of requirements. Namely, for each $d \geq 2$, one seeks a sound and complete *finite*

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set of circuit-specific axiom schemata for the PROP of unitary d -level circuits over a fixed universal gate signature, with axioms given by local schemata of bounded wire arity and schematic form independent of d . This contrasts with the qubit case [6, 8, 9, 14] and, as a result, leaves qudit equational reasoning without a uniform completeness theory. Moreover, beyond foundational interest, the absence of such a fixed rewrite system makes it harder to justify rewriting-based optimisation and verification of native qudit circuits without resorting to ad hoc matrix calculations.

At a high level, we treat control as a primitive operation in a polycontrolled PROP. Control is fundamental in circuit design and compilation: given a subcircuit $f : n \rightarrow n$, one forms a controlled version by adjoining a new wire and applying f conditionally; importantly, this construction must be coherent with sequential composition and wire permutations. Rather than treating each controlled- U gate as an ad hoc derived primitive, we build control directly into the categorical structure itself [14, 17]. Concretely, we equip the circuit language with a family of control functors $(C_k)_{k=0}^{d-1}$. Each C_k sends an endomorphism $f : n \rightarrow n$ to an endomorphism $C_k(f) : 1 + n \rightarrow 1 + n$, which applies f exactly when the added control qudit is in computational basis state $|k\rangle$ (and otherwise acts as the identity on the targets). Multi-controlled operations are obtained systematically by iterating these functors. Treating control as a first-class constructor is what makes bounded-arity axioms possible: local schemata can refer to $C_k(-)$ itself, rather than quantifying over arbitrarily many explicit control wires.

Main result. For each finite $d \geq 2$, we define a polycontrolled circuit PROP $\mathbf{CQC}_d$. Moreover, we introduce a finite equational theory $\mathbf{QC}_d$ generated by local axiom schemata whose schematic form is uniform in d and whose instance involve at most three wires. Theorem 4.19 states that $\mathbf{QC}_d$ is sound and complete for the standard unitary semantics $\llbracket - \rrbracket : \mathbf{CQC}_d \rightarrow \mathbf{Qudit}_d$: for any n -qudit circuits $C_1, C_2 : n \rightarrow n$,

$$\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket \iff \mathbf{QC}_d \vdash C_1 = C_2$$

Here, “uniformity” has three aspects: (i) bounded wire arity; (ii) a fixed, finite list of axiom *schema families* whose diagrammatic shape does not depend on d ; and (iii) only indices (in $[d]$) vary with d .

Our completeness proof proceeds by reduction to the linear-optical calculus **LOPP**. In particular, following Clément et al. [8], we transfer completeness from **LOPP** [7, 18]. Using a reflected d -ary Gray code, we encode an n -qudit circuit as a d^n -mode single-photon linear-optical circuit. This encoding preserves the intended unitary semantics, up to conjugation by a fixed Gray-code basis permutation. A key technical point is that $\mathbf{CQC}_d$ and **LOPP** interpret $\otimes$ differently (tensor product versus direct sum under single-photon semantics). Consequently, the translation cannot be strictly monoidal; instead, we fix a deterministic serialisation of $\otimes$ and prove an encoding/decoding retraction. We then transport completeness of **LOPP** back to $\mathbf{CQC}_d$ via decoding and a “mimicking” argument. When $d = 2$, the reflected d -ary Gray code specialises to the binary reflected Gray code; correspondingly, our reduction to **LOPP** recovers the qubit-level completeness-transfer pattern of [8].

For each finite $d \geq 2$ we obtain:

- (1) *A polycontrolled circuit PROP.* We define the endomorphism PROP $\mathbf{CQC}_d$ of d -level unitary circuits, equipped with control functors $(C_k)_{k=0}^{d-1}$ whose iteration yields multi-controlled operations.
- (2) *A finite bounded-arity axiom system.* We present a finite set of local axiom schemata $\mathbf{QC}_d$ (each instance uses at most three wires) of schematic form uniform in d , and show that it is sound and complete for unitary d -level circuit equality. This yields a fixed, syntax-directed rewrite system suitable as a foundation for rewriting-based optimisation and verification of native qudit circuits.
- (3) *A completeness transfer via Gray-code embedding.* Using a d -ary Gray code, we encode n -qudit circuits as d^n -mode **LOPP** circuits and decode **LOPP** derivations back into $\mathbf{CQC}_d$, yielding a modular completeness proof via [7, 18].
- (4) *Expressing standard qudit gate families.* Our generators and axioms express adjacent two-level gates and basis-controlled phases. Consequently, one can assemble common universal qudit gate families (as recalled in §2.4). This keeps the presentation circuit-level and dimension-uniform, whilst remaining compatible with standard synthesis practice.

Throughout, we work with *exact* unitary equality: global phases are *not* quotiented away and therefore appear as scalar ($0 \rightarrow 0$) morphisms. Our goal is a circuit-specific presentation of the PROP of unitary circuits over a fixed signature; unlike the uniform-in- d completeness results for finite-dimensional ZX-/ZW-/ZXW-calculi [13, 26, 34], we do not aim to axiomatise arbitrary linear maps.

The paper is organised as follows. Section 2 defines $\mathbf{CQC}_d$ and its unitary semantics; Section 3 presents the axiom schemata generating $\mathbf{QC}_d$; and Section 4 give the Gray-code translation to **LOPP** and the completeness transfer culminating in Theorem 4.19.

2 Qudit circuits as a polycontrolled PROP

In this section we set up the circuit language used throughout this work. Fix a dimension $d \geq 2$. Our goal is to work in a *circuit-level* setting: objects are wire counts, morphisms are unitary d -level circuits, and circuit equality will ultimately be captured by a finite, local axiom system. The key design choice is that *control* is *primitive*. Rather than treating each controlled gate as a separate derived gadget, we build control into the categorical structure: for every computational basis value $k \in [d]$ we have a constructor $C_k(-)$ that turns a circuit $f : n \rightarrow n$ into its k -controlled version $C_k(f) : 1 + n \rightarrow 1 + n$. Multi-controlled operations are then obtained simply by iterating C_k .

2.1 PROPs and (poly)controlled PROPs

We work in a strict symmetric monoidal setting, so wire counts form the objects, parallel composition is strictly associative, and diagrams are taken modulo the strict coherence laws.

Definition 2.1 ([23]). A PROP $\mathbf{P}$ is a strict symmetric monoidal category whose objects are the natural numbers, generated by a distinguished object 1 with tensor $m \otimes n = m + n$ and unit object 0.

We write $\mathbf{P}(m, n)$ for the hom-set $m \rightarrow n$. Composition is written $\circ$, tensor (parallel composition) as $\otimes$, and $\sigma_{m,n} : m + n \rightarrow n + m$ denotes the symmetry (wire swap).

Since our circuits are unitary, we only consider endomorphisms. For a PROP $\mathbf{P}$, write $\mathbf{P}_{\text{endo}}$ for its wide endomorphism subcategory: $\mathbf{P}_{\text{endo}}(n, n) = \mathbf{P}(n, n)$ and $\mathbf{P}_{\text{endo}}(m, n) = \emptyset$ for $m \neq n$.

A permutation is any morphism $\pi : n \rightarrow n$ obtained from identities and symmetries $\sigma_{m,n}$ using only $\circ$ and $\otimes$. Such morphisms are canonically invertible; we write π^{-1} for the inverse permutation.

Informally, “adding a control” is an operation that, given an endomorphism $f : n \rightarrow n$, produces an endomorphism on one additional wire that applies f conditionally. Categorically, we package this as an endofunctor on $\mathbf{P}_{\text{endo}}$ that adjoins a distinguished wire and behaves well with respect to the PROP structure.

Definition 2.2 ([14]). Let $\mathbf{P}$ be a PROP and $\mathbf{P}_{\text{endo}}$ its endomorphism PROP. A *control functor* is a functor $C : \mathbf{P}_{\text{endo}} \rightarrow \mathbf{P}_{\text{endo}}$ such that:

- on objects, $C(n) = 1 + n$ (it adjoins one distinguished *control wire*);
- on morphisms, for every $f : n \rightarrow n$, every $m \in \mathbb{N}$, and every permutation $\pi : n \rightarrow n$, the morphism $C(f) : 1 + n \rightarrow 1 + n$ satisfies:

$$C(f \otimes \text{id}_m) = C(f) \otimes \text{id}_m, \quad (1)$$

$$C(C(f)) \circ (\sigma_{1,1} \otimes \text{id}_n) = (\sigma_{1,1} \otimes \text{id}_n) \circ C(C(f)), \quad (2)$$

$$C(\pi^{-1} \circ f \circ \pi) = (\text{id}_1 \otimes \pi^{-1}) \circ C(f) \circ (\text{id}_1 \otimes \pi). \quad (3)$$

The three coherence laws can be read as follows. Equation (1) says that unused wires remain untouched. Equation (2) says that nesting control twice is independent of the order of the two control wires, up to swapping them. Equation (3) says that controlling is invariant under permuting the target wires: if you relabel the target register, controlling commutes with that relabelling.

Definition 2.3 ([14]). A *polycontrolled PROP* is a PROP $\mathbf{P}$ equipped with a family of control functors $(C_k)_{k \in K}$ on $\mathbf{P}_{\text{endo}}$, indexed by a set K . When $K = [d] = \{0, \dots, d-1\}$ we speak of a *d-ary polycontrolled PROP*.

We use the usual string-diagram notation for PROPs: diagrams are read left-to-right; sequential composition $D \circ C$ is plugging; tensor $C \otimes D$ is juxtaposition; and symmetries $\sigma_{m,n}$ are wire swaps. In a polycontrolled PROP, $C_k(f)$ is depicted by adjoining a distinguished control wire (drawn above the targets) labelled by k .

$$\begin{array}{ccccccc} \boxed{} & \text{---} & \text{X} & \text{---} & \boxed{C} \boxed{D} & \text{---} & \boxed{C} \boxed{D} \\ \text{id}_0 & \text{id}_1 & \sigma_{1,1} & D \circ C & C \otimes D & C_k(C) \end{array}$$

Fix $d \geq 2$, and let $\mathbf{P}$ be a d -ary polycontrolled PROP with control functors $(C_k)_{k \in [d]}$. For a word $u = u_1 \dots u_m \in [d]^*$ and an endomorphism $f : n \rightarrow n$, we write $C_u(f) : |u| + n \rightarrow |u| + n$ for the iterated control of f by u , defined by $C_\epsilon(f) := f$ and $C_{ku}(f) := C_k(C_u(f))$. Equivalently, $C_u(f) = C_{u_1}(\dots(C_{u_m}(f))\dots)$.

Definition 2.4. Let $d \geq 2$ and let $\mathbf{P}$ be a d -ary polycontrolled PROP equipped with $(C_k)_{k=0}^{d-1}$.

- (1) The family (C_k) is *compatible* if for all $n \geq 0$, all $f : n \rightarrow n$, and all $a, b \in \{0, \dots, d-1\}$, nesting a -control and b -control commutes up to swapping the resulting control wires:

$$\begin{array}{ccc} \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ C_{ab}(f) \circ (\sigma_{1,1} \otimes \text{id}_n) & = & (\sigma_{1,1} \otimes \text{id}_n) \circ C_{ba}(f) \end{array} \quad (\text{COMPAT})$$

- (2) The family (C_k) is *commutative* if for all $n \geq 0$, all $f, g : n \rightarrow n$, and all distinct $a, b \in \{0, \dots, d-1\}$, operations controlled on distinct basis values commute:

$$\begin{array}{ccc} \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ C_a(f) \circ C_b(g) & = & C_b(g) \circ C_a(f) \end{array} \quad (\text{COMM})$$

- (3) The family (C_k) is *exhaustive* if for all $n \geq 0$ and all $f : n \rightarrow n$, controlling f on every basis value collapses to an unconditional application of f on the targets (leaving the control wire intact):

$$\begin{array}{ccc} \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ \text{---} \text{---} \text{---} \text{---} & \text{---} \text{---} \text{---} \text{---} & \\ C_0(f) \circ C_1(f) \circ \dots \circ C_{d-1}(f) & = & \text{id}_1 \otimes f \end{array} \quad (\text{EXH})$$

In the circuit PROP $\mathbf{CQC}_d$ defined below, we *do not* impose any equations relating C_a and C_b at the level of structural equality. Instead, the finite equational theory QC_d will be used to *derive* the global control-algebra laws in Section 3.3.

2.2 Raw circuits and the PROP $\mathbf{CQC}_d$

We now fix a dimension $d \geq 2$ and define a circuit syntax with explicit control constructors. We deliberately separate two layers:

- *raw circuits*, which are syntax trees built from generators using term-formers $\circ$, $\otimes$, and $C_k(-)$; and
- *circuits in the PROP $\mathbf{CQC}_d$* , obtained by quotienting raw syntax by the PROP coherence laws and the coherence laws that make each C_k into a control functor.

Keeping the raw syntax is useful later, because our Gray-code encoding/decoding maps are defined by structural induction on syntax.

Let Σ_{QC_d} consist of the following endomorphism generators:

$$\begin{array}{lll} \boxed{} : 0 \rightarrow 0, & \text{---} : 1 \rightarrow 1, & \text{X} : 2 \rightarrow 2, \\ \boxed{H^{(r,r+1)}} : 1 \rightarrow 1 \quad (r \in [d-1]), & \text{---} \text{---} \text{---} \text{---} : 0 \rightarrow 0 \quad (\theta \in \mathbb{R}). \end{array}$$

In particular, Σ_{QC_d} is infinite because it contains a generator for every real parameter θ .

Definition 2.5. Fix $d \geq 2$. For each $n \in \mathbb{N}$, let $\mathbf{CQC}_d^{\text{raw}}(n, n)$ be the set of *raw n-wire circuits*, defined as the smallest family of sets $(\mathbf{CQC}_d^{\text{raw}}(n, n))_{n \in \mathbb{N}}$ closed under:

- (1) *Generators.* For every generator $g : n \rightarrow n$ of Σ_{QC_d} , g is in $\mathbf{CQC}_d^{\text{raw}}(n, n)$.
- (2) *Sequential composition.* If $C, D \in \mathbf{CQC}_d^{\text{raw}}(n, n)$ then $D \circ C \in \mathbf{CQC}_d^{\text{raw}}(n, n)$.
- (3) *Parallel composition.* If $C \in \mathbf{CQC}_d^{\text{raw}}(m, m)$ and $D \in \mathbf{CQC}_d^{\text{raw}}(n, n)$ then $C \otimes D \in \mathbf{CQC}_d^{\text{raw}}(m+n, m+n)$.
- (4) *Control.* For each $k \in [d]$, if $C \in \mathbf{CQC}_d^{\text{raw}}(n, n)$ then $C_k(C) \in \mathbf{CQC}_d^{\text{raw}}(1+n, 1+n)$.

Thus $\mathbf{CQC}_d^{\text{raw}}$ is the free term algebra generated by Σ_{QC_d} using $\circ$, $\otimes$, and the unary constructors $C_k(-)$. A raw term records the syntactic structure of a controlled circuit.

We will occasionally need explicit raw representatives for identities and permutations (e.g. when stating coherence axioms). For each $n \geq 0$ let $\text{id}_n \in \mathbf{CQC}_d^{\text{raw}}(n, n)$ be the canonical raw identity: $\text{id}_0 := \boxed{}$ and $\text{id}_{n+1} := \text{id}_n \otimes \text{---}$.

Permutation terms are raw circuits generated from identities and the swap $\bowtie$ using only $\circ$ and $\otimes$; they denote wire permutations and are canonically invertible. In particular, for each $m, n \geq 0$ we fix a canonical *block-swap* permutation term $\sigma_{m,n} : m + n \rightarrow n + m$, defined recursively by $\sigma_{0,n} := \text{id}_n$, $\sigma_{m,0} := \text{id}_m$, $\sigma_{1,1} := \bowtie$, $\sigma_{m,1} := (\sigma_{m-1,1} \otimes \text{id}_1) \circ (\text{id}_{m-1} \otimes \sigma_{1,1})$, and $\sigma_{m,n} := (\text{id}_1 \otimes \sigma_{m,n-1}) \circ (\sigma_{m,1} \otimes \text{id}_{n-1})$.

The raw syntax carries more parentheses and wiring bookkeeping than we want to treat as meaningful. We therefore quotient by the least congruence that forces: (i) the strict PROP laws; and (ii) the axioms saying that each C_k really is a control functor.

Definition 2.6. Let $\equiv$ be the least congruence on $\mathbf{CQC}_d^{\text{raw}}$ —that is, an equivalence relation on each $\mathbf{CQC}_d^{\text{raw}}(n, n)$, closed under all contexts built from $\circ$, $\otimes$, and each C_k , that contains all well-typed instances of:

- the strict PROP equations [23] (associativity/unit for $\circ$ and $\otimes$, interchange, symmetry naturality and involutivity); and
- for each $k \in [d]$, the functoriality and coherence equations for C_k from Definition 2.2: $C_k(\text{id}_n) \equiv \text{id}_{1+n}$, $C_k(D \circ C) \equiv C_k(D) \circ C_k(C)$, and the coherence schemata (1)–(3) (with π ranging over permutation terms).

Thus $C \equiv C'$ means that C and C' differ only by the PROP coherence and by the coherence laws that make each C_k a well-behaved control constructor. No gate-specific equations are imposed here; those enter later via the equational theory $\mathbf{QC}_d$.

Definition 2.7. Define the *endomorphism PROP* $\mathbf{CQC}_d$ by

$$\mathbf{CQC}_d(n, n) := \mathbf{CQC}_d^{\text{raw}}(n, n) / \equiv, \quad \mathbf{CQC}_d(m, n) := \emptyset \text{ if } m \neq n,$$

with $\circ$ and $\otimes$ induced by the term constructors on representatives, and with symmetry $\sigma_{m,n}$ induced by the corresponding raw permutation terms.

Write $[-]_{\equiv} : \mathbf{CQC}_d^{\text{raw}}(n, n) \rightarrow \mathbf{CQC}_d(n, n)$ for the quotient map. By construction, each C_k respects $\equiv$ and therefore descends to an endofunctor on $(\mathbf{CQC}_d)_{\text{endo}}$ satisfying Definition 2.2. We call $(\mathbf{CQC}_d, (C_k)_{k=0}^{d-1})$ the *polycontrolled PROP of d -level quantum circuits*.

2.3 Qudit semantics

We now fix the intended semantics: circuits denote unitaries on $(\mathbb{C}^d)^{\otimes n}$. Fix $d \geq 2$ and recall $[d] := \{0, 1, \dots, d-1\}$. Let $\mathbb{C}^d$ be the d -dimensional Hilbert space with computational basis $\{|0\rangle, \dots, |d-1\rangle\}$. For $n \geq 0$, set $\mathcal{H}_n := (\mathbb{C}^d)^{\otimes n} \cong \mathbb{C}^{d^n}$, with basis $\{|x\rangle \mid x \in [d]^n\}$, and write $I_{\mathcal{H}_n}$ for the identity on $\mathcal{H}_n$.

Recall that $U : \mathcal{H}_n \rightarrow \mathcal{H}_n$ is *unitary* if $U^\dagger U = UU^\dagger = I_{\mathcal{H}_n}$, equivalently $U^{-1} = U^\dagger$. We write $U(N) := \{U \in \mathbb{C}^{N \times N} \mid U^\dagger U = UU^\dagger = I_N\}$.

Definition 2.8. Let $\mathbf{Qudit}_d$ be the *endomorphism PROP* defined by the hom-sets

$$\mathbf{Qudit}_d(n, n) := U(d^n), \quad \mathbf{Qudit}_d(m, n) := \emptyset \text{ if } m \neq n,$$

where we identify $\mathcal{H}_n \cong \mathbb{C}^{d^n}$ via the computational basis. Composition is ordinary composition of linear maps (matrix multiplication),

tensor on morphisms is the tensor product of operators, and the symmetry $\sigma_{m,n} : m + n \rightarrow n + m$ is the swap unitary determined by

$$\sigma_{m,n}(|x\rangle \otimes |y\rangle) := |y\rangle \otimes |x\rangle \quad (x \in [d]^m, y \in [d]^n).$$

In particular, $\mathbf{Qudit}_d(0, 0) \cong U(1) = \{z \in \mathbb{C} \mid |z| = 1\}$ consists of global phases.

Given any unitary $U \in \mathbf{Qudit}_d(n, n)$ and a basis value $k \in [d]$, define its *k-controlled extension* $C_k(U) \in \mathbf{Qudit}_d(n+1, n+1)$ by

$$C_k(U) := |k\rangle \langle k| \otimes U + \sum_{\ell \in [d], \ell \neq k} |\ell\rangle \langle \ell| \otimes I_{\mathcal{H}_n}. \quad (4)$$

Semantically, $C_k(U)$ applies U to the target register iff the added control qudit is in state $|k\rangle$, and otherwise acts as the identity.

Definition 2.9. Define, by structural induction, a family of maps

$$[\cdot] : \mathbf{CQC}_d^{\text{raw}}(n, n) \longrightarrow \mathbf{Qudit}_d(n, n) \quad (n \in \mathbb{N})$$

as follows.

- *Generators.* On the generators of $\Sigma_{\mathbf{QC}_d}$ set:

$$\begin{aligned} - \llbracket \text{---} \rrbracket &:= 1, \quad \llbracket \Theta \rrbracket := e^{i\theta}, \quad \llbracket \text{---} \rrbracket := \sum_{k \in [d]} |k\rangle \langle k|, \\ - \llbracket \bowtie \rrbracket &:= \sum_{i,j \in [d]} |j, i\rangle \langle i, j|, \\ - \llbracket H^{(r,r+1)} \rrbracket &:= \sum_{k \in [d] \setminus \{r, r+1\}} |k\rangle \langle k| \\ &\quad + \frac{1}{\sqrt{2}} (|r\rangle + |r+1\rangle) \langle r| \\ &\quad + \frac{1}{\sqrt{2}} (|r\rangle - |r+1\rangle) \langle r+1|. \end{aligned}$$

- *Sequential and parallel composition.* For raw circuits C, D , set

$$\llbracket D \circ C \rrbracket := \llbracket D \rrbracket \circ \llbracket C \rrbracket, \quad \llbracket C \otimes D \rrbracket := \llbracket C \rrbracket \otimes \llbracket D \rrbracket.$$

- *Control.* For $C \in \mathbf{CQC}_d^{\text{raw}}(n, n)$ and $k \in [d]$, set

$$\llbracket C_k(C) \rrbracket := C_k(\llbracket C \rrbracket) \in \mathbf{Qudit}_d(1+n, 1+n). \quad (5)$$

Proposition 2.10. The interpretation $[\cdot] : \mathbf{CQC}_d^{\text{raw}}(n, n) \rightarrow \mathbf{Qudit}_d(n, n)$ respects the structural congruence $\equiv$ of Definition 2.6. Hence it factors uniquely through the quotient to a well-defined strict symmetric monoidal functor

$$[\cdot] : \mathbf{CQC}_d \longrightarrow \mathbf{Qudit}_d,$$

satisfying $\llbracket [C]_{\equiv} \rrbracket = \llbracket C \rrbracket$ for every raw circuit C .

PROOF. The strict PROP equations are respected because $\circ$ is interpreted as composition of linear maps, $\otimes$ as tensor product, and the swap generator as the swap unitary. For control, the functoriality of $C_k(-)$ together with (4) yields $\llbracket C_k(\text{id}_n) \rrbracket = I_{\mathcal{H}_{n+1}}$ and $\llbracket C_k(D \circ C) \rrbracket = \llbracket C_k(D) \rrbracket \circ \llbracket C_k(C) \rrbracket$. The coherence axioms (1)–(3) hold because $C_k(-)$ acts only on the additional control tensor factor and is therefore insensitive to unused wires and to permutations of the target wires. $\square$

2.4 Universality

Fix $d \geq 2$. We now record that our gate signature $\Sigma_{\mathbf{QC}_d}$ is universal for d -level unitary computation: every unitary $U \in U(d^n)$ is denoted by some circuit in $\mathbf{CQC}_d(n, n)$.

We rely on two standard synthesis facts. First, every 2×2 unitary admits an Euler decomposition in terms of a Hadamard and computational-basis phases [2, 24]: for all $U \in U(2)$ there exist $\theta, \alpha, \beta, \gamma \in \mathbb{R}$ such that

$$U = e^{i\theta} Z(\alpha) H Z(\beta) H Z(\gamma),$$

where $H \in U(2)$ is the Hadamard and $Z(\phi) = \text{diag}(1, e^{i\phi})$.

Second, $U(d)$ is generated by *adjacent two-level* unitaries [10, 16, 24, 29]: every $V \in U(d)$ is a product of unitaries, each supported on $\text{span}\{|r\rangle, |r+1\rangle\}$ for some $r \in [d-1]$.

For convenience, define

$$Z_d^{(j)}(\theta) := \llbracket C_j(\textcircled{\theta}) \rrbracket, \quad H_d^{(r,r+1)} := \llbracket -\boxed{H^{(r,r+1)}}- \rrbracket.$$

Lemma 2.11. *Fix $r \in [d-1]$. Let $W \in U(d)$ act as the identity on the orthogonal complement of $\text{span}\{|r\rangle, |r+1\rangle\}$. Then there exists a 1-qudit circuit $C_W \in \mathbf{CQC}_d(1, 1)$ such that $\llbracket C_W \rrbracket = W$.*

PROOF. Restrict W to $\text{span}\{|r\rangle, |r+1\rangle\} \cong \mathbb{C}^2$. By the Euler decomposition, this restriction factors (up to a global phase) as a product of 2×2 Hadamards and computational-basis phases. Embedding back into $\mathbb{C}^d$, each factor becomes either an adjacent Hadamard $H_d^{(r,r+1)}$, a level-phase $Z_d^{(j)}(\cdot)$, or a global phase. Each is realised by a circuit in $\mathbf{CQC}_d$, and composing them yields C_W . $\square$

Define the two-qudit unitary $CZ_d := I_{\mathcal{H}_2} - 2|0, 0\rangle\langle 0, 0| \in U(d^2)$. Equivalently, CZ_d sends $|0, 0\rangle \mapsto -|0, 0\rangle$ and fixes all other computational basis states. In $\mathbf{CQC}_d$, $\llbracket C_0(C_0(\textcircled{\pi})) \rrbracket = CZ_d$.

We use the standard universality fact that for all $n \geq 0$, arbitrary single-qudit unitaries together with CZ_d generate $U(d^n)$ [3].

Proposition 2.12. *For every $n \geq 0$, the interpretation map $\llbracket \cdot \rrbracket : \mathbf{CQC}_d(n, n) \rightarrow \mathbf{Qudit}_d(n, n) = U(d^n)$ is surjective.*

PROOF. Let $U \in U(d^n)$.

If $n = 0$, then $U \in U(1)$ is a scalar phase $e^{i\theta}$, denoted by $\textcircled{\theta}$.

Assume $n \geq 1$. By adjacent two-level generation of $U(d)$, every single-qudit unitary can be written as a product of unitaries supported on $\text{span}\{|r\rangle, |r+1\rangle\}$. By Lemma 2.11, each such factor is realised in $\mathbf{CQC}_d(1, 1)$, hence arbitrary single-qudit unitaries are realised.

Together with CZ_d , these single-qudit unitaries generate $U(d^n)$. The gate CZ_d is realised in $\mathbf{CQC}_d$, and the PROP structure (tensoring with identities and permutations) allows placing these realised 1- and 2-qudit circuits on any desired wire(s). Composing the resulting circuits yields some $C \in \mathbf{CQC}_d(n, n)$ with $\llbracket C \rrbracket = U$. $\square$

2.5 Equational theories on $\mathbf{CQC}_d$

The PROP $\mathbf{CQC}_d$ quotients raw circuits only by *structural* bookkeeping: PROP coherence plus the coherence laws of control functors. To reason about genuine circuit identities (e.g. moving phases, rewriting controlled patterns, decomposing swaps), we will work with an additional equational theory Γ on $\mathbf{CQC}_d$.

It is useful to keep two notions of equality separate:

- *structural equality* $C \equiv C'$ on raw syntax (Definition 2.6); and
- *content equality* generated by extra circuit equations in a chosen equational theory Γ on $\mathbf{CQC}_d$.

Definition 2.13. An *equational theory* Γ on $\mathbf{CQC}_d$ is a collection of well-typed equations $f = g$ between morphisms of $\mathbf{CQC}_d$.

For such a Γ , let $(\approx_\Gamma)_n$ be the least family of relations on the hom-sets $\mathbf{CQC}_d(n, n)$ such that:

- (1) every axiom $f = g$ in Γ satisfies $f \approx_\Gamma g$;
- (2) each $\approx_\Gamma$ is an equivalence relation;
- (3) $f_1 \approx_\Gamma g_1$ and $f_2 \approx_\Gamma g_2$ imply $f_2 \circ f_1 \approx_\Gamma g_2 \circ g_1$;

- (4) $f \approx_\Gamma g$ and $h \approx_\Gamma k$ imply $f \otimes h \approx_\Gamma g \otimes k$;
- (5) $f \approx_\Gamma g$ implies $C_k(f) \approx_\Gamma C_k(g)$ for every $k \in [d]$.

We write $\Gamma \vdash f = g$ to mean $f \approx_\Gamma g$. When C, C' are raw terms, the judgement $\Gamma \vdash C = C'$ is shorthand for $\Gamma \vdash [C]_\equiv = [C']_\equiv$ in $\mathbf{CQC}_d$. Intuitively, $\Gamma \vdash f = g$ is the congruence closure of Γ under sequential composition, parallel composition, and adding controls.

Definition 2.14. For each $d \geq 2$, let $\mathbf{QC}_d$ be the equational theory on $\mathbf{CQC}_d$ generated by the axiom *schemata* displayed in Figures 1 and 2.

We will show in Theorem 4.19 that $\mathbf{QC}_d$ is sound and complete for the unitary semantics $\llbracket - \rrbracket : \mathbf{CQC}_d \rightarrow \mathbf{Qudit}_d$. Some schemata in Figures 1–2 use derived-gate abbreviations, defined next in §3.1. Having fixed $\mathbf{CQC}_d$ and its intended semantics, we can now state the finite set of local schemata that axiomatise equality in $\mathbf{CQC}_d$.

3 A Finite Axiom System for Qudit Circuits

Fix $d \geq 2$. This section presents the finite set of *local* axiom schemata that generates our equational theory $\mathbf{QC}_d$ on the circuit PROP $\mathbf{CQC}_d$ (Definition 2.7). These schemata are the non-structural rewrite principles used later in the completeness transfer of Section 4. Throughout, all equations are taken in $\mathbf{CQC}_d$, i.e. modulo the PROP laws and the control-coherence equations already built into $\mathbf{CQC}_d$.

3.1 Derived gates used in the axiom schemata

The generators of $\mathbf{CQC}_d$ are the structural PROP circuits, the adjacent two-level Hadamards $\llbracket -\boxed{H^{(r,r+1)}}- \rrbracket$ ($r \in [d-1]$), the scalar phases $\textcircled{\theta}$ ($\theta \in \mathbb{R}$), and the control constructors $C_k(-)$. The axiom schemata in Figures 1–2 are written using a small family of *derived* one-qudit gates (and derived two-level Hadamards), matching the decoding notation of §4.4. These are purely *abbreviations* for composite circuits in $\mathbf{CQC}_d$: they introduce no new generators; instead, they package common subcircuits so that the schemata remain readable.

For readability, we present these derived gates in algebraic form in this subsection only. Crucially, symbols such as $H^{(i,j)}$, $X^{(i,j)}$, and $R_x^{(i,j)}(\theta)$ denote *morphisms in $\mathbf{CQC}_d$* (i.e. circuit diagrams modulo the structural congruence), *not* matrices. Only after applying the interpretation functor $\llbracket - \rrbracket : \mathbf{CQC}_d \rightarrow \mathbf{Qudit}_d$ do they correspond to the intended unitary operators. We write $H^{(r,r+1)}$ for the adjacent Hadamard generator $\llbracket -\boxed{H^{(r,r+1)}}- \rrbracket$.

We will use three families of derived one-qudit morphisms (all abbreviations in $\mathbf{CQC}_d$):

- (1) $X^{(i,j)}$ for $i, j \in [d]$;
- (2) $H^{(i,j)}$ for distinct $i, j \in [d]$;
- (3) $R_x^{(i,j)}(\theta)$ for distinct $i, j \in [d]$ and $\theta \in \mathbb{R}$.

For each adjacent pair $(r, r+1)$ we define the adjacent swap $X^{(r,r+1)}$ by

$$X^{(r,r+1)} := H^{(r,r+1)} \circ C_{r+1}(\textcircled{\pi}) \circ H^{(r,r+1)} \quad (6)$$

For notational uniformity, we set $X^{(i,i)} := \text{id}_1$. For distinct $i, j \in [d]$, we define the two-level swap $X^{(i,j)}$ as a composite of adjacent swaps. Assuming $i < j$, set $S_{i,j} := \prod_{k=0}^{j-i-1} X^{(i+k, i+k+1)}$ where $\prod_{k=0}^m f_k := f_m \circ \dots \circ f_0$, then

$$X^{(i,j)} := S_{i+1,j}^{-1} \circ X^{(i,i+1)} \circ S_{i+1,j}, \quad (7)$$

and for $i > j$ set $X^{(i,j)} := X^{(j,i)}$.

We define the two-level Hadamard $H^{(i,j)}$ on levels (i, j) by conjugating an adjacent Hadamard by the corresponding swap that “slides” the relevant level into position. For $i = j$, set $H^{(i,i)} := \text{id}_1$. If $i < j$, set

$$H^{(i,j)} := X^{(j,i+1)} \circ H^{(i,i+1)} \circ X^{(j,i+1)}, \quad (8)$$

and for $i > j$ define the ordered version by swapping the two levels:

$$H^{(i,j)} := X^{(j,i)} \circ H^{(j,i)} \circ X^{(j,i)} \quad (9)$$

Finally, for distinct $i, j \in [d]$ and $\theta \in \mathbb{R}$, we define the two-level beam-splitter rotation $R_x^{(i,j)}(\theta)$ by

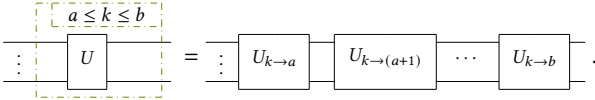
$$R_x^{(i,j)}(\theta) := H^{(i,j)} \circ C_i(\bigcirc) \circ C_j(\bigcirc) \circ H^{(i,j)} \quad (10)$$

Under $\llbracket - \rrbracket$, these families act as follows:

- $\llbracket X^{(i,j)} \rrbracket$ swaps $|i\rangle$ and $|j\rangle$ and fixes all other computational basis states.
- $\llbracket H^{(i,j)} \rrbracket$ acts as the usual 2×2 Hadamard on $\text{span}\{|i\rangle, |j\rangle\}$ (in the ordered basis $(|i\rangle, |j\rangle)$) and as the identity on the orthogonal complement.
- $\llbracket R_x^{(i,j)}(\theta) \rrbracket$ acts on $\text{span}\{|i\rangle, |j\rangle\}$ as $\begin{pmatrix} \cos(\theta) & i \sin(\theta) \\ i \sin(\theta) & \cos(\theta) \end{pmatrix}$ (in the ordered basis $(|i\rangle, |j\rangle)$) and as the identity elsewhere.

To state several families of axioms systematically in d , we use a compact meta-notation in circuit diagrams. An *indexed box* is a dash-dotted rectangle surrounding a subdiagram. It is *not* part of the raw syntax of Definition 2.5; it abbreviates a finite iterated composite of the enclosed circuit.

An indexed box is annotated by an index variable k together with an optional unary predicate $\varphi(k)$, where k is the only free variable. The intended reading is that k ranges over integers with $0 \leq k < d$ that satisfy $\varphi(k)$ (for instance $\varphi(k)$ may be $a \leq k < b$, or $k \neq 0$). The box denotes the sequential composite of the enclosed subcircuit, taken in increasing order of k :



If the predicate is omitted (we write just “ k ”), this abbreviates the default predicate $0 \leq k < d$.

For instance, if $d = 3$, then $\llbracket H^{(i,j)} \rrbracket = \llbracket H^{(0,j)} \rrbracket \llbracket H^{(1,j)} \rrbracket \llbracket H^{(2,j)} \rrbracket$.

Note that indexed boxes expand only by *sequential* composition along the same wires, so they do not increase wire arity (all schemata still involve at most three wires).

3.2 The axiom schemata and how to read them

This subsection fixes the axiom schemata generating QC_d , and explains the intended roles among them. The rules are presented as local equalities between circuit diagrams; later, we use them to derive the global control-algebra principles (compatibility, commutativity, and exhaustivity) needed for the encoding/decoding arguments.

We define QC_d as the congruence generated by the schematic equations in Figures 1–2 and closed under $\circ$, $\otimes$, and all C_k (Definition 2.13). As always, these are equations in CQC_d ; they are stated

modulo the PROP law and the control functoriality/coherence already built into CQC_d .

A useful reading is that Figure 1 collects the one- and two-qudit identities that form the main “gate algebra” of the theory, while Figure 2 supplies the finite gate-level ingredients from which the global control principles of Definition 2.4 become derivable.

We split the non-structural axioms into two groups according to their role in the development.

- *Core axioms* (Figure 1) form the main algebraic interface of the circuit theory: phase arithmetic, basic two-level identities, the swap decomposition, and the local decomposition/rotation principles.
- *Control-algebra support axioms* (Figure 2) are included so that the global control-algebra principles—namely *compatibility*, *commutativity*, and *exhaustivity* from Definition 2.4—are *derivable inside* QC_d , rather than assumed as part of the underlying structure. Concretely, one could strengthen the definition of CQC_d by quotienting the raw syntax additionally by compatibility, commutativity and exhaustivity of control (or by taking these as global rewrite schemata). We deliberately do not do this: these principles are semantically sound for the intended projector-based control interpretation, but they are not consequences of control functoriality alone. Keeping them out of the structural congruence makes the ambient PROP CQC_d conservative, and ensure that the completeness theorem is genuinely a statement about what follows from the finite axiom set QC_d .

3.3 Derived control-algebra laws

The three control-algebra equations of Definition 2.4 are derivable in the circuit PROP CQC_d from the axiom system QC_d . The proofs are purely syntactic and are deferred to Appendices C, D and E. The present subsection records the results and the common reduction step used throughout.

Lemma 3.1. *Every morphism of CQC_d is provably equal in QC_d to a circuit generated (using only sequential and parallel composition) by*

$$\mathcal{G} = \{\text{id}_n \mid n \in \mathbb{N}\} \cup \{\bigcirc(\theta) \mid \theta \in \mathbb{R}\} \cup \{\llbracket H^{(i,j)} \rrbracket \mid 0 \leq i < j < d\} \\ \cup \{C_k(\bigcirc(\theta)) \mid k \in [d], \theta \in \mathbb{R}\} \cup \{C_k(C_\ell(\pi)) \mid k, \ell \in [d]\}.$$

Lemma 3.1 is obtained by a normalisation procedure that pushes controls down to a bounded set of controlled “layers” (Appendix C.1). Once circuits have been reduced to this form, each control-algebra statement can be proved by tracking how the outer controls interact with layers from $\mathcal{G}$. Compatibility reduces to exchanging the (unique) layers that simultaneously involve both outer control wires (Appendix C); commutativity reduces to commuting pairs of differently-controlled layers from $\mathcal{G}$ (Appendix D); and exhaustivity is then proved by induction, using the totalisation axioms together with the derived compatibility and commutativity steps (Appendix E).

Theorem 3.2. *For any $n \geq 0$, any $C : n \rightarrow n$ in CQC_d , and any $a, b \in [d]$,*

$$\text{QC}_d \vdash C_a(C_b(C)) \circ (\bigwedge \otimes \text{id}_n) = (\bigwedge \otimes \text{id}_n) \circ C_b(C_a(C)).$$

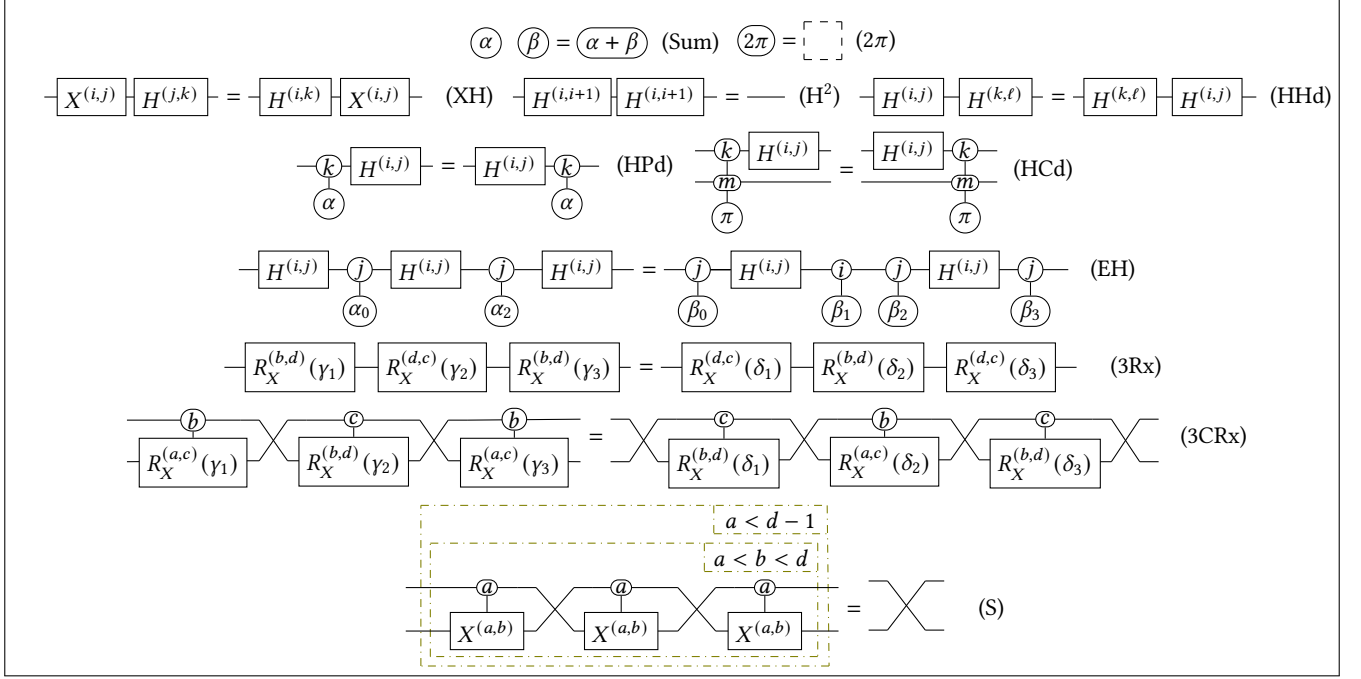


Figure 1: Core axiom schemata for QC_d . In (EH), (3Rx) and (3CRx) the right-hand-side angles are determined by the explicit relations recorded in Appendix A. Rules (XH), (HPd), and (HCd) apply when the indices i, j, k are pairwise distinct. In (HHd) the indices i, j, k, ℓ are pairwise distinct.

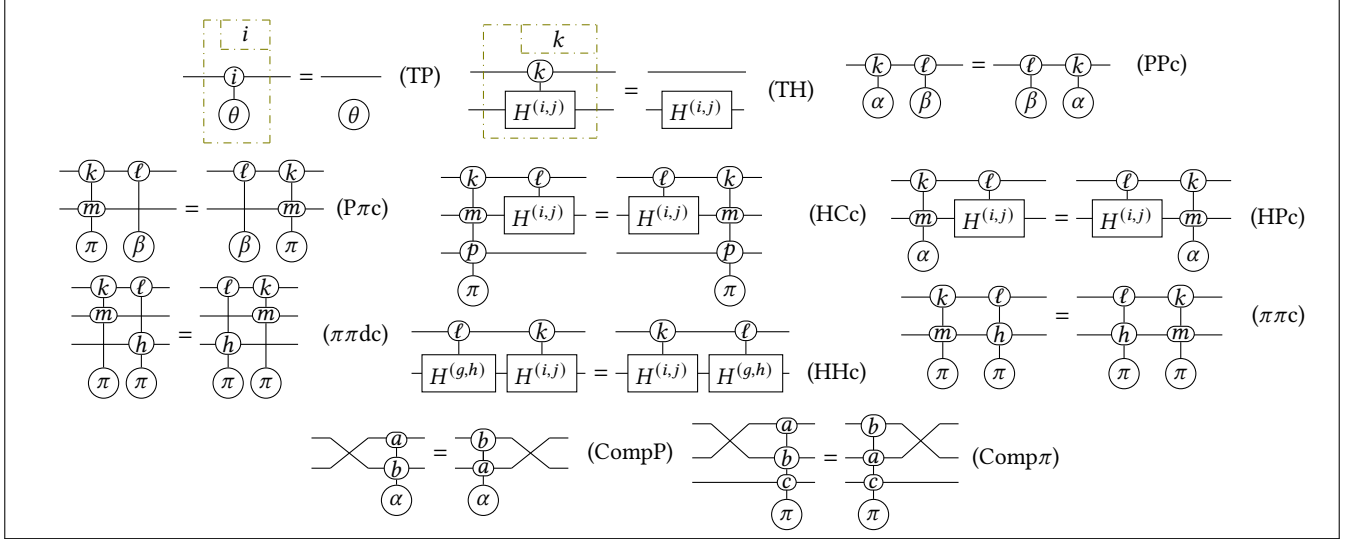


Figure 2: Auxiliary axiom schemata for QC_d . (1) “Totalisation” rules (TP) and (TH), used to recover exhaustivity of control by a structural induction on circuits. If one takes exhaustivity as an admissible rule schema, then (TP) and (TH) are immediate instances and can be omitted. (2) Local commutation/conjugation schemata for *differently-controlled* operations: (PPc), (Pπc), (ππdc), (HPc), (HCc), (HHc) require $k \neq \ell$. These patterns are redundant once one is allowed to use distinct-control commutativity as a derived rule. (3) The mixed-control swap schemata (CompP) and (Compπ). If compatibility were assumed as an ambient (structural) principle of CQC_d , then (CompP) and (Compπ) would be derivable and could be omitted.

Theorem 3.3. For any $n \geq 0$, any $C_1, C_2 : n \rightarrow n$ in CQC_d , and any distinct $k, \ell \in [d]$,

$$QC_d \vdash C_k(C_1) \circ C_\ell(C_2) = C_\ell(C_2) \circ C_k(C_1).$$

Theorem 3.4. For every $C : n \rightarrow n$ in $\mathbf{CQC}_d$,

$$\mathbf{QC}_d \vdash \prod_{k=0}^{d-1} C_k(C) = \text{---} \otimes C.$$

4 Embedding Qudit Circuits into LOPP Circuits

Our completeness proof for $\mathbf{QC}_d$ proceeds by reduction to linear optics. We translate qudit circuits in the PROP $\mathbf{CQC}_d$ into circuits in the linear-optical PROP **LOPP**, appeal to the sound and complete equational theory LOPP for **LOPP** [7, 18], and then transport derivations back to $\mathbf{CQC}_d$. The bridge is a fixed reflected d -ary Gray ordering of the computational basis (§4.2): it identifies the d^n optical modes with the n -qudit basis states in such a way that *consecutive* modes differ in exactly one digit by ± 1 . As a result, each local **LOPP** generator (acting on one mode or two consecutive modes) decodes to either a basis-controlled phase or a (multi-)controlled adjacent two-level qudit gate.

A key technical point (already present for qubits [8]) is that, although both $\mathbf{CQC}_d$ and **LOPP** are PROPs, their monoidal products have different semantics: $\otimes$ in $\mathbf{CQC}_d$ is interpreted by tensor product on Hilbert spaces, whereas $\otimes$ in **LOPP** is interpreted by direct sum under the single-photon model (Remark 1). Consequently, our translations preserve sequential composition $\circ$ but cannot be strict monoidal PROP morphisms. We therefore work on *raw* syntax and fix a deterministic serialisation of $\otimes$.

Concretely, we define two context-sensitive translations:

- a *contextual encoding* E_b^a (Definition 4.8) sending a raw n -qudit circuit to a raw **LOPP** circuit on d^{a+n+b} modes, intended to represent the circuit acting on the *middle* n qudits of an $(a+n+b)$ -qudit register; and
- a *contextual decoding* D_n^t (Definition 4.11) sending a raw **LOPP** circuit supported on consecutive modes $t, t+1, \dots, t+\ell-1 \subseteq \{0, \dots, d^n-1\}$ to a raw n -qudit circuit, by interpreting each local optical generator as a suitably (multi-)controlled two-level qudit gate in Gray order.

The completeness transfer uses three ingredients:

- (1) ENC (Lemma 4.10): E is semantics-preserving for the Gray-ordered LOPP semantics $\llbracket - \rrbracket_{\text{LOPP}}$ (Definition 4.7);
- (2) RETR (Theorem 4.17): decoding the encoding recovers the original circuit (padded by identities), provably in $\mathbf{QC}_d$; and
- (3) MIMIC (Theorem 4.16): any LOPP derivation decodes to a $\mathbf{QC}_d$ derivation.

Together with completeness of LOPP, this yields completeness of $\mathbf{QC}_d$.

4.1 The LOPP calculus as a semantic model

This subsection recalls the linear-optical circuit PROP **LOPP** and its standard *single-photon* unitary semantics. These are the only facts about **LOPP** we use in the proof transfer: the raw syntax and quotient PROP, the interpretation $\llbracket - \rrbracket_{\text{sp}}$, and the existence of a fixed finite sound and complete equational theory.

Let Σ^{LOPP} be the (typed) set of generators

$$\boxed{} : 0 \rightarrow 0, \quad \text{---} : 1 \rightarrow 1, \quad \text{---} \times \text{---} : 2 \rightarrow 2,$$

together with the optical generators (each indexed by $\theta \in \mathbb{R}$)

$$\boxed{\theta} : 1 \rightarrow 1, \quad \text{---} \times \text{---}^\theta : 2 \rightarrow 2.$$

Intuitively, these are the empty diagram, the identity wire, a swap of two adjacent modes, and the two basic optical components: a phase shifter and a beam splitter.

Definition 4.1. Let $\mathbf{LOPP}^{\text{raw}}$ be the free *endomorphism* PROP on Σ^{LOPP} (so $\mathbf{LOPP}^{\text{raw}}(m, n) = \emptyset$ for $m \neq n$). Let $\equiv$ be the least congruence on $\mathbf{LOPP}^{\text{raw}}$ containing all instances of the strict PROP equations (identities, associativity/unit for $\circ$ and $\otimes$, interchange, symmetry naturality and involutivity), as in Definition 2.6.

Define the endomorphism PROP **LOPP** by

$$\mathbf{LOPP}(m, m) := \mathbf{LOPP}^{\text{raw}}(m, m) / \equiv, \quad \mathbf{LOPP}(m, n) := \emptyset \text{ if } m \neq n,$$

with $\circ$ and $\otimes$ induced by the term constructors.

Thus, a morphism in $\mathbf{LOPP}(m, m)$ is an m -mode linear-optical circuit modulo strict symmetric-monoidal bookkeeping.

For $m \geq 0$, let $\mathcal{M}_m := \mathbb{C}^m$ with its standard *mode basis* $\{|0\rangle, \dots, |m-1\rangle\}$. We identify unitaries $U \in U(\mathcal{M}_m)$ with their $m \times m$ matrices in this basis, i.e. with elements of $U(m)$. For disjoint mode blocks of sizes m and n , parallel composition corresponds to the *block-diagonal direct sum*: for $U \in U(m)$ and $V \in U(n)$,

$$U \oplus V := \begin{pmatrix} U & 0 \\ 0 & V \end{pmatrix} \in U(m+n),$$

using the canonical identification $\mathcal{M}_{m+n} \cong \mathcal{M}_m \oplus \mathcal{M}_n$.

Definition 4.2. Working in the standard mode bases of $\mathcal{M}_m = \mathbb{C}^m$, we interpret every raw **LOPP** term on m modes as an $m \times m$ unitary matrix.

The *single-photon interpretation* of **LOPP** is the family of maps

$$\llbracket \cdot \rrbracket_{\text{sp}} : \mathbf{LOPP}^{\text{raw}}(m, m) \longrightarrow U(m) \quad (m \in \mathbb{N})$$

defined by structural induction on raw terms:

- *Generators.*

$$\begin{aligned} \llbracket \boxed{} \rrbracket_{\text{sp}} &:= 1, & \llbracket \text{---} \rrbracket_{\text{sp}} &:= (1), & \llbracket \boxed{\theta} \rrbracket_{\text{sp}} &:= (e^{i\theta}), \\ \llbracket \text{---} \times \text{---} \rrbracket_{\text{sp}} &:= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, & \llbracket \text{---} \times \text{---}^\theta \rrbracket_{\text{sp}} &:= \begin{pmatrix} \cos(\theta) & i \sin(\theta) \\ i \sin(\theta) & \cos(\theta) \end{pmatrix}. \end{aligned}$$

- *Sequential composition.* For $C_1, C_2 \in \mathbf{LOPP}^{\text{raw}}(m, m)$,

$$\llbracket C_2 \circ C_1 \rrbracket_{\text{sp}} := \llbracket C_2 \rrbracket_{\text{sp}} \circ \llbracket C_1 \rrbracket_{\text{sp}}.$$

- *Parallel composition.* For $C_1 \in \mathbf{LOPP}^{\text{raw}}(m, m)$ and $C_2 \in \mathbf{LOPP}^{\text{raw}}(n, n)$, $\llbracket C_1 \otimes C_2 \rrbracket_{\text{sp}} := \llbracket C_1 \rrbracket_{\text{sp}} \oplus \llbracket C_2 \rrbracket_{\text{sp}}.$

The interpretation $\llbracket - \rrbracket_{\text{sp}}$ respects the structural congruence $\equiv$ of Definition 4.1, hence factors through the quotient to a well-defined family of maps $\llbracket - \rrbracket_{\text{sp}} : \mathbf{LOPP}(m, m) \rightarrow U(\mathcal{M}_m)$, which we again denote by $\llbracket - \rrbracket_{\text{sp}}$.

Remark 1. Under the single-photon interpretation, the monoidal product of **LOPP** is *block-diagonal sum*:

$$\llbracket C_1 \otimes C_2 \rrbracket_{\text{sp}} = \llbracket C_1 \rrbracket_{\text{sp}} \oplus \llbracket C_2 \rrbracket_{\text{sp}}.$$

In contrast, the monoidal product of $\mathbf{CQC}_d$ is interpreted by the *Kronecker tensor product*:

$$\llbracket C_1 \otimes C_2 \rrbracket = \llbracket C_1 \rrbracket \otimes \llbracket C_2 \rrbracket.$$

Because $U \oplus V$ and $U \otimes V$ are fundamentally different operations, our translations preserve sequential composition but must serialise $\otimes$; they therefore cannot be strict monoidal PROP morphisms.

The LOPP signature contains no primitive notion of (poly)control; in our reduction, control arises only from the Gray-coded identification between optical modes and computational basis states.

Theorem 4.3 ([7, 18]). *Let LOPP denote the equational theory consisting of the structural PROP equations together with the non-structural axiom schemata of Figure 9 (listed explicitly in Appendix I). Then LOPP is sound and complete for the single-photon semantics: for any circuits $C_1, C_2 : m \rightarrow m$,*

$$\llbracket C_1 \rrbracket_{\text{sp}} = \llbracket C_2 \rrbracket_{\text{sp}} \implies \text{LOPP} \vdash C_1 = C_2.$$

Fix $d \geq 2$. Since an n -qudit system has Hilbert space dimension d^n , our encoding and decoding only use **LOPP** circuits on d^n modes. We write $\mathbf{LOPP}_d$ for the full subcategory of **LOPP** on objects $\{d^n \mid n \geq 0\}$. As noted in Remark 1, $\mathbf{LOPP}_d$ is not closed under the monoidal product on objects, but Theorem 4.3 applies *arity-wise* to each fixed object d^n .

4.2 Gray codes for mode ordering

LOPP generators are local in the mode index: they act on one mode, or on two *consecutive* modes. To decode such local optical steps as local qudit operations, we fix a Gray ordering of the d^n computational basis states. In this ordering, consecutive mode labels correspond to basis words that differ in exactly one digit, by ± 1 . We will use this neighbour property in §4.4 to interpret two-mode LOPP generators as (multi-)controlled adjacent two-level qudit gates.

Fix $d \geq 2$ and write $[d] = \{0, 1, \dots, d-1\}$. An n -qudit register has Hilbert space $\mathcal{H}_n = (\mathbb{C}^d)^{\otimes n}$ with computational basis $\{|x\rangle \mid x \in [d]^n\}$. In the single-photon optical model, a d^n -mode interferometer acts on $\mathcal{M}_{d^n}$ with basis $\{|t\rangle \mid 0 \leq t < d^n\}$, where $|t\rangle$ means “one photon in mode t ”. We now choose an explicit identification between these bases that interacts well with consecutive mode indices.

We use the standard reflected d -ary Gray code, which specialises to the usual binary reflected Gray code when $d = 2$.

Definition 4.4. Let $[d] = \{0, 1, \dots, d-1\}$. For each $n \geq 0$ define a map

$$G_n^d : \{0, 1, \dots, d^n - 1\} \longrightarrow [d]^n$$

recursively by:

- $G_0^d(0) = \epsilon$;
- for $n \geq 1$, write $0 \leq k < d^n$ uniquely as $k = q \cdot d^{n-1} + r$ with $0 \leq q < d$ and $0 \leq r < d^{n-1}$, and set

$$G_n^d(k) := \begin{cases} q \cdot G_{n-1}^d(r) & \text{if } q \text{ is even,} \\ q \cdot G_{n-1}^d(d^{n-1} - 1 - r) & \text{if } q \text{ is odd.} \end{cases}$$

Here ϵ is the empty word, and $q \cdot w$ denotes prefixing the word w by the digit q . The next lemma records the operational fact we need: in Gray order, incrementing the mode index changes exactly one computational digit, and only by an adjacent step.

For $d = 3$ and $n = 2$ the reflected Gray order is

$$00, 01, 02, 12, 11, 10, 20, 21, 22,$$

so successive words differ in exactly one digit, and that digit changes by ± 1 .

Lemma 4.5. *For all $n \geq 1$, we have $G_n^d(0) = (0, \dots, 0)$. Moreover, for each $0 \leq k < d^n - 1$ the words $G_n^d(k)$ and $G_n^d(k+1)$ differ in exactly one position, and in that position the digit changes by one: if*

$$G_n^d(k) = (a_1, \dots, a_n) \quad \text{and} \quad G_n^d(k+1) = (b_1, \dots, b_n),$$

then there is a unique $t \in \{1, \dots, n\}$ such that $a_t \neq b_t$ and $|a_t - b_t| = 1$, while $a_j = b_j$ for all $j \neq t$.

We package the resulting basis reordering as a permutation unitary on $\mathcal{H}_n$.

Definition 4.6. For each $n \geq 0$, define the unitary $\mathfrak{G}_{n,d} : \mathcal{M}_{d^n} \rightarrow \mathcal{H}_n$ as the *permutation matrix* implementing the Gray reordering:

$$\mathfrak{G}_{n,d} := \sum_{t=0}^{d^n-1} |G_n^d(t)\rangle \langle t|,$$

equivalently $\mathfrak{G}_{n,d}(|t\rangle) = |G_n^d(t)\rangle$ for $0 \leq t < d^n$. In particular, $\mathfrak{G}_{n,d}^{-1} = \mathfrak{G}_{n,d}^\dagger$.

Operationally, $\mathfrak{G}_{n,d}$ is the fixed change-of-basis map identifying the mode basis with the computational basis in Gray order.

Definition 4.7. For each $n \geq 0$ define a map on raw terms

$$\llbracket \cdot \rrbracket_{\text{LOPP}} : \mathbf{LOPP}^{\text{raw}}(d^n, d^n) \longrightarrow \mathbf{Qudit}_d(n, n)$$

by conjugating the single-photon matrix semantics with the Gray permutation matrix:

$$\llbracket C \rrbracket_{\text{LOPP}} := \mathfrak{G}_{n,d} \circ \llbracket C \rrbracket_{\text{sp}} \circ \mathfrak{G}_{n,d}^\dagger,$$

where $\llbracket - \rrbracket_{\text{sp}}$ is the single-photon interpretation of Definition 4.2.

Thus $\llbracket - \rrbracket_{\text{LOPP}}$ is the single-photon semantics transported to the qudit computational basis using the fixed Gray identification. For each $n \geq 0$, the map $\llbracket - \rrbracket_{\text{LOPP}}$ respects the structural congruence $\equiv$ on $\mathbf{LOPP}^{\text{raw}}(d^n, d^n)$, and hence factors through the quotient to a well-defined function

$$\llbracket - \rrbracket_{\text{LOPP}} : \mathbf{LOPP}(d^n, d^n) \longrightarrow \mathbf{Qudit}_d(n, n).$$

Moreover, the family $(\llbracket - \rrbracket_{\text{LOPP}})_{n \geq 0}$ is compatible with identities and sequential composition, i.e. for all $C_1, C_2 : d^n \rightarrow d^n$ in **LOPP**,

$$\llbracket \text{id}_{d^n} \rrbracket_{\text{LOPP}} = I_{\mathcal{H}_n}, \quad \llbracket C_2 \circ C_1 \rrbracket_{\text{LOPP}} = \llbracket C_2 \rrbracket_{\text{LOPP}} \circ \llbracket C_1 \rrbracket_{\text{LOPP}}.$$

Equivalently, $\llbracket - \rrbracket_{\text{LOPP}}$ is a well-defined functor $\mathbf{LOPP}_d \rightarrow \mathbf{Qudit}_d$ (with object map $d^n \mapsto n$).

Consequently, for $n \geq 0$ and $C_1, C_2 : d^n \rightarrow d^n$ in **LOPP**, if $\llbracket C_1 \rrbracket_{\text{LOPP}} = \llbracket C_2 \rrbracket_{\text{LOPP}}$ in $\mathbf{Qudit}_d(n, n)$, then $\text{LOPP} \vdash C_1 = C_2$.

4.3 Contextual encoding of quantum circuits

Fix $d \geq 2$. This subsection defines the *contextual encoding* $E_b^a : \mathbf{CQC}_d^{\text{raw}}(n, n) \rightarrow \mathbf{LOPP}^{\text{raw}}(d^{a+n+b}, d^{a+n+b})$, which translates an n -qudit circuit into a single-photon linear-optical circuit on d^{a+n+b} modes. The parameters (a, b) describe surrounding context: $E_b^a(C)$ is intended to represent C acting on the *middle* n qudits of an $(a+n+b)$ -qudit register.

We work on raw syntax, and we interpret **LOPP** circuits on d^n modes via the Gray-ordered semantics $\llbracket - \rrbracket_{\text{LOPP}}$ of Definition 4.7. As

noted in Remark 1, the translation preserves sequential composition $\circ$ but must *serialise* $\otimes$.

The encoding uses three kinds of operations on raw **LOPP** syntax: reordering Gray-ordered mode blocks to place an encoded subcircuit in context; lifting a local d^n -mode circuit across surrounding Gray branches; and lifting only along a designated branch to model control. Because reflected Gray order reverses the traversal of blocks with odd leading digit, the lifting gadgets sometimes mirror a local subcircuit; this is built into the constructions below.

We therefore fix the following raw **LOPP** constructions (defined explicitly in Appendix G and Section J):

- For all $a, n, b \geq 0$, a raw permutation circuit

$$\sigma_{a,n,b}^d \in \mathbf{LOPP}^{\text{raw}}(d^{a+n+b}, d^{a+n+b}),$$

intended to implement, at the level of Gray-ordered mode labels, the block symmetry that swaps the middle n -qudit block with the lower b -qudit block.

- For all $p, n \geq 0$, an operator

$$\text{GrayLift}_d^p(-) : \mathbf{LOPP}^{\text{raw}}(d^n, d^n) \longrightarrow \mathbf{LOPP}^{\text{raw}}(d^{n+p}, d^{n+p}),$$

which lifts a d^n -mode circuit uniformly across the d^p Gray blocks determined by p surrounding qudits.

- For all $p, n \geq 0$ and $j \in [d]$, an operator

$$j\text{GrayLift}_d^p(-) : \mathbf{LOPP}^{\text{raw}}(d^n, d^n) \longrightarrow \mathbf{LOPP}^{\text{raw}}(d^{n+1+p}, d^{n+1+p}),$$

which performs the same lift, but only on the branch where a designated control digit equals j (and acts as the identity on the other branches).

In the main text we use only how these gadgets interact with decoding, stated in Lemmas 4.14 and 4.15.

Remark 2. Under the single-photon semantics, $\text{GrayLift}_d^p(C)$ is (up to Gray-induced permutations of the mode basis) a block-diagonal direct sum of d^p copies of $\llbracket C \rrbracket_{\text{sp}}$, one for each fixed choice of the surrounding p Gray digits. The reflected Gray traversal reverses some suffix blocks (precisely when the prefix digit is odd); the GrayLift implementation accounts for this by conjugating the affected blocks by a mode-reversal permutation.

Definition 4.8. Fix $d \geq 2$. For each $n, a, b \geq 0$ define a function

$$E_b^a : \mathbf{CQC}_d^{\text{raw}}(n, n) \longrightarrow \mathbf{LOPP}^{\text{raw}}(d^{a+n+b}, d^{a+n+b})$$

by structural induction on raw circuits.

- (1) *Sequential composition.* For $C_1, C_2 \in \mathbf{CQC}_d^{\text{raw}}(n, n)$,

$$E_b^a(C_2 \circ C_1) := E_b^a(C_2) \circ E_b^a(C_1).$$

- (2) *Parallel composition.* Let $C_1 \in \mathbf{CQC}_d^{\text{raw}}(n_1, n_1)$ and $C_2 \in \mathbf{CQC}_d^{\text{raw}}(n_2, n_2)$, and view $C_1 \otimes C_2$ as acting on $n_1 + n_2$ qudits with C_1 on the upper block and C_2 on the lower block. Set

$$E_b^a(C_1 \otimes C_2) := E_b^{a+n_1}(C_2) \circ E_b^a(C_1).$$

This fixes our serialisation convention: we encode the lower block first and then the upper block, matching the decoding convention of Definition 4.11.

- (3) *Identities.* For the canonical identity $\text{id}_n \in \mathbf{CQC}_d^{\text{raw}}(n, n)$, set

$$E_b^a(\text{id}_n) := \text{id}_{d^{a+n+b}}.$$

- (4) *Swap.* For the swap generator, set

$$E_b^a(\text{X}) := \sigma_{a,b,2}^d \circ \sigma_{a+b,1,1}^d \circ \sigma_{a,2,b}^d.$$

- (5) *Global phases.* For the scalar phase generator with parameter θ , set

$$E_b^a(\text{G}(\theta)) := \left(-\text{G}(\theta) \right)^{\otimes d^{a+b}}.$$

(Under the single-photon semantics, applying the same phase shifter on every mode yields a global phase.)

- (6) *Two-level Hadamards.* For each $0 \leq i < d - 1$, fix a d -mode **LOPP** circuit $B_d^{(i,i+1)} : 1^{\otimes d} \rightarrow 1^{\otimes d}$ whose single-photon semantics is $H_d^{(i,i+1)}$. (Existence follows from standard linear-optical decompositions; a concrete choice is given in Appendix G.) Define

$$E_b^a\left(-\text{H}^{(r,r+1)}\right) := \sigma_{a,1,b}^d \circ \text{GrayLift}_d^{a+b}\left(B_d^{(i,i+1)}\right) \circ \sigma_{a,b,1}^d.$$

- (7) *Control.* For $C \in \mathbf{CQC}_d^{\text{raw}}(n, n)$ and $j \in [d]$, define

$$E_b^a(C_j(C)) := \sigma_{a,n+1,b}^d \circ j\text{GrayLift}_d^{a+b}(E_0^0(C)) \circ \sigma_{a,b,n+1}^d.$$

Finally, for $C \in \mathbf{CQC}_d^{\text{raw}}(n, n)$ we abbreviate $E(C) := E_0^0(C) \in \mathbf{LOPP}^{\text{raw}}(d^n, d^n)$.

We record two basic properties of the encoding. First, it respects the structural congruence on raw qudit circuits and therefore descends to the quotient PROP.

Lemma 4.9. Fix $d \geq 2$. For all $n, a, b \geq 0$ and all raw circuits $C, C' \in \mathbf{CQC}_d^{\text{raw}}(n, n)$, if $C \equiv C'$ in the structural congruence of Definition 2.6, then

$$E_b^a(C) \equiv E_b^a(C')$$

in the strict-PROP congruence on raw **LOPP** circuits of Definition 4.1.

Hence E_b^a induces a well-defined function

$$E_b^a : \mathbf{CQC}_d(n, n) \longrightarrow \mathbf{LOPP}(d^{a+n+b}, d^{a+n+b}).$$

Second, the encoding preserves the intended unitary semantics, where **LOPP** circuits are interpreted through the Gray identification.

Lemma 4.10. For every $n \geq 0$ and every circuit $C : n \rightarrow n$ in $\mathbf{CQC}_d$,

$$\llbracket E(C) \rrbracket_{\text{LOPP}} = \llbracket C \rrbracket. \quad (\text{ENC})$$

4.4 Contextual decoding of LOPP circuits

This subsection defines a syntax-directed *decoding* map from raw **LOPP** circuits back to raw qudit circuits. The key point is locality: $-\text{G}(\theta)-$ acts on one mode, while X and $\text{X}^{(i,j)}$ act on two *consecutive* modes. Under the reflected Gray ordering (Definition 4.4), two consecutive modes correspond to computational basis words that differ in exactly one digit by ± 1 (Lemma 4.5). Decoding therefore produces either a basis-state phase, or an adjacent two-level single-qudit gate controlled on the surrounding digits.

We reuse the derived one-qudit gate families $-\text{X}^{(i,j)}-$ (two-level swaps) and $-\text{R}_X^{(i,j)}(\theta)-$ (two-level beam-splitter rotations) from §3.1. Recall also that for a word $u \in [d]^*$ and a circuit f we write $C_u(f)$ for iterated control by u .

Given $n \geq 1$ and words $u, w \in [d]^*$ with $|u| + 1 + |w| = n$, we define for any one-qudit gate $g : 1 \rightarrow 1$ the n -qudit circuit

$$\Lambda_w^u(g) := \sigma_{|u|,|w|,1} \circ C_{uw}(g) \circ \sigma_{|u|,1,|w|},$$

where uw is concatenation and $\sigma_{a,b,c} := \text{id}_a \otimes \sigma_{b,c}$. Semantically, $\Lambda_w^u(g)$ applies g to qudit $|u| + 1$ if and only if the surrounding qudits are in basis state $|u\rangle \otimes |w\rangle$, and acts as the identity otherwise.

Definition 4.11. Fix $d \geq 2$ and $n \geq 0$. For integers t, ℓ with $0 \leq t$ and $t + \ell \leq d^n$, define a map

$$D_n^t : \mathbf{LOPP}^{\text{raw}}(\ell, \ell) \longrightarrow \mathbf{CQC}_d^{\text{raw}}(n, n)$$

by structural induction on the raw **LOPP** term, viewing it as occupying the consecutive global modes $t, \dots, t + \ell - 1$:

(1) *Sequential composition.* If $C_1, C_2 \in \mathbf{LOPP}^{\text{raw}}(\ell, \ell)$, then

$$D_n^t(C_2 \circ C_1) := D_n^t(C_2) \circ D_n^t(C_1).$$

(2) *Parallel composition (serialised).* If $C_1 \in \mathbf{LOPP}^{\text{raw}}(\ell_1, \ell_1)$ and $C_2 \in \mathbf{LOPP}^{\text{raw}}(\ell_2, \ell_2)$, placed side by side so that C_1 occupies modes $t, \dots, t + \ell_1 - 1$ and C_2 occupies modes $t + \ell_1, \dots, t + \ell_1 + \ell_2 - 1$, then

$$D_n^t(C_1 \otimes C_2) := D_n^{t+\ell_1}(C_2) \circ D_n^t(C_1).$$

(3) *Structural generators.* For the empty diagram and the identity wire,

$$D_n^t \left(\begin{array}{c} \text{---} \\ \text{---} \end{array} \right) := \text{id}_n, \quad D_n^t(\text{---}) := \text{id}_n.$$

(4) *Phase shifters.* Let $\boxed{\theta}$ act on the (global) mode t , and write $G_n^d(t) = u \in [d]^n$. Define

$$D_n^t \left(\boxed{\theta} \right) := C_u \left(\bigotimes \right).$$

(5) *Two-mode generators on consecutive modes.* Assume $n \geq 1$ and consider a generator acting on consecutive global modes t and $t + 1$ (so $t + 1 < d^n$). By Lemma 4.5 there are unique words $u, w \in [d]^*$, a unique digit $v \in [d]$, and a unique sign $\varepsilon \in \{-1, +1\}$ (with $v + \varepsilon \in [d]$) such that

$$G_n^d(t) = uvw, \quad G_n^d(t + 1) = u(v + \varepsilon)w, \quad |u| + 1 + |w| = n.$$

• *Swap of modes.* For the **LOPP** swap generator, set

$$D_n^t \left(\begin{array}{c} \diagup \\ \diagdown \end{array} \right) := \Lambda_w^u \left(\boxed{X^{(v, v+\varepsilon)}} \right).$$

• *Beam splitter.* For the **LOPP** beam splitter generator, set

$$D_n^t \left(\begin{array}{c} \theta \\ \diagup \quad \diagdown \end{array} \right) := \Lambda_w^u \left(\boxed{R_X^{(v, v+\varepsilon)}(\theta)} \right).$$

Finally, for a raw **LOPP** circuit C on d^n modes we abbreviate $D(C) := D_n^0(C) \in \mathbf{CQC}_d^{\text{raw}}(n, n)$.

Lemma 4.12. *Each decoded generator acts non-trivially on at most one two-dimensional computational subspace (determined by one or two consecutive Gray indices) and as the identity on the orthogonal complement:*

- $D_n^t(\boxed{\theta})$ applies a phase only on the basis vector $|G_n^d(t)\rangle$;
- $D_n^t(\begin{smallmatrix} \diagup \\ \diagdown \end{smallmatrix})$ and $D_n^t(\begin{smallmatrix} \theta \\ \diagup \quad \diagdown \end{smallmatrix})$ act only on $\text{span}\{|G_n^d(t)\rangle, |G_n^d(t + 1)\rangle\}$.

Lemma 4.13. *Let C_1, C_2 be raw **LOPP** circuits on d^n modes. If $C_1 \equiv C_2$ in the structural congruence generated by the strict PROP equations (as in Definition 4.1), then*

$$\mathbf{QC}_d \vdash D(C_1) = D(C_2).$$

PROOF. See Appendix I. The proof is by induction on strict-PROP contexts. The only non-trivial case is the mixed-product axiom; it is handled because decoding of subcircuits on disjoint mode blocks yields (multi-)controlled gates that commute in $\mathbf{QC}_d$ (using derived distinct-control commutativity). $\square$

We now record the two interface facts about the **LOPP** gadgets used in the encoding (introduced in §4.3) that we will need for the encoding/decoding retraction argument.

Lemma 4.14. *Let D_n^t be the contextual decoding map of Definition 4.11. For every $n, p \geq 0$ and every raw **LOPP** circuit $C \in \mathbf{LOPP}^{\text{raw}}(d^n, d^n)$ we have*

$$\mathbf{QC}_d \vdash D_{n+p}^0 \left(\text{GrayLift}_d^p(C) \right) = \text{id}_p \otimes D_n^0(C).$$

Moreover, for each $j \in [d]$,

$$\mathbf{QC}_d \vdash D_{n+1+p}^0 \left(j\text{GrayLift}_d^p(C) \right) = \text{id}_p \otimes C_j \left(D_n^0(C) \right).$$

PROOF. See Appendix H, in particular Lemmas H.4 and H.5. $\square$

Remark 3. Intuitively, decoding a GrayLift produces a product over all control branches. Regrouping branches uses derived commutativity of distinct control (Appendix D), and collapsing the total product uses derived exhaustivity (Appendix E).

Lemma 4.15. *For all $a, n, b \geq 0$,*

$$\mathbf{QC}_d \vdash D_{a+n+b}^0 \left(\sigma_{a,n,b}^d \right) = \sigma_{a,n,b},$$

where $\sigma_{a,n,b} : a + n + b \rightarrow a + b + n$ denotes the corresponding qudit-wire symmetry $\sigma_{a,n,b} := \text{id}_a \otimes \sigma_{n,b}$.

PROOF. See Appendix J. $\square$

Theorem 4.16. *Let $n \geq 0$ and let $C_1, C_2 : d^n \rightarrow d^n$ be circuits in **LOPP**. If $\mathbf{LOPP} \vdash C_1 = C_2$, then $\mathbf{QC}_d \vdash D(C_1) = D(C_2)$.*

PROOF. This follows by induction on **LOPP**-derivations, using that decoding is compatible with the strict PROP congruence (Lemma 4.13) and by checking, axiom-by-axiom, that each non-structural **LOPP** axiom decodes to a derivable equation in $\mathbf{QC}_d$. Full details are given in Appendix I. $\square$

Thus every rewrite used in an **LOPP** derivation can be replayed, after decoding, inside $\mathbf{QC}_d$. We will combine this mimicking step with the encoding/decoding retraction in the next subsection to transfer completeness from **LOPP** to $\mathbf{QC}_d$.

4.5 Completeness via the linear-optical reduction

This subsection assembles the translation results into a single completeness argument for $\mathbf{QC}_d$. The main point is that encoding into **LOPP** and then decoding back is a retraction provable in $\mathbf{QC}_d$; combined with mimicking (Theorem 4.16) and completeness of **LOPP**, this transfers semantic equality to derivability in $\mathbf{QC}_d$.

Theorem 4.17. *For all $n, a, b \geq 0$ and all circuits $C : n \rightarrow n$ in $\mathbf{CQC}_d$,*

$$\mathbf{QC}_d \vdash D_{a+n+b}^0 \left(E_b^a(C) \right) = \text{id}_a \otimes C \otimes \text{id}_b.$$

PROOF. Pick a raw representative of C and argue by structural induction. The cases for $\circ$ and the (serialised) $\otimes$ follow immediately from Definitions 4.8 and 4.11. All generator cases are handled uniformly by eliminating the block-permutation wrapper via Lemma 4.15, and discharging the remaining GrayLift steps via Lemma 4.14; the full bookkeeping is given in Appendix H.

We only expand the key control step. Let $C = C_j(C')$ with $C' : m \rightarrow m$. By Definition 4.8 we have

$$E_b^a(C) = \sigma_{a,m+1,b}^d \circ j\text{GrayLift}_d^{a+b}(E_0^0(C')) \circ \sigma_{a,b,m+1}^d.$$

Applying $D_{a+m+1+b}^0$ and using Lemma 4.15 (on the outer permutations) together with the controlled GrayLift interface in Lemma 4.14 yields, in QC_d ,

$$\begin{aligned} D_{a+m+1+b}^0(E_b^a(C_j(C'))) &= \sigma_{a,m+1,b} \circ (\text{id}_{a+b} \otimes C_j(D_m^0(E_0^0(C')))) \\ &\quad \circ \sigma_{a,b,m+1} \\ &= \text{id}_a \otimes C_j(D_m^0(E_0^0(C'))) \otimes \text{id}_b \\ &= \text{id}_a \otimes C_j(C') \otimes \text{id}_b, \end{aligned}$$

where the middle equality is strict PROP coherence and the last equality uses the induction hypothesis for C' at context $(0, 0)$. $\square$

Corollary 4.18. *For every $n \geq 0$ and every circuit $C : n \rightarrow n$ in CQC_d , $\text{QC}_d \vdash D(E(C)) = C$.*

PROOF. Set $a = b = 0$ in Theorem 4.17. $\square$

Theorem 4.19. *For every $d \geq 2$ the equational theory QC_d of CQC_d is sound and complete for d -dimensional unitary circuit. That is, for any n -qudit circuits $C_1, C_2 : n \rightarrow n$ we have*

$$\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket \iff \text{QC}_d \vdash C_1 = C_2.$$

PROOF. Soundness is immediate: each axiom schema is valid under $\llbracket - \rrbracket$, and $\llbracket - \rrbracket$ is a strict symmetric monoidal functor (Proposition 2.10).

For completeness, if $\llbracket C_1 \rrbracket = \llbracket C_2 \rrbracket$ then by Lemma 4.10 we have $\llbracket E(C_1) \rrbracket_{\text{LOPP}} = \llbracket E(C_2) \rrbracket_{\text{LOPP}}$, hence completeness of LOPP gives $\text{LOPP} \vdash E(C_1) = E(C_2)$. Decode and mimic (Theorem 4.16) to obtain $\text{QC}_d \vdash D(E(C_1)) = D(E(C_2))$, and conclude by Corollary 4.18. $\square$

5 Conclusion

We presented a dimension-uniform, circuit-level equational theory for unitary qudit circuits. Our starting point is the polycontrolled circuit PROP CQC_d , where control is primitive: for each $k \in [d]$, the constructor $C_k(-)$ functorially adjoins a distinguished control wire and applies a circuit exactly when that wire is in basis state $|k\rangle$. Over a fixed universal gate signature (adjacent two-level Hadamards, basis-controlled phases, and the structural PROP generator), we defined QC_d : a *finite* family of local axiom schemata whose instances involve at most three wires and whose diagrammatic shape is independent of d . Our main theorem establishes soundness and completeness for the standard semantics $\llbracket - \rrbracket : \text{CQC}_d \rightarrow \text{Qudit}_d$: QC_d derives exactly the valid equalities of *exact* unitary semantics, including global phases as $0 \rightarrow 0$ scalars. A notable consequence is that the expected *global* control algebra (compatibility, distinct-control commutativity, and exhaustivity) is derivable within QC_d , rather than being imposed as part of the ambient structural congruence.

Completeness is obtained by a modular proof transfer to linear optics. Using the reflected d -ary Gray code, we encode n -qudit circuits as d^n -mode single-photon **LOPP** circuits (up to conjugation by a fixed Gray-basis permutation), and we handle the tensor/direct-sum mismatch by a deterministic serialisation of $\otimes$. We then prove an encoding/decoding retraction inside QC_d and a mimicking theorem showing that every LOPP rewrite step decodes to a derivation in QC_d . In this way, completeness of LOPP yields completeness of QC_d .

Several directions remain open. First, it is natural to extend beyond unitary endomorphisms to circuits with ancilla initialisation, discard, and measurement, aiming for finitary, bounded-arity axiomatisations of qudit *channels*. Second, on the algorithmic side, it would be valuable to extract practical rewriting methods from QC_d (normal forms, decision procedures, or terminating/confluent fragments) and to connect them to certified optimisation and verification pipelines for native qudit hardware. Finally, exploring alternative embeddings or gate signatures that better align monoidal structure, may simplify both the metatheory and its compiler-facing interfaces.

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A Angle relations for rotation axioms

This appendix fixes the *meta-level parameter functions* used in the rotation axioms (EH), (3Rx) and (3CRx). Each of these axioms is a schematic equation in which the *left-hand side* contains freely chosen real parameters, while the *right-hand side* parameters are *deterministic functions* of the left-hand side. The purpose of recording these functions is purely bookkeeping: it makes each axiom schema fully explicit, and (when needed) supports a uniqueness convention for rewriting.

In Delorme's one-qubit Euler parametrisations, the decomposition of a two-level unitary naturally comes with an overall scalar factor (*global phase* in $U(2)$). In the present paper we apply such decompositions *inside a fixed two-dimensional subspace* $\text{span}\{|i\rangle, |j\rangle\}$ of a d -level system. A phase factor that is "global" for the 2×2 block is *not* a scalar on the full d -dimensional space: rather, it is the operation that multiplies both $|i\rangle$ and $|j\rangle$ by the same phase while leaving the other basis levels unchanged. In our syntax this is realised by applying the same level-phase on both levels, i.e. by $C_i(\alpha) \circ C_j(\alpha)$. Crucially, on the (i, j) -subspace this acts as $e^{i\alpha} I_2$, hence it *commutes* with every gate supported on (i, j) (in particular with $H^{(i,j)}$ and with all level phases). As a result, whenever the qubit formulas isolate a scalar phase, we are free to slide the corresponding two-level phase and absorb it into neighbouring level phases. This is the only reason our explicit angle formulas do not coincide syntactically with the qubit case; semantically they implement the same underlying $U(2)$ parametrisation on $\text{span}\{|i\rangle, |j\rangle\}$.

Throughout, equalities between angles are understood modulo 2π . We write $\arg(z)$ for the principal argument of a non-zero complex number $z \in \mathbb{C}$, taking values in $(-\pi, \pi]$. When we restrict an angle to a range (e.g. $\beta \in [0, 2\pi)$), we always mean its canonical representative in that interval. We also use the standard two-argument arctangent $\text{atan2}(y, x)$ with values in $(-\pi, \pi]$.

Euler-type relation (EH)

The axiom (EH) is a two-level Euler-type identity (applied in the paper on a subspace $\text{span}\{|i\rangle, |j\rangle\}$ of a qudit). Syntactically, the left-hand side is parameterised by two real angles α_0, α_2 , while the right-hand side uses four angles $\beta_0, \beta_1, \beta_2, \beta_3$. The intended meaning is:

for every choice of (α_0, α_2) , there is a *unique* choice of $(\beta_0, \beta_1, \beta_2, \beta_3)$ in the prescribed ranges that makes (EH) sound (on the relevant two-dimensional subspace), and hence sound as a qudit equation by acting as the identity outside that subspace.

We follow Delorme's explicit Euler parametrisation (stated for qubits) and use the same extraction function, with the "two-level global phase" convention explained above.

Given $\alpha_0, \alpha_2 \in \mathbb{R}$, define

$$z = -\sin\left(\frac{\alpha_0 + \alpha_2}{2}\right) + i \cos\left(\frac{\alpha_0 - \alpha_2}{2}\right), \quad z' = \cos\left(\frac{\alpha_0 + \alpha_2}{2}\right) - i \sin\left(\frac{\alpha_0 - \alpha_2}{2}\right).$$

We then define $\beta_0, \beta_1, \beta_2, \beta_3 \in [0, 2\pi)$ by cases. The degenerate cases $z = 0$ or $z' = 0$ are treated separately to avoid dividing by 0; the generic case uses the ratio z/z' .

- If $z' = 0$, set

$$\beta_0 = 2 \arg(z), \quad \beta_1 = \frac{\pi + \alpha_0 + \alpha_2}{2} - \arg(z), \quad \beta_2 = \frac{\pi + \alpha_0 + \alpha_2}{2} - \arg(z), \quad \beta_3 = 0.$$

- If $z = 0$, set

$$\beta_0 = 2 \arg(z'), \quad \beta_1 = \frac{\alpha_0 + \alpha_2}{2} - \arg(z'), \quad \beta_2 = \pi + \frac{\alpha_0 + \alpha_2}{2} - \arg(z'), \quad \beta_3 = 0.$$

- Otherwise (when $z \neq 0$ and $z' \neq 0$), set

$$\begin{aligned} \beta_0 &= \arg(z) + \arg(z'), & \beta_1 &= -\arg\left(i + \left|\frac{z}{z'}\right|\right) + \frac{\pi + \alpha_0 + \alpha_2}{2} - \arg(z), \\ \beta_2 &= \arg\left(i + \left|\frac{z}{z'}\right|\right) + \frac{\pi + \alpha_0 + \alpha_2}{2} - \arg(z), & \beta_3 &= \arg(z) - \arg(z'). \end{aligned}$$

With this choice, Equation (EH) holds on the relevant two-dimensional subspace. Since every occurrence of (EH) in the main development is applied to a two-level gate supported on $\text{span}\{|i\rangle, |j\rangle\}$ and acts as the identity on the other computational levels, these formulas can be used systematically for all $d \geq 2$.

Three-rotation identities (3Rx and 3CRx)

The axioms (3Rx) and (3CRx) express the standard fact that the same real 3×3 rotation (hence an element of $\text{SO}(3)$) admits both a ZXZ and an XZX Euler decomposition. They are the qudit/circuit-side reflections of the "(E3)" principle used in the linear-optical setting: (3Rx) is the uncontrolled identity, and (3CRx) is its controlled analogue. The parameter relations are identical in both cases.

Concretely, let $R_x(\theta)$ and $R_z(\theta)$ be the standard real rotation matrices about the x - and z -axes, and write

$$R_{E3} = \begin{pmatrix} r_{1,1} & r_{1,2} & r_{1,3} \\ r_{2,1} & r_{2,2} & r_{2,3} \\ r_{3,1} & r_{3,2} & r_{3,3} \end{pmatrix} \in \text{SO}(3).$$

The angles $\gamma_1, \gamma_2, \gamma_3$ (left-hand side of (3Rx)) and $\delta_1, \delta_2, \delta_3$ (right-hand side) are chosen so that

$$R_{E3} = R_z(\gamma_1) R_x(\gamma_2) R_z(\gamma_3) = R_x(\delta_1) R_z(\delta_2) R_x(\delta_3). \quad (11)$$

Thus $(\gamma_1, \gamma_2, \gamma_3)$ are the ZZZ Euler angles of R_{E3} and $(\delta_1, \delta_2, \delta_3)$ are the corresponding XZX Euler angles.

The extraction formulas below are exactly those used in the linear-optical literature; they are stated with explicit “generic” and “gimbal-lock” cases, since Euler angle are not unique in degenerate configurations.

Assume first that we are not in a gimbal-lock configuration for XZX , i.e. $-1 < r_{1,1} < 1$. Then

$$\delta_2 = \arccos(r_{1,1}), \quad \delta_1 = \text{atan2}(r_{3,1}, r_{2,1}), \quad \delta_3 = \text{atan2}(r_{1,3}, -r_{1,2}). \quad (12)$$

The remaining cases correspond to gimbal lock for XZX , i.e. $\delta_2 \in \{0, \pi\}$, where only a constraint on $\delta_1 \pm \delta_3$ is determined:

- If $r_{1,1} = 1$, then necessarily $\delta_2 = 0$, and all solutions satisfy

$$\delta_2 = 0, \quad \delta_1 + \delta_3 = \text{atan2}(r_{3,2}, r_{3,3}). \quad (13)$$

- If $r_{1,1} = -1$, then necessarily $\delta_2 = \pi$, and all solutions satisfy

$$\delta_2 = \pi, \quad \delta_1 - \delta_3 = \text{atan2}(r_{3,2}, r_{3,3}). \quad (14)$$

In gimbal-lock cases there remains one degree of freedom in (δ_1, δ_3) ; any choice satisfying (13) or (14) yields the same rotation matrix. For definiteness one may choose canonical representatives by setting the last X -rotation to 0.

Dually, if $-1 < r_{3,3} < 1$, then

$$\gamma_2 = \arccos(r_{3,3}), \quad \gamma_3 = \text{atan2}(r_{1,3}, -r_{2,3}), \quad \gamma_1 = \text{atan2}(r_{3,1}, r_{3,2}). \quad (15)$$

The degenerate ZZZ cases are:

- If $r_{3,3} = 1$, then $\gamma_2 = 0$ and all solutions satisfy

$$\gamma_2 = 0, \quad \gamma_1 + \gamma_3 = \text{atan2}(-r_{1,2}, r_{1,1}). \quad (16)$$

- If $r_{3,3} = -1$, then $\gamma_2 = \pi$ and all solutions satisfy

$$\gamma_2 = \pi, \quad \gamma_1 - \gamma_3 = \text{atan2}(-r_{1,2}, r_{1,1}). \quad (17)$$

Again one may choose canonical representatives (e.g. by setting the last Z -rotation to 0), but any satisfying pair give the same rotation.

B Auxiliary diagrammatic derivations

Throughout this appendix we work in the equational theory QC_d , and write $QC_d \vdash C_1 = C_2$ when the circuit identity $C_1 = C_2$ is derivable from the axioms of Figures 1–2

B.1 Catalogue of derived rules

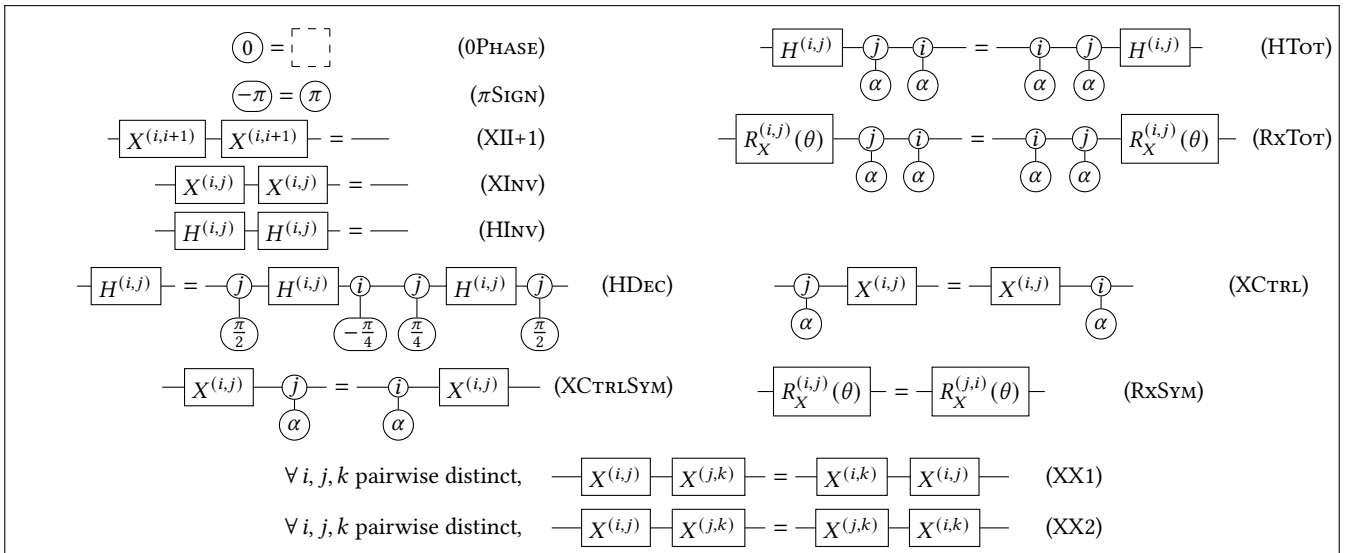
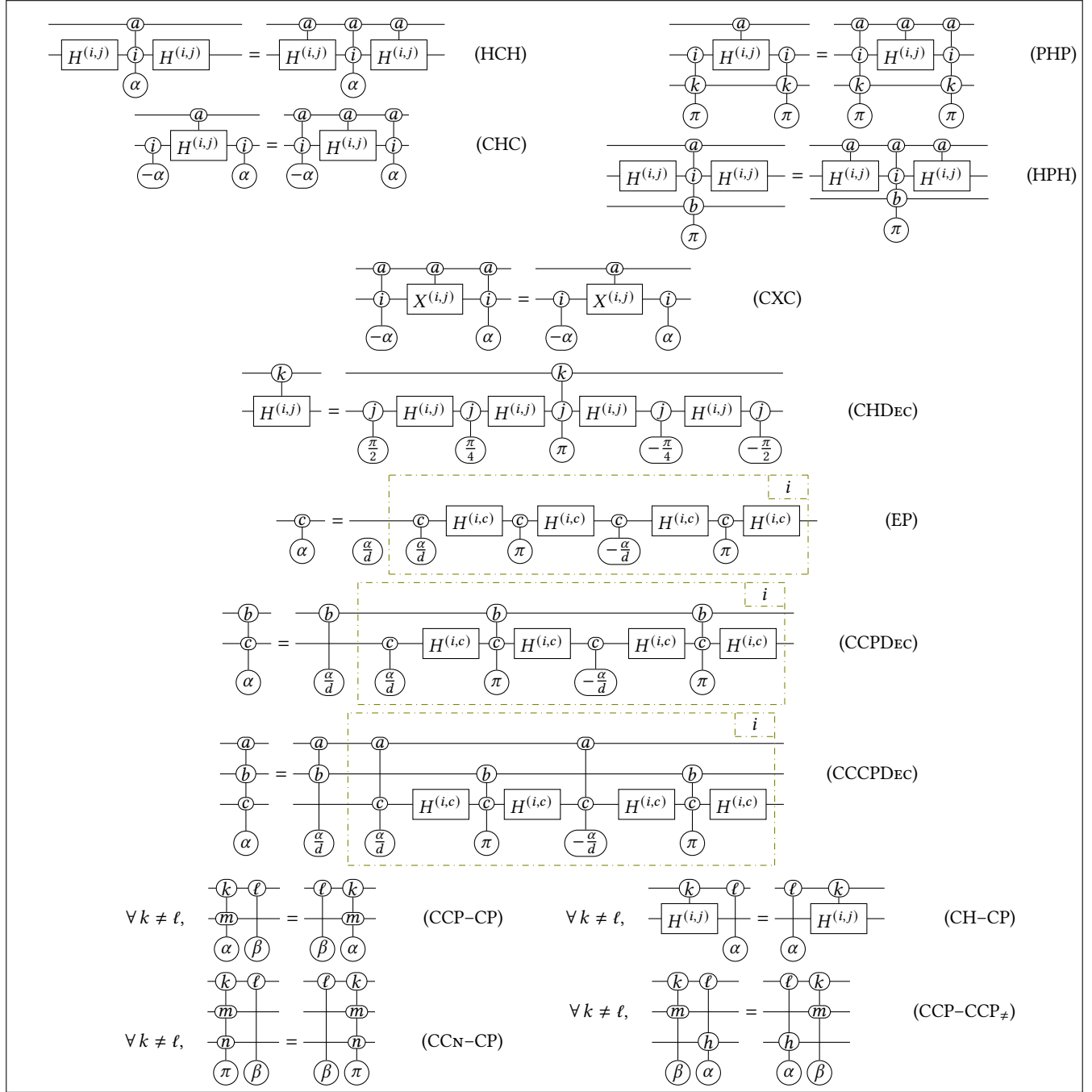


Figure 3: Derived rules in QC_d (batch 1).

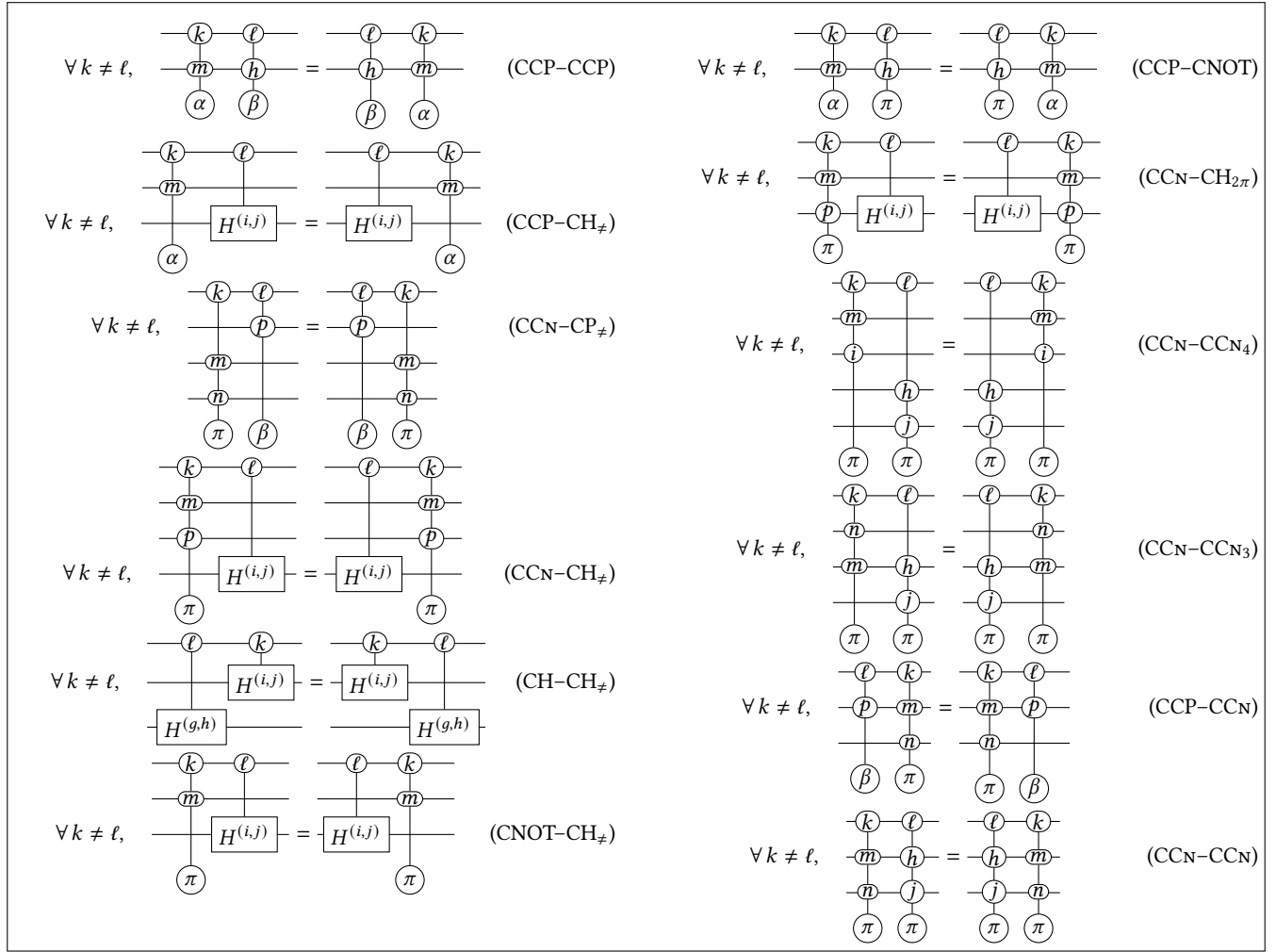
Figure 4: Derived rules in QC_d (batch 2).

B.2 Proofs of the derived rules

PROOF OF (0PHASE).

$$\textcircled{0} \stackrel{(2\pi)}{=} \textcircled{0} \textcircled{2\pi} \stackrel{(\text{Sum})}{=} \textcircled{2\pi} \stackrel{(2\pi)}{=} \textcircled{2\pi} \textcircled{2\pi} \stackrel{(\text{Sum})}{=} \textcircled{4\pi} \stackrel{(2\pi)}{=} \textcircled{0}$$

□

Figure 5: Derived rules in QC_d (batch 3).PROOF OF (πSIGN) .

$$\begin{array}{c} \textcircled{-\pi} \stackrel{(2\pi)}{=} \textcircled{-\pi} \textcircled{2\pi} \stackrel{(\text{Sum})}{=} \textcircled{\pi} \end{array}$$

□

PROOF OF $(\text{XII}+1)$.

$$\begin{array}{c} \boxed{X^{(i,i+1)}} \boxed{X^{(i,i+1)}} = \boxed{H^{(i,i+1)}} \textcircled{i+1} \boxed{H^{(i,i+1)}} \boxed{H^{(i,i+1)}} \textcircled{i+1} \boxed{H^{(i,i+1)}} \stackrel{(H^2)}{=} \boxed{H^{(i,i+1)}} \textcircled{i+1} \textcircled{i+1} \boxed{H^{(i,i+1)}} \\ \stackrel{(\text{Sum})(2\pi)}{=} \boxed{H^{(i,i+1)}} \boxed{H^{(i,i+1)}} \stackrel{(H^2)}{=} \text{---} \end{array}$$

□

PROOF OF (XInv) . If $i = j$, this is trivial.

PROOF OF (XCTRL).

$$\begin{aligned}
& \text{---} \begin{array}{c} \textcircled{j} \\ \alpha \end{array} \text{---} X^{(i,j)} \text{---} = \text{---} \begin{array}{c} \textcircled{j} \\ \alpha \end{array} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} H^{(i,j)} \text{---} \stackrel{(\text{HInv})}{=} \text{---} H^{(i,j)} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \alpha \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} H^{(i,j)} \text{---} \\
& \stackrel{(\text{EH})}{=} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ -\frac{\pi}{2} \end{array} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \alpha - \frac{\pi}{4} \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \alpha + \frac{\pi}{4} \end{array} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \frac{\pi}{2} - \alpha \end{array} \text{---} \stackrel{(\text{PPc}),(\text{HTot}),(\text{Sum})}{=} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ -\frac{\pi}{2} \end{array} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ -\frac{\pi}{4} \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \frac{\pi}{4} \end{array} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \frac{\pi}{2} \end{array} \text{---} \begin{array}{c} \textcircled{i} \\ \alpha \end{array} \text{---} \\
& \stackrel{(\text{Sum}),(\pi\text{SIGN})}{=} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \frac{\pi}{2} \end{array} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ -\frac{\pi}{4} \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \frac{\pi}{4} \end{array} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \frac{\pi}{2} \end{array} \text{---} \begin{array}{c} \textcircled{i} \\ \alpha \end{array} \text{---} \stackrel{(\text{HDEC})}{=} \text{---} X^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \alpha \end{array} \text{---}
\end{aligned}$$

□

PROOF OF (XX1).

$$\begin{aligned}
& \text{---} X^{(i,j)} \text{---} X^{(j,k)} \text{---} = \text{---} X^{(i,j)} \text{---} H^{(j,k)} \text{---} \begin{array}{c} \textcircled{k} \\ \pi \end{array} \text{---} H^{(j,k)} \text{---} \stackrel{(\text{XH})}{=} \text{---} H^{(i,k)} \text{---} X^{(i,j)} \text{---} \begin{array}{c} \textcircled{k} \\ \pi \end{array} \text{---} H^{(j,k)} \text{---} \\
& \stackrel{(\text{HPd})}{=} \text{---} H^{(i,k)} \text{---} \begin{array}{c} \textcircled{k} \\ \pi \end{array} \text{---} X^{(i,j)} \text{---} H^{(j,k)} \text{---} \stackrel{(\text{HInv})}{=} \text{---} H^{(i,k)} \text{---} \begin{array}{c} \textcircled{k} \\ \pi \end{array} \text{---} H^{(i,k)} \text{---} X^{(i,j)} \text{---} = \text{---} X^{(i,k)} \text{---} X^{(i,j)} \text{---}
\end{aligned}$$

□

PROOF OF (XX2).

$$\begin{aligned}
& \text{---} X^{(i,j)} \text{---} X^{(j,k)} \text{---} \stackrel{(\text{XInv})}{=} \text{---} X^{(j,k)} \text{---} X^{(j,k)} \text{---} X^{(i,j)} \text{---} X^{(j,k)} \text{---} \\
& \stackrel{(\text{XX1})}{=} \text{---} X^{(j,k)} \text{---} X^{(i,k)} \text{---} X^{(j,k)} \text{---} X^{(j,k)} \text{---} \stackrel{(\text{XInv})}{=} \text{---} X^{(j,k)} \text{---} X^{(i,k)} \text{---}
\end{aligned}$$

□

PROOF OF (XCTRLSYM).

$$\begin{aligned}
& \text{---} X^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \alpha \end{array} \text{---} \stackrel{(\text{XInv})}{=} \text{---} X^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \alpha \end{array} \text{---} X^{(i,j)} \text{---} X^{(i,j)} \text{---} \stackrel{(\text{XCTRL})}{=} \text{---} X^{(i,j)} \text{---} X^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \alpha \end{array} \text{---} X^{(i,j)} \text{---} \stackrel{(\text{XInv})}{=} \text{---} \begin{array}{c} \textcircled{i} \\ \alpha \end{array} \text{---} X^{(i,j)} \text{---}
\end{aligned}$$

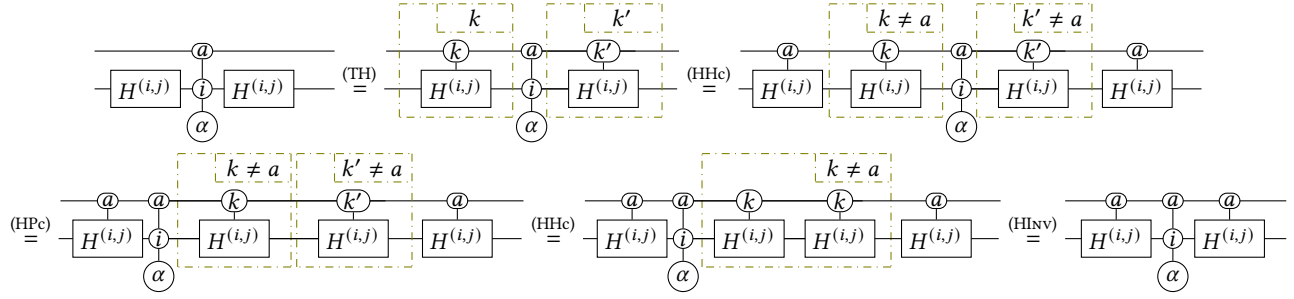
□

PROOF OF (RXSYM).

$$\begin{aligned}
& \text{---} R_X^{(i,j)}(\theta) \text{---} = \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ -\theta \end{array} \text{---} H^{(i,j)} \text{---} \stackrel{(\text{Sum})(2\pi)}{=} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ -\theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} H^{(i,j)} \text{---} \\
& \stackrel{(\text{PPc})}{=} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} \begin{array}{c} \textcircled{i} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ -\theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} H^{(i,j)} \text{---} \\
& \stackrel{(\text{HInv})}{=} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} H^{(i,j)} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ -\theta \end{array} \text{---} H^{(i,j)} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{j} \\ \pi \end{array} \text{---} H^{(i,j)} \text{---} \\
& = \text{---} X^{(i,j)} \text{---} H^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ -\theta \end{array} \text{---} H^{(i,j)} \text{---} X^{(i,j)} \text{---} \stackrel{(\text{XH})}{=} \text{---} H^{(j,i)} \text{---} X^{(i,j)} \text{---} \begin{array}{c} \textcircled{i} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{j} \\ -\theta \end{array} \text{---} H^{(i,j)} \text{---} X^{(i,j)} \text{---} \\
& \stackrel{(\text{XCTRL}),(\text{XCTRLSYM})}{=} \text{---} H^{(j,i)} \text{---} \begin{array}{c} \textcircled{j} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{i} \\ -\theta \end{array} \text{---} X^{(i,j)} \text{---} H^{(i,j)} \text{---} X^{(i,j)} \text{---} \stackrel{(\text{XH})}{=} \text{---} H^{(j,i)} \text{---} \begin{array}{c} \textcircled{j} \\ \theta \end{array} \text{---} \begin{array}{c} \textcircled{i} \\ -\theta \end{array} \text{---} H^{(j,i)} \text{---} X^{(i,j)} \text{---} X^{(i,j)} \text{---} \\
& = \text{---} R_X^{(j,i)}(\theta) \text{---} X^{(i,j)} \text{---} X^{(i,j)} \text{---} \stackrel{(\text{XInv})}{=} \text{---} R_X^{(j,i)}(\theta) \text{---}
\end{aligned}$$

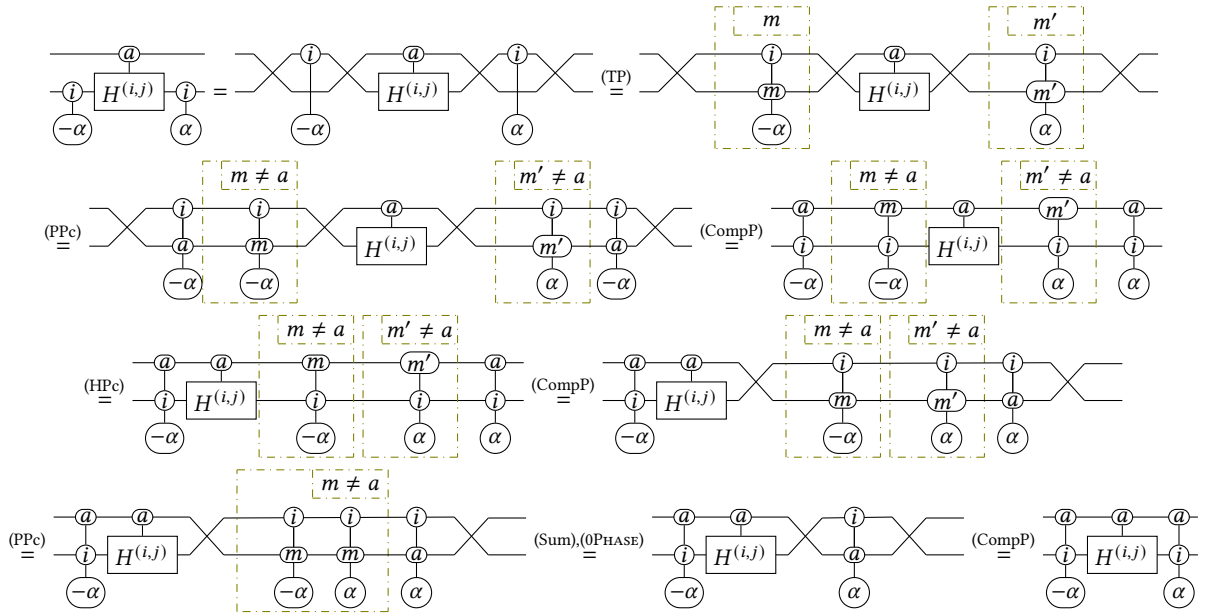
□

PROOF OF (HCH).



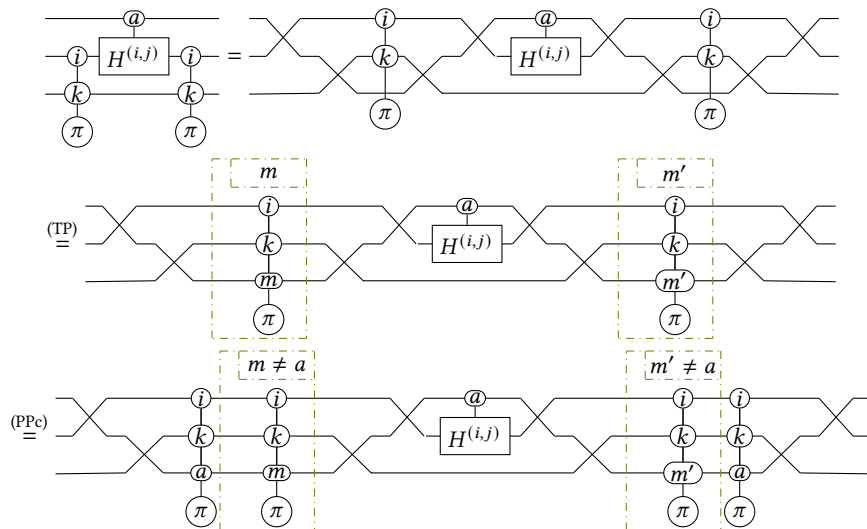
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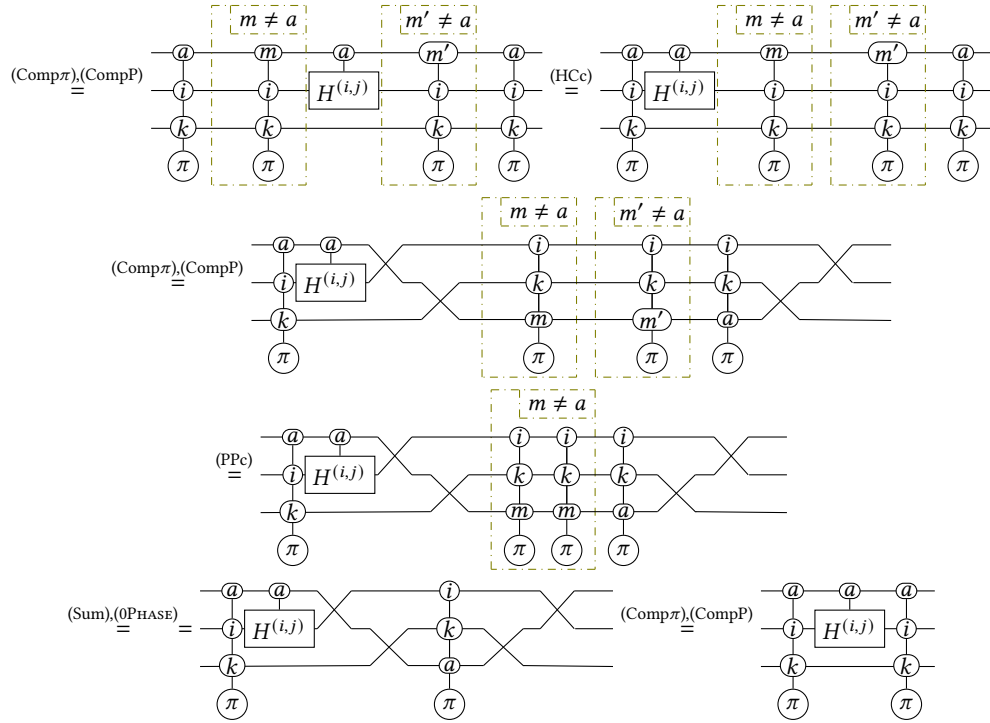
PROOF OF (CHC).



□

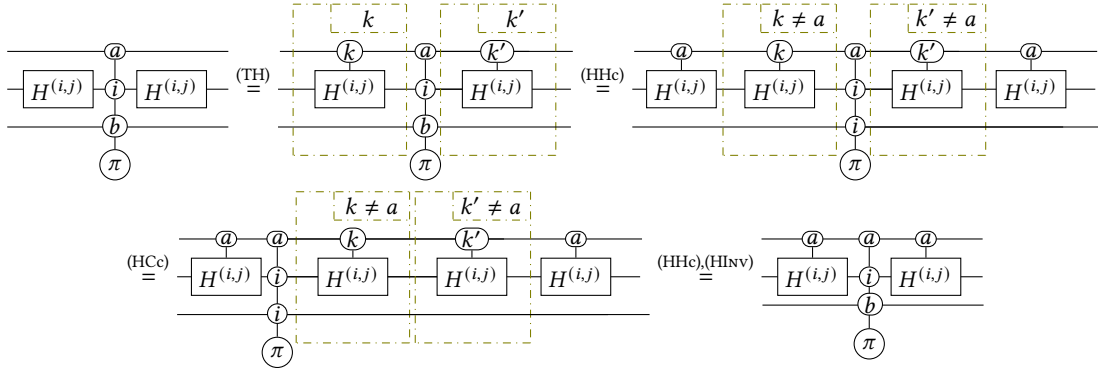
PROOF OF (PHP).





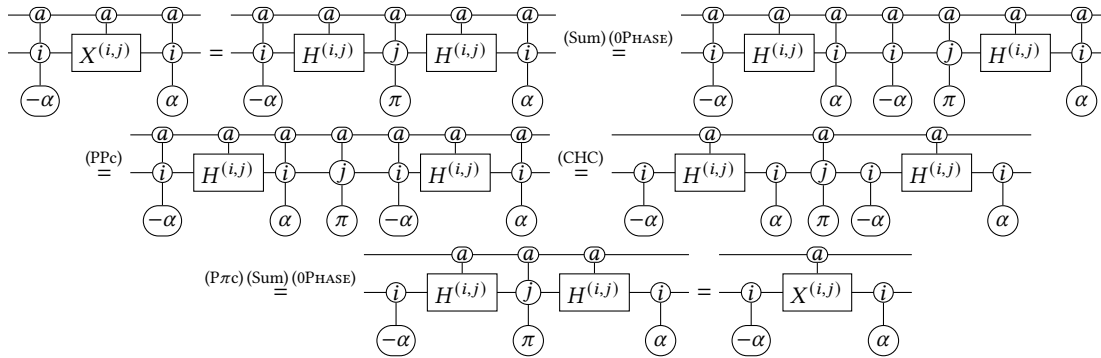
□

PROOF OF (HPH).



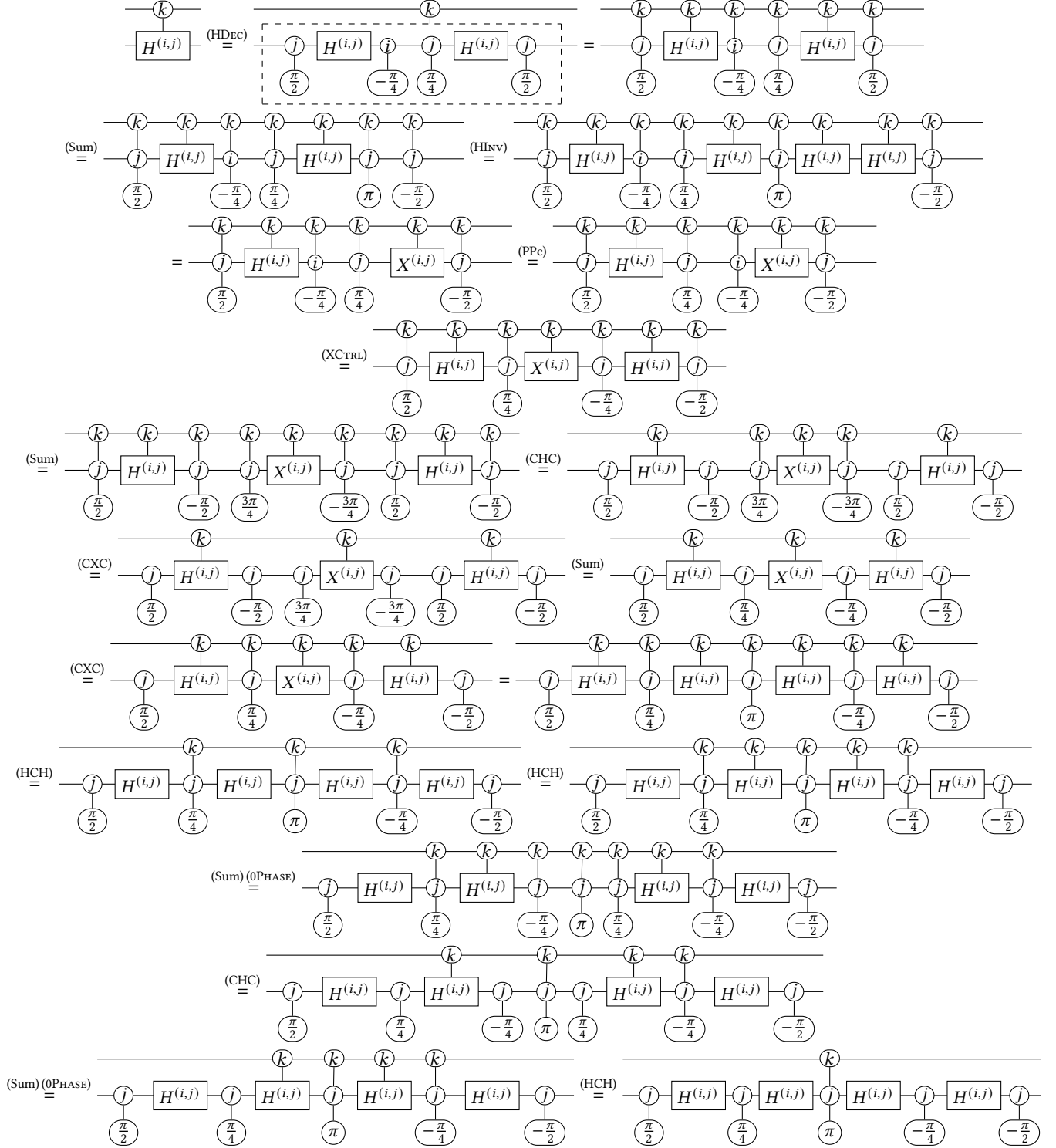
□

PROOF OF (CXC).

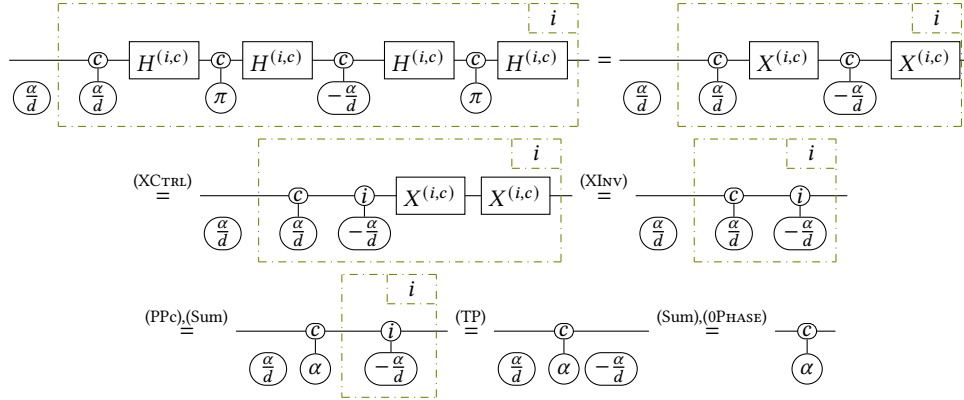


□

PROOF OF (CHDEC).

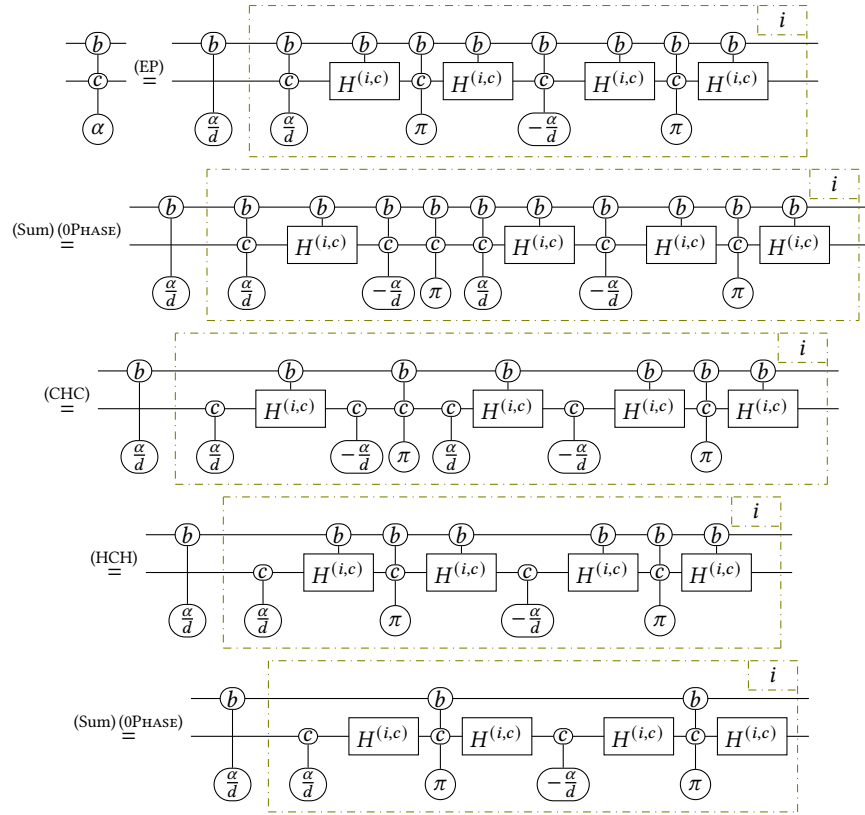


PROOF OF (EP).



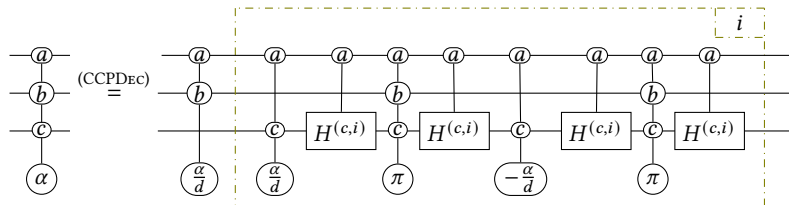
□

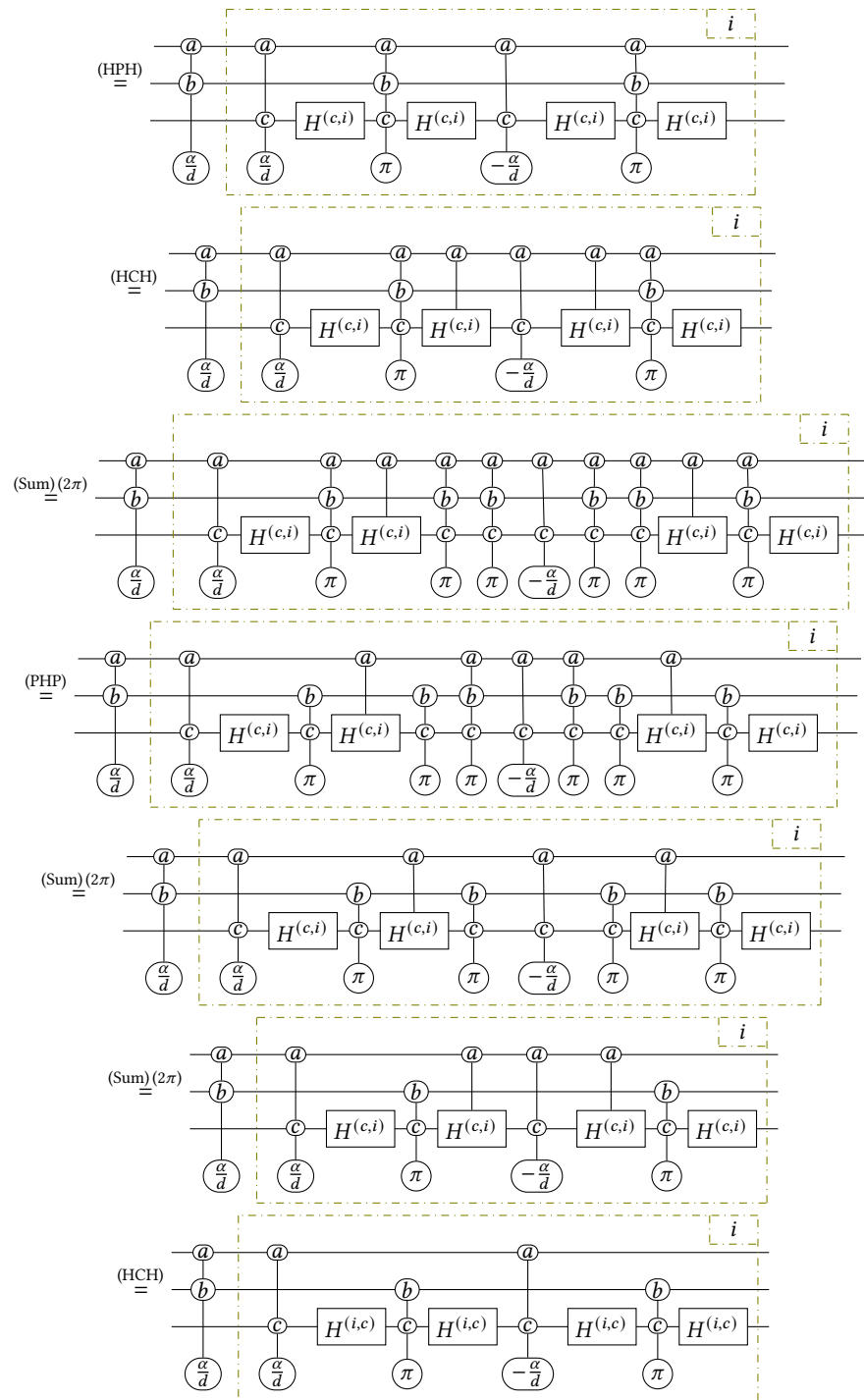
PROOF OF (CCPDEC).



□

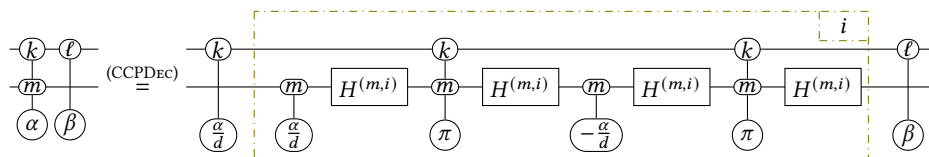
PROOF OF (CCCPDEC).

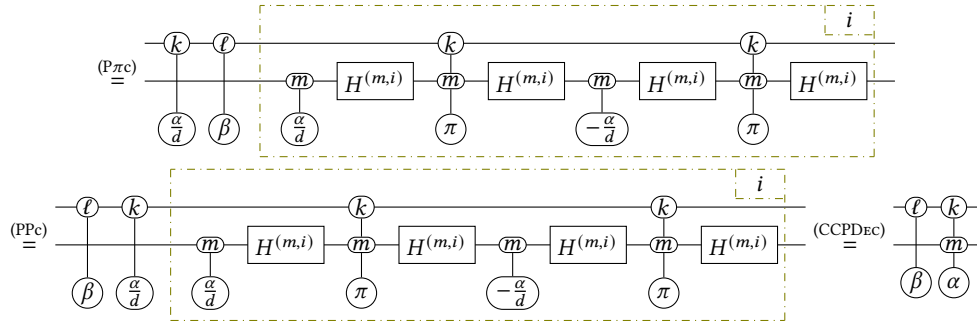




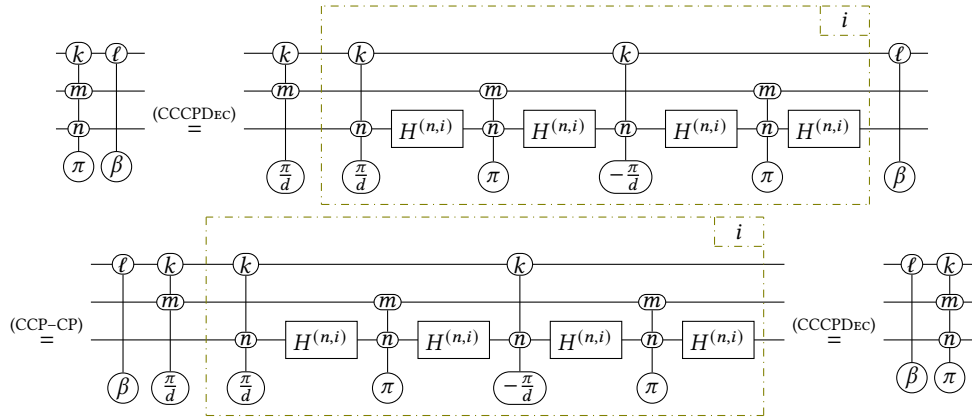
9

PROOF OF (CCP-CP).

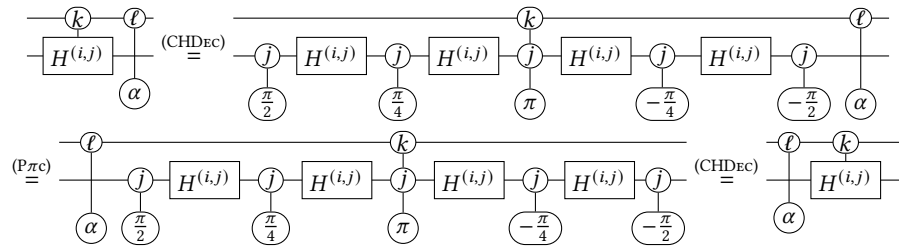
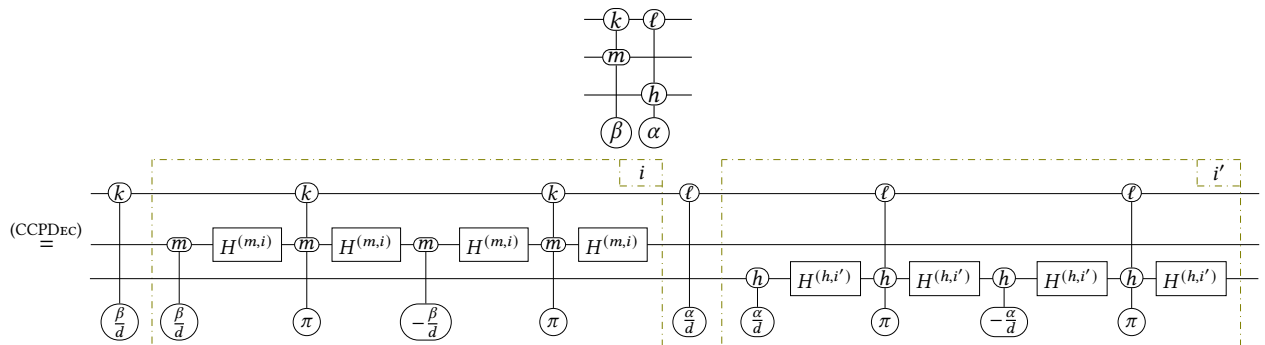


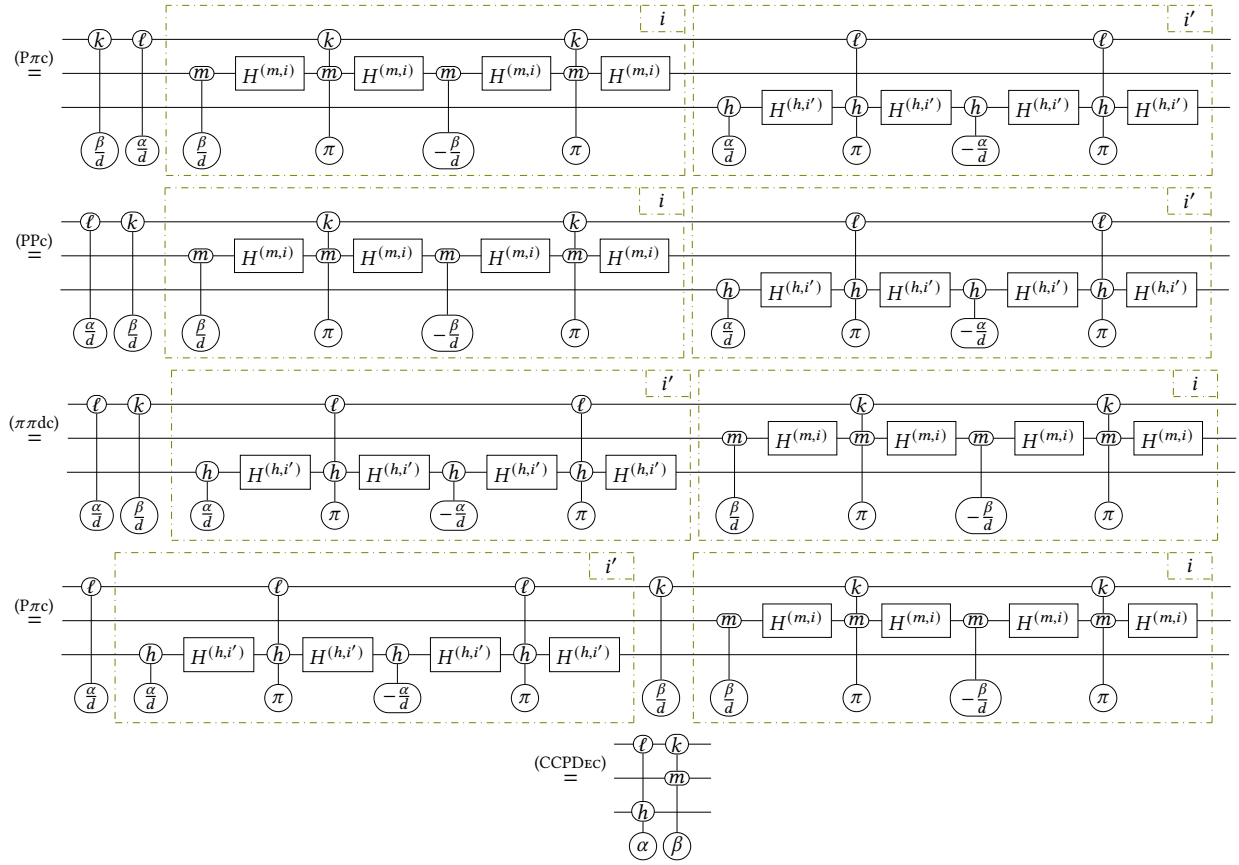


PROOF OF (CCN-CP).



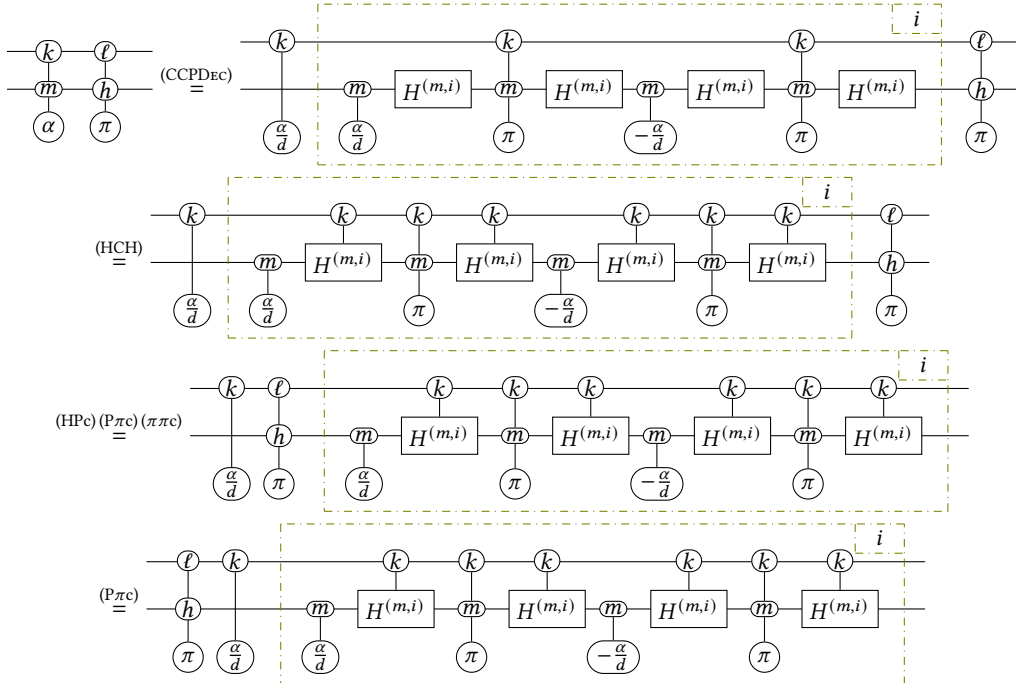
PROOF OF (CH-CP).

PROOF OF (CCP-CCP_≠).



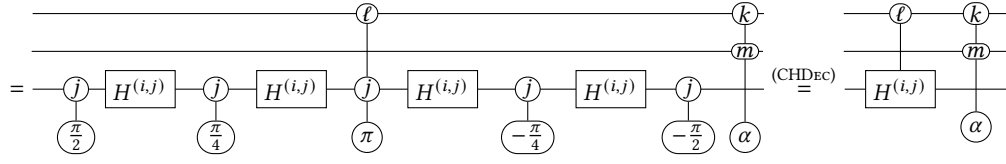
□

PROOF OF (CCP-CNOT).



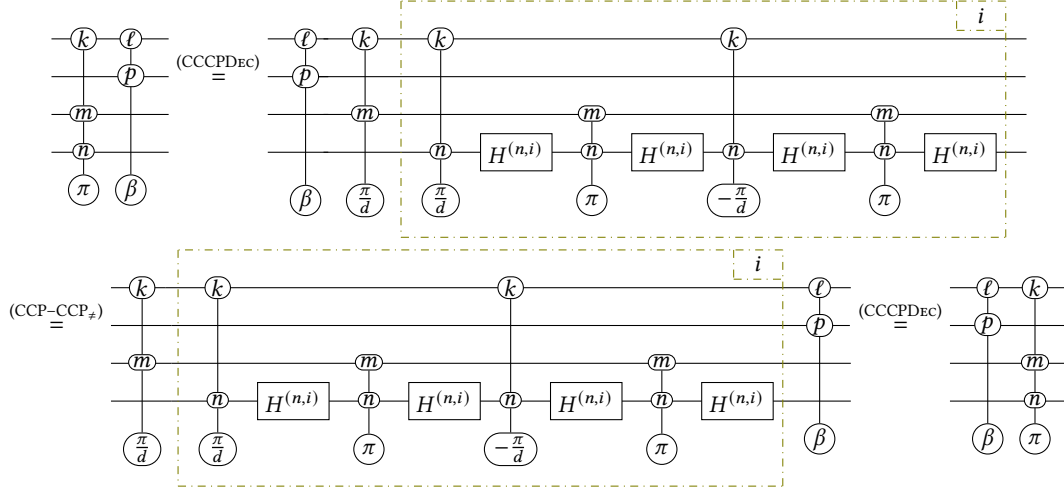


9



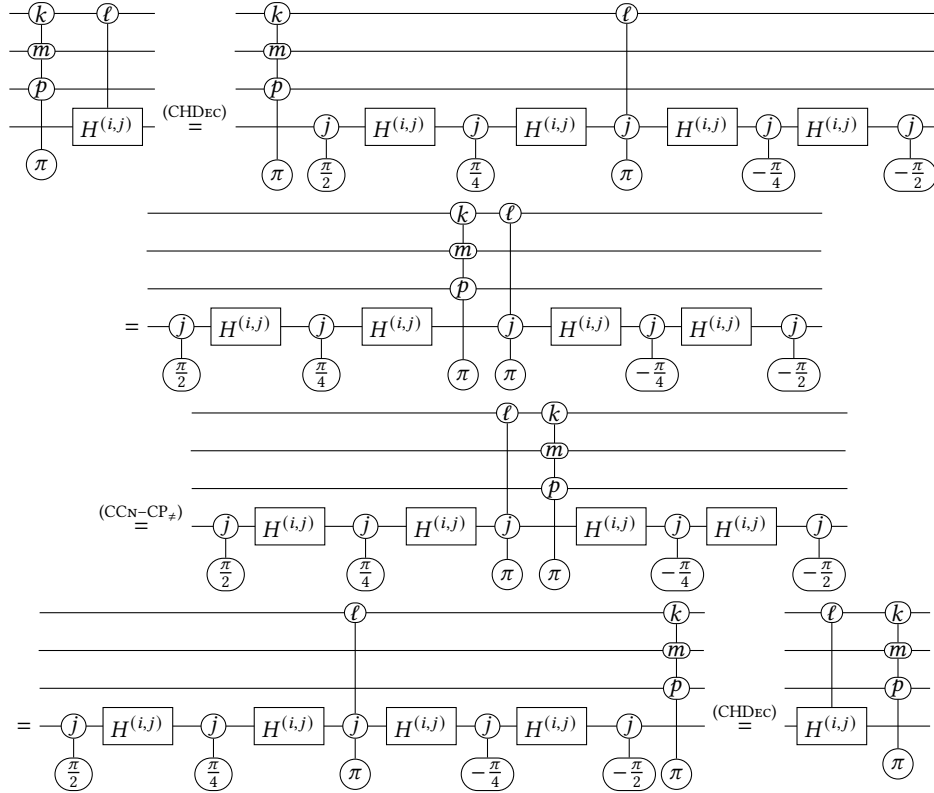
□

PROOF OF $(CCN-CP_{\neq})$.

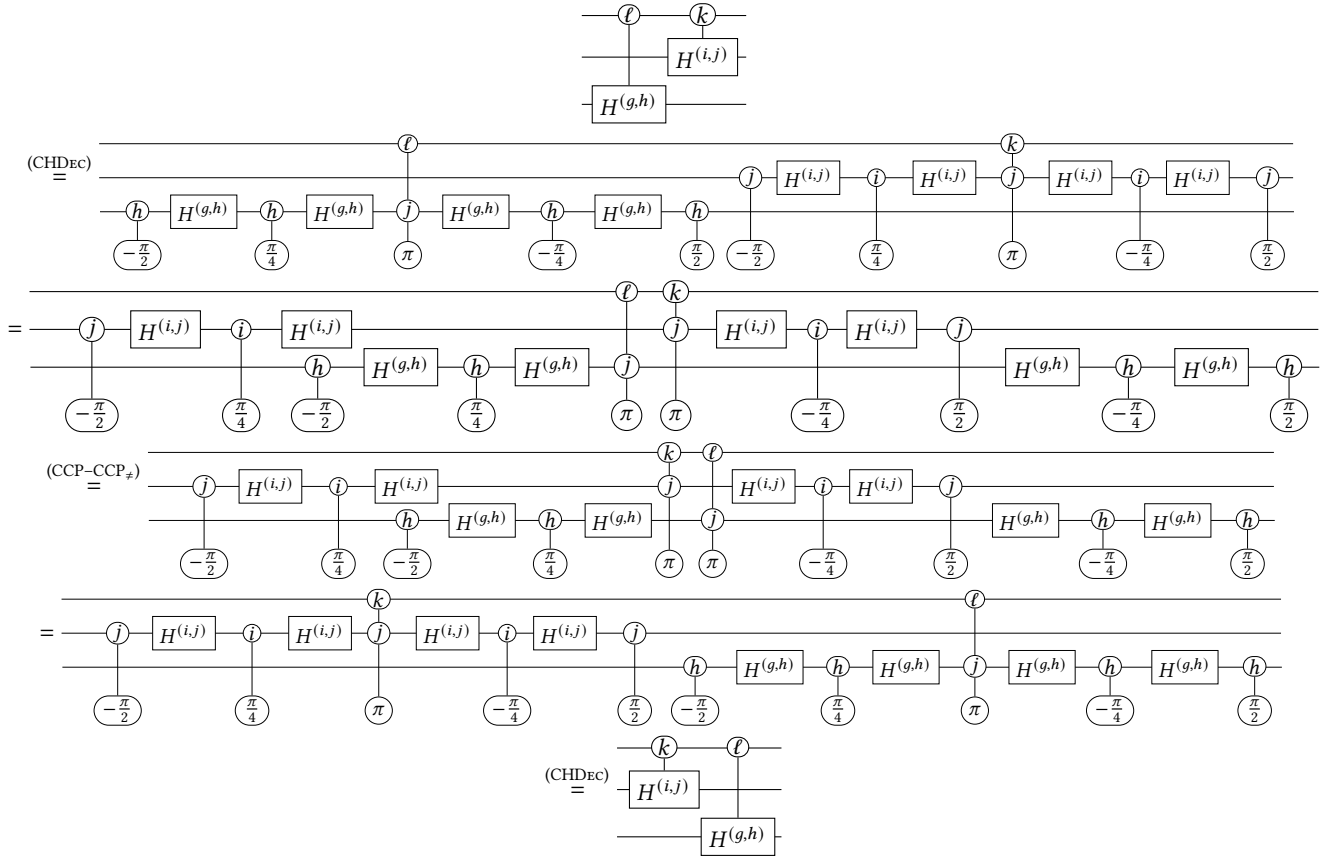


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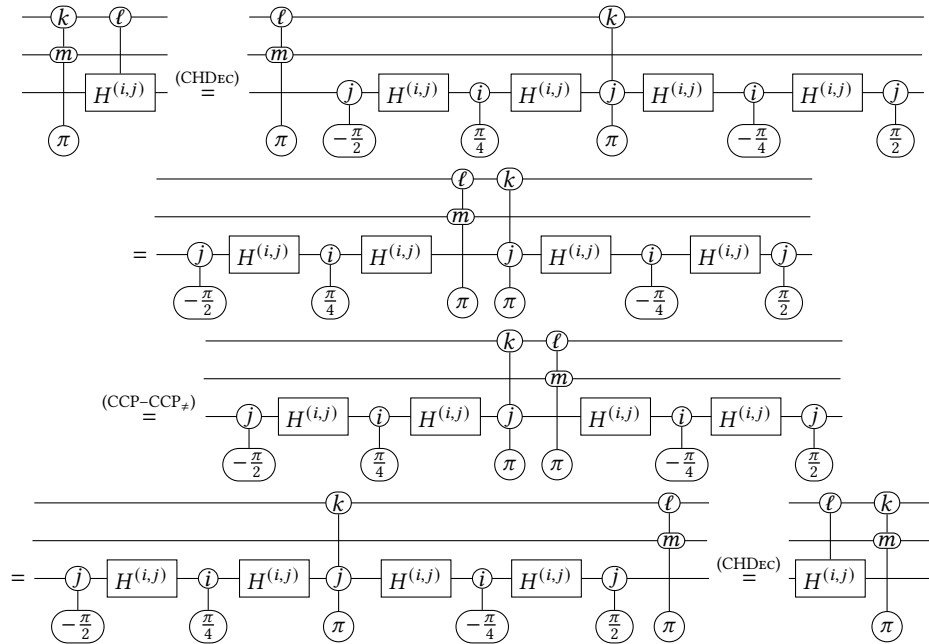
PROOF OF $(CCN-CH_{\neq})$.



□

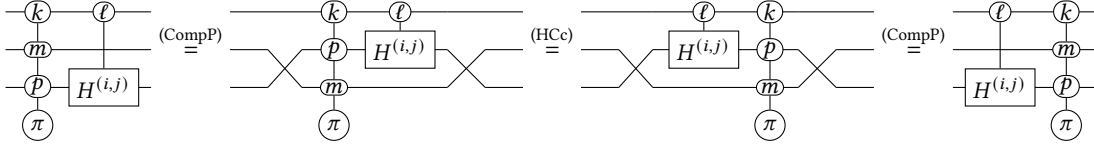
PROOF OF (CH-CH_≠).

□

PROOF OF (CNOT-CH_≠).

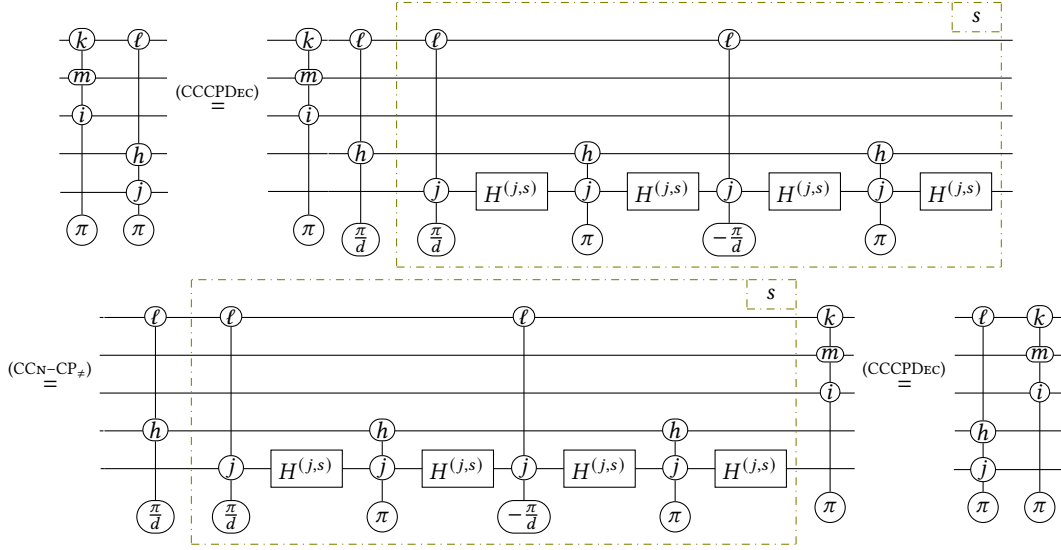
□

PROOF OF (CCN-CH_{2π}).



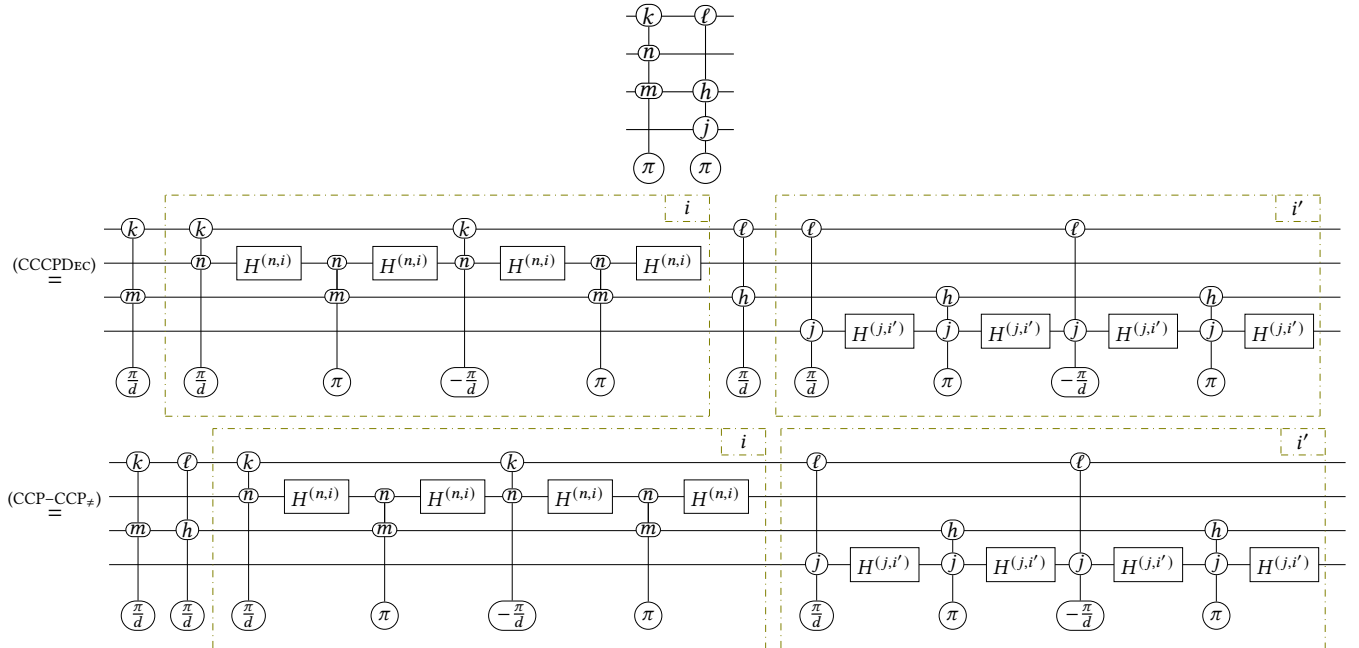
□

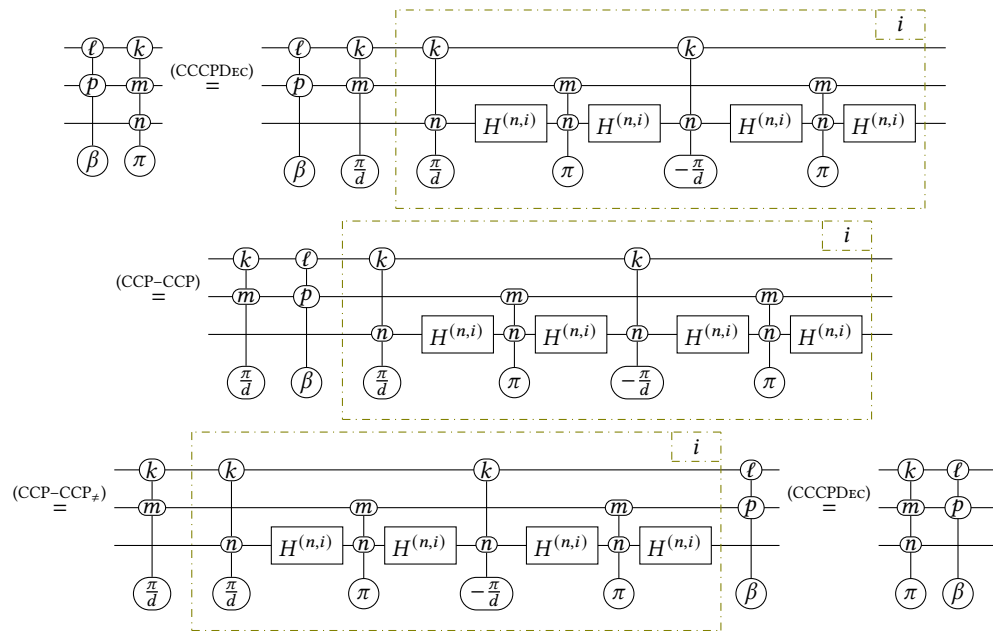
PROOF OF (CCN-CCN₄).



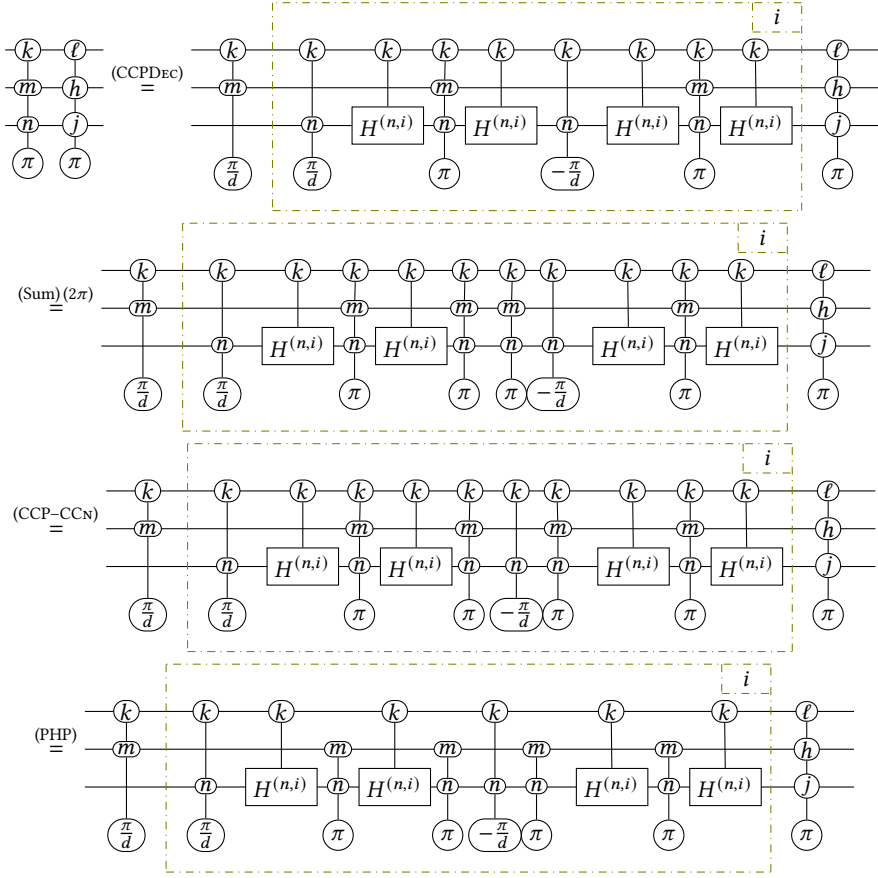
□

PROOF OF (CCN-CCN₃).

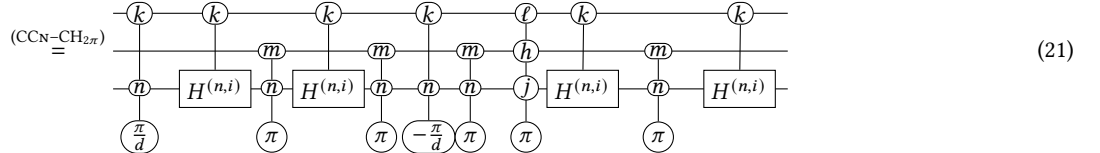
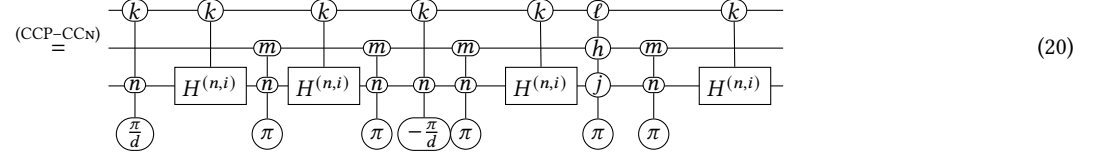
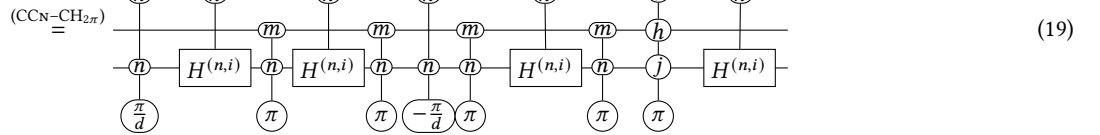
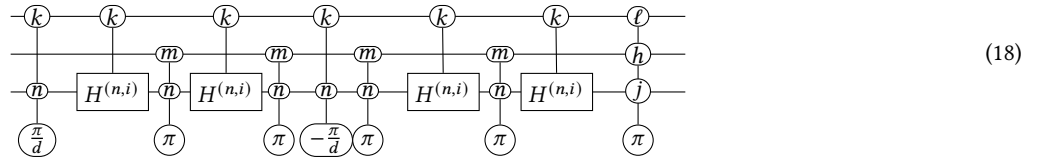


☐

PROOF OF (CCN-CCN).



For any i we have



(CCP-CCN) =  (22)

$$\begin{array}{c}
\text{(CCP-CCN)} \equiv (\text{CCN-CH}_{2\pi}) \\
\begin{array}{cccccccccccccccc}
\text{---} \ell & \text{---} k & & k & & k & & k & & k & & k & & k \\
\text{---} h & & & & & m & & m & & m & & m & & m \\
\text{---} j & \text{---} n & \boxed{H^{(n,i)}} & \text{---} n & \boxed{H^{(n,i)}} & \text{---} n & \text{---} n & \text{---} n & \boxed{H^{(n,i)}} & \text{---} n & \boxed{H^{(n,i)}} & & & \\
\text{---} \pi & \text{---} \frac{\pi}{d} & & \pi & & \pi & \text{---} \frac{\pi}{d} & \text{---} \pi & & \pi & & \pi & & \pi
\end{array}
\end{array} \quad (23)$$

and hence, applying this for all i ,

Figure 1 displays four quantum circuit diagrams, labeled (CCP-CCN), (PHP), (CCP-CCN)(Sum) (2\pi), and (CCP-DEC), illustrating the proposed CCP-CCN and its variants. Each circuit consists of three horizontal lines representing qubits, with various gates and phases. The (CCP-CCN) circuit includes a dashed box labeled 'i' containing a sequence of $H(n,i)$ gates and phase rotations. The (PHP) circuit is similar but with different phase rotations. The (CCP-CCN)(Sum) (2\pi) circuit includes a sum of phases. The (CCP-DEC) circuit is a simplified version of the others.

☐

C Compatibility

Fix $d \geq 2$ and the polycontrolled dagger PROP $\mathbf{QCQ}_d$ of §2, equipped with control functor C_k for each $k \in \{0, \dots, d-1\}$. In this appendix we prove Theorem 3.2 (§3.3), i.e. the mixed-control *compatibility* principle (COMPAT in Definition 2.4), inside the equational theory $\mathbf{QCQ}_d$.

Our proof is purely syntactic and uses a normalisation step that will also be reused in Appendix D (commutativity of distinct controls). Throughout we work modulo:

- the strict symmetric monoidal coherence in $\mathbf{CQC}_d$ (associativity, unit, symmetry, interchange), so wire permutations can be freely inserted and removed; and
- the coherence laws for each control functor C_k (functoriality, strength, and naturality), so controls can be pushed through structural contexts.

Algorithm 1 SEPARATE: push parallelism out, isolate single-gate layers with explicit controls

```

1: function SEPARATE( $f$ )
2:   let  $f' := f$  where we removed swaps using SMC coherence and (S)
3:   match  $f'$  with
4:     case  $f_1 \otimes f_2$ :
5:       return (SEPARATE( $f_1$ )  $\otimes$   $\text{id}_{|f_2|}$ )  $\circ$  ( $\text{id}_{|f_1|} \otimes$  SEPARATE( $f_2$ ))
6:     case  $f_1 \circ f_2$ :
7:       return SEPARATE( $f_1$ )  $\circ$  SEPARATE( $f_2$ )
8:     case  $C_m(f_1)$ :
9:       match  $f_1$  with
10:        case  $g \circ h$ : return SEPARATE( $C_m(g)$ )  $\circ$  SEPARATE( $C_m(h)$ )
11:        case  $g \otimes h$ :
12:          return (SEPARATE( $C_m(g)$ )  $\otimes$   $\text{id}_{|h|}$ )  $\circ$  ( $\text{id}_1 \otimes \sigma_{|g|,|h|}$ )  $\circ$ 
13:            (SEPARATE( $C_m(h)$ )  $\otimes$   $\text{id}_{|g|}$ )  $\circ$  ( $\text{id}_1 \otimes \sigma_{|h|,|g|}$ )
14:        case  $\boxed{H^{(i,j)}}$ : return SEPARATE(Apply CHDec to rewrite  $f'$ )
15:        case  $\alpha$ : return  $f'$ 
16:        case  $\text{id}_n$ : return  $\text{id}_{n+1}$ 
17:        case  $C_n(g)$ :
18:          match  $g$  with
19:            case  $\pi$ : return  $f'$ 
20:            case  $\beta$ : return SEPARATE(Apply CCPDec to rewrite  $f'$ )
21:            case  $C_k(\alpha)$ : return SEPARATE(Apply CCCPDec to rewrite  $f'$ )
22:            otherwise return SEPARATE( $C_m(\text{SEPARATE}(f_1))$ )
23:   otherwise: return  $f'$ 

```

C.1 Normalisation into controlled layers

The point of this subsection is to isolate a small family of “one-gate layers”, each equipped with an explicit control word. Concretely, we show that every circuit can be rewritten (provably in QC_d) into a sequential composition of such layers, with all wire permutations made explicit. This normal form is what allows the compatibility argument to reduce to the single axiom $(C\pi S)$, and the commutativity argument to reduce to a finite table of generator commutations.

Definition C.1. An atomic layer is a morphism of the form

$$L := (\text{id}_p \otimes G \otimes \text{id}_q) : n \rightarrow n, \quad G \in \{\text{id}_n \mid n \in \mathbb{N}\} \cup \{(\theta) \mid \theta \in \mathbb{R}\} \cup \{\boxed{H^{(i,j)}} \mid 0 \leq i < j < d\} \cup \{C_k((\theta)) \mid k, \theta\} \cup \{C_k(C_\ell((\pi))) \mid k, \ell\}.$$

A *controlled layer* is any circuit of the form $C_u(L)$ for a control word u .

If a circuit C is written as a sequential composition of controlled layers

$$C = L_m \circ \dots \circ L_1, \quad L_t = C_{u_t}(\text{id}_{p_t} \otimes G_t \otimes \text{id}_{q_t}),$$

its *control-depth multiset* is $\text{cd}(C) := \{|u_t| \mid t = 1, \dots, m\}$. We order such multisets by the usual multiset extension of the natural order on $\mathbb{N}$.

We now give an explicit normalisation procedure. Informally, it: (i) eliminates implicit swaps using strict monoidal coherence and the swap decomposition axiom, (ii) pushes $\otimes$ outward by serialising it into a composition of “upper then lower” actions, and (iii) pushes controls down to generators, expanding controlled composites using the bounded-arity axioms of QC_d .

Lemma C.2. SEPARATE terminates on all inputs f and returns a sequential composition

$$\text{SEPARATE}(f) = L_m \circ \dots \circ L_1,$$

where each L_t is an atomic layer and all permutations appear only as explicit wire-swaps between such layers.

PROOF. Define a well-founded measure on circuits by

$$\mu(f) := (\# \text{of non-identity gates in } f, \sum_{g \text{ gate in } f} \text{control-depth}(g)),$$

ordered lexicographically. Each recursive clause of the definition of SEPARATE either:

- eliminates a tensor node in favour of sequential composition of smaller circuits (strictly reducing the number of non-identity gates in the first component), or

- pushes a control one step closer to a generator and immediately expands a controlled generator using Lemmas CHDEC, CCPDEC and CCCPDEC, which strictly reduces the total nesting depth of non-generator gates.

Functoriality, strength and swap laws are used only to rearrange the diagram and do not increase μ . Hence μ decreases along every recursive branch, so SEPARATE terminates. By construction, the output is a composition of controlled layers with generator drawn from the stated finite set and with all permutations explicit. $\square$

Corollary C.3. *Every morphism of $\mathbf{CQC}_d$ is provably equal in $\mathbf{QC}_d$ to a circuit generated (using only sequential and parallel composition) by*

$$\mathcal{G} = \{ id_n \mid n \in \mathbb{N} \} \cup \{ \langle \theta \rangle \mid \theta \in \mathbb{R} \} \cup \{ -\boxed{H^{(i,j)}} \mid 0 \leq i < j < d \} \cup \{ C_k(\langle \theta \rangle) \mid k, \theta \} \cup \{ C_k(C_\ell(\langle \pi \rangle)) \mid k, \ell \}$$

C.2 Compatibility of mixed controls

We now derive mixed-control compatibility (COMPAT in Definition 2.4) from the finite axiom system $\mathbf{QC}_d$, using the normalisation machinery above.

PROOF OF THEOREM 3.2. Write $S := \sigma_{1,1} \otimes id_n : 2 + n \rightarrow 2 + n$. Consider the two $(2 + n)$ -wire circuits

$$F_{ab} := C_a(C_b(f)), \quad F_{ba} := C_b(C_a(f)).$$

Apply the normalisation procedure SEPARATE (Algorithm 1) to F_{ab} and F_{ba} . By construction, each rewrite step performed by SEPARATE is justified either by strict PROP coherence, by control-functoriality/strength/naturality, or by one of the bounded-arity axiom schemata of Figures 1–2. Hence $\mathbf{QC}_d \vdash F_{ab} = \text{SEPARATE}(F_{ab})$ and $\mathbf{QC}_d \vdash F_{ba} = \text{SEPARATE}(F_{ba})$.

Inspecting the clauses of SEPARATE, the *only* situation in which two nested controls are left *undecomposed* is the branch

$$\text{case } C_m(C_n(g)) \text{ with } g = \langle \pi \rangle : \text{return } f',$$

i.e. nested control on the scalar π -phase. All other nested-control cases are immediately expanded using Lemmas CCPDEC and CCCPDEC (and the controlled-Hadamard decompositions), so that in $\text{SEPARATE}(F_{ab})$ and $\text{SEPARATE}(F_{ba})$ the only atomic layers that can simultaneously involve the two *outer* control wires are instances of the doubly-controlled π -phase $C_a(C_b(\langle \pi \rangle))$ (possibly tensored with identity on untouched wires).

We now push the swap S through the sequential decomposition $\text{SEPARATE}(F_{ab})$ from right to left. For any layer that acts on at most one of the two outer control wires, this is pure strict symmetric-monoidal coherence (naturality of σ). For a layer equal to $C_a(C_b(\langle \pi \rangle))$ on the two outer control wires, the required step is exactly the axiom schema (C π S), i.e. (CompP). Thus we can rewrite

$$\text{SEPARATE}(F_{ab}) \circ S = S \circ \text{SEPARATE}(F_{ba})$$

in $\mathbf{QC}_d$. Finally, replacing $\text{SEPARATE}(F_{ab})$ by F_{ab} and $\text{SEPARATE}(F_{ba})$ by F_{ba} using the equalities above yields $F_{ab} \circ S = S \circ F_{ba}$, as required. $\square$

D Commutativity

Fix $d \geq 2$ and the polycontrolled dagger PROP $\mathbf{CQC}_d$ of §2. In this appendix we prove Theorem 3.3 (§3.3), i.e. that circuits controlled on *distinct* computational-basis values commute.

As in Appendix C, we work modulo strict symmetric monoidal coherences and the coherence laws of each control functor. We will use the finite generating family $\mathcal{G}$ obtained by the normalisation procedure SEPARATE (Corollary C.3) to reduce the commutativity statement to a finite list of generator-level commutations.

PROOF OF THEOREM 3.3. By Corollary C.3 and SMC coherence we may assume that

$$f = id_p \otimes f' \otimes id_{n-n_f-p}, \quad g = id_q \otimes g' \otimes id_{n-n_g-q},$$

with $f' : n_f \rightarrow n_f$ and $g' : n_g \rightarrow n_g$ belonging to $\mathcal{G}$, and where $0 \leq p \leq q$. If $q - p > n_f$, we can insert a symmetry $id_p \otimes \sigma_{n_f, q-p-n_f} \otimes id_{n-q}$ on both sides of the desired equation to reduce to the case $q - p \leq n_f$. Thus we only need to consider the relative offset $i := q - p \in \{0, \dots, n_f\}$. If $g' = id_{n_g}$ then $C_\ell(g)$ is the identity and the statement is trivial, so we exclude this case in the table below.

The remaining possibilities are summarised in the following table, where each non-trivial entry cites the derived commutation lemma used to swap the two layers in question (all equations are proved in Appendix B).

	$f' = id_n$	$f' = (\theta)$	$f' = C_i((\theta))$	$f' = C_i(C_j((\pi)))$	$f' = \boxed{H^{(i,j)}} -$
$g' = id_m$	trivial	trivial	trivial	trivial	trivial
$g' = (\gamma)$		$i = 0: (PPc)$	$i = 0: (CCP-CP)$	$i = 0: (CCN-CP)$	$i = 0: (CH-CP)$
$g' = C_k((\gamma))$			$i = 0: (CCP-CCP)$ $i = 1: (CCP-CCP_{\neq})$	$i = 0: (CCP-CCN)$ $i = 1: (CCN-CP_{\neq})$	$i = 0: (HPc)$ $i = 1: (CCP-CH_{\neq})$
$g' = C_k(C_{\ell}((\pi)))$				$i = 0: (CCN-CCN)$ $i = 1: (CCN-CCN_3)$ $i = 2: (CCN-CCN_4)$	$i = 0: (HCc)$ $i = 1: (CCN-CH_{2\pi})$ $i = 2: (CCN-CH_{\neq})$
$g' = \boxed{H^{(k,\ell)}} -$					$i = 0: (HHc)$ $i = 1: (CH-CH_{\neq})$

Each cited lemma states precisely that the corresponding pair of controlled generators commutes for the indicated value of the offset i . Tensoring with identities on the untouched wires and inserting coherence isomorphisms as needed yields

$$C_k(f) \circ C_{\ell}(g) = C_{\ell}(g) \circ C_k(f)$$

for each of the finitely many generator pair. Since arbitrary circuits are built from these generators by sequential and parallel composition, and C_k is a strict monoidal functor, the general case follows. $\square$

E Exhaustivity

In this appendix we prove Theorem 3.4 (§3.3), i.e. that the family of control functors $(C_k)_{k=0}^{d-1}$ is *exhaustive*.

PROOF OF THEOREM 3.4. Using the swap decomposition axiom (S) together with strict symmetric-monoidal coherence, we may assume f is presented without explicit swap generators. We then prove the claim by structural induction on f .

Let $E(f)$ denote the equation

$$\prod_{k=0}^{d-1} C_k(f) = \text{---} \otimes f.$$

We prove $E(f)$ for all circuits f by structural induction on f using the generators and operations of $\mathbf{CQC}_d$.

Base cases. For the monoidal unit and a single wire we have

$$\prod_{k=0}^{d-1} C_k(\text{---}) = \text{---} = \text{---} \otimes \text{---}, \quad \prod_{k=0}^{d-1} C_k(\text{---}) = \text{id}_2 = \text{---} \otimes \text{---},$$

so $E(\text{---})$ and $E(\text{---})$ hold. Since $\text{id}_n \cong \text{---}^{\otimes n}$, the general identity case follows from the tensor step below.

For a scalar phase (θ) , the *total phase* axiom (TP) gives

$$\prod_{k=0}^{d-1} C_k((\theta)) \stackrel{(TP)}{=} \text{---} \otimes (\theta),$$

so $E((\theta))$ holds.

For a two-level Hadamard $\boxed{H^{(i,j)}} -$ the *total Hadamard* equation (TH) yields

$$\prod_{k=0}^{d-1} C_k(\boxed{H^{(i,j)}} -) \stackrel{(TH)}{=} \text{---} \otimes \boxed{H^{(i,j)}} -,$$

so $E(\boxed{H^{(i,j)}} -)$ holds.

Sequential composition. Assume $E(g)$ and $E(h)$. By functoriality of each C_k we have $C_k(g \circ h) = C_k(g) \circ C_k(h)$, hence

$$\prod_{k=0}^{d-1} C_k(g \circ h) = \prod_{k=0}^{d-1} (C_k(g) \circ C_k(h)).$$

Using commuting controls (Theorem 3.3), we can reorder the factors so that all $C_k(g)$ appear to the left of all $C_k(h)$:

$$\prod_{k=0}^{d-1} (C_k(g) \circ C_k(h)) \stackrel{\text{Theorem 3.3}}{=} \prod_{k=0}^{d-1} C_k(g) \circ \prod_{k=0}^{d-1} C_k(h).$$

By the induction hypothesis,

$$\prod_{k=0}^{d-1} C_k(g) = \text{---} \otimes g, \quad \prod_{k=0}^{d-1} C_k(h) = \text{---} \otimes h,$$

and therefore

$$\prod_{k=0}^{d-1} C_k(g \circ h) = (\text{---} \otimes g) \circ (\text{---} \otimes h) = \text{---} \otimes (g \circ h),$$

so $E(g \circ h)$ holds.

Parallel composition. Assume $E(g)$ and $E(h)$. Using the definition of C_k as a strict monoidal endofunctor together with the strength/coherence laws, a controlled tensor can be written (up to symmetry) as

$$C_k(g \otimes h) = (C_k(g) \otimes \text{id}_{|h|}) \circ (\text{---} \otimes \sigma_{|g|,|h|}) \circ (C_k(h) \otimes \text{id}_{|g|}) \circ (\text{---} \otimes \sigma_{|h|,|g|}).$$

Hence

$$\begin{aligned} \prod_{k=0}^{d-1} C_k(g \otimes h) &= \prod_{k=0}^{d-1} \left((C_k(g) \otimes \text{id}_{|h|}) \circ (\text{---} \otimes \sigma_{|g|,|h|}) \circ (C_k(h) \otimes \text{id}_{|g|}) \circ (\text{---} \otimes \sigma_{|h|,|g|}) \right) \\ &\stackrel{\text{Theorem 3.3}}{=} \prod_{k=0}^{d-1} (C_k(g) \otimes \text{id}_{|h|}) \circ (\text{---} \otimes \sigma_{|g|,|h|}) \circ \prod_{k=0}^{d-1} (C_k(h) \otimes \text{id}_{|g|}) \circ (\text{---} \otimes \sigma_{|h|,|g|}). \end{aligned}$$

Applying the induction hypothesis pointwise and using monoidality of $\text{---} \otimes \text{---}$,

$$\prod_{k=0}^{d-1} C_k(g \otimes h) = (\text{---} \otimes g \otimes \text{id}_{|h|}) \circ (\text{---} \otimes \sigma_{|g|,|h|}) \circ (\text{---} \otimes h \otimes \text{id}_{|g|}) \circ (\text{---} \otimes \sigma_{|h|,|g|}) = \text{---} \otimes (g \otimes h),$$

so $E(g \otimes h)$ holds.

Nested control. Finally, suppose $E(g)$ holds and consider $C_m(g)$. Using compatibility of mixed controls (Theorem 3.2), nested controls may be exchanged at the cost of a swap:

$$C_k(C_m(g)) = (\sigma_{1,1} \otimes \text{id}_{|g|}) \circ C_m(C_k(g)) \circ (\sigma_{1,1} \otimes \text{id}_{|g|}).$$

Hence

$$\begin{aligned} \prod_{k=0}^{d-1} C_k(C_m(g)) &= (\sigma_{1,1} \otimes \text{id}_{|g|}) \circ \prod_{k=0}^{d-1} C_m(C_k(g)) \circ (\sigma_{1,1} \otimes \text{id}_{|g|}) \\ &= (\sigma_{1,1} \otimes \text{id}_{|g|}) \circ C_m \left(\prod_{k=0}^{d-1} C_k(g) \right) \circ (\sigma_{1,1} \otimes \text{id}_{|g|}) \\ &\stackrel{E(g)}{=} (\sigma_{1,1} \otimes \text{id}_{|g|}) \circ C_m(\text{---} \otimes g) \circ (\sigma_{1,1} \otimes \text{id}_{|g|}). \end{aligned}$$

By functoriality and coherence of C_m , the right-hand side is precisely $\text{---} \otimes C_m(g)$, so $E(C_m(g))$ holds.

Corollary E.1. *Since we can write the swap gate using the generators above (Equation (S)), $E(\sigma_{m,n})$ also holds, and the previous lemma extends to all circuits in $\mathbf{CQC}_d$.*

Since every circuit in $\mathbf{CQC}_d$ is built from the above generators by sequential composition, tensor product, and (nested) control, the property $E(f)$ holds for all f , completing the proof. $\square$

The exhaustivity property says that the collection of projectors $(|k\rangle\langle k|)_{k=0}^{d-1}$ on the control wire forms a partition of the identity, which is reflected syntactically by the equation of Theorem 3.4. As a useful consequence, controlled operations are stable under conjugation on the target system.

F Further derived identities

Throughout this appendix we work in the equational theory $\mathbf{QC}_d$, and write $\mathbf{QC}_d \vdash C_1 = C_2$ when the circuit identity $C_1 = C_2$ is derivable from the axioms of Figures 1–2

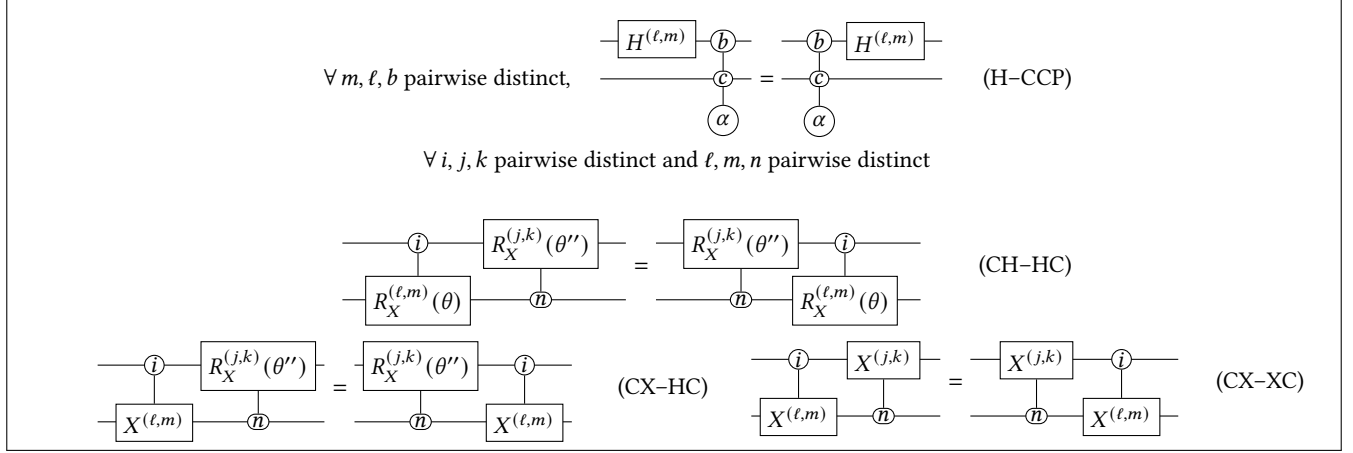


Figure 6: Derived rules in QC_d (batch 4).

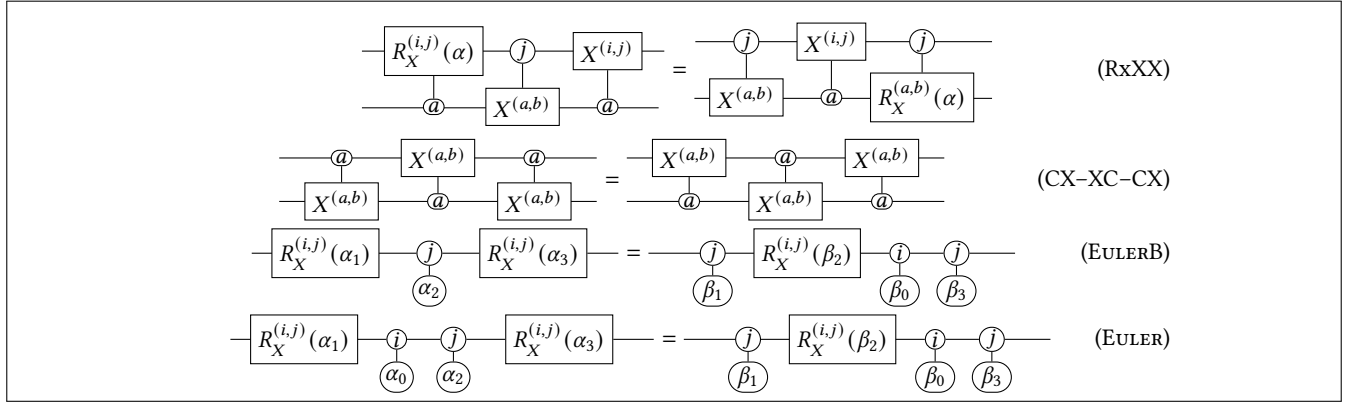
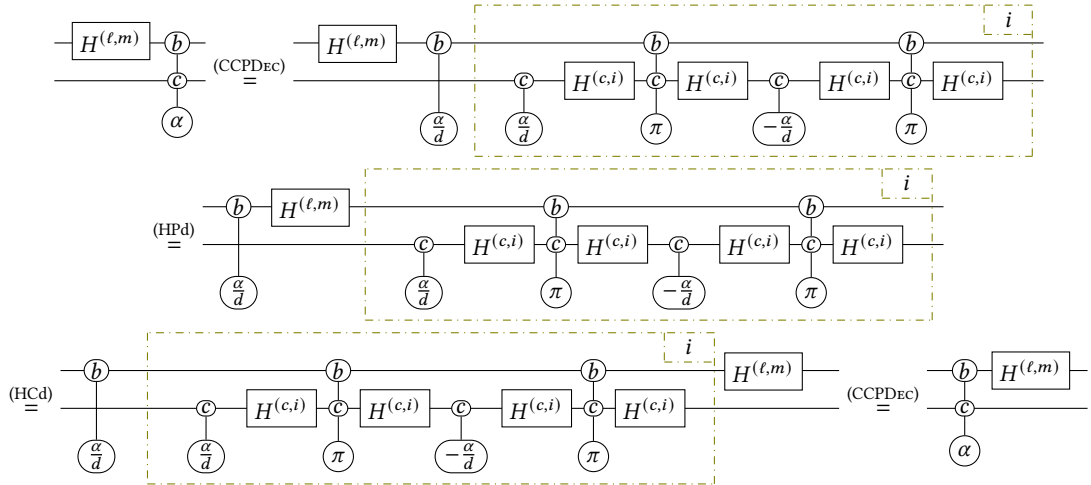


Figure 7: Derived rules in QC_d (batch 5).

F.1 Catalogue of derived rules

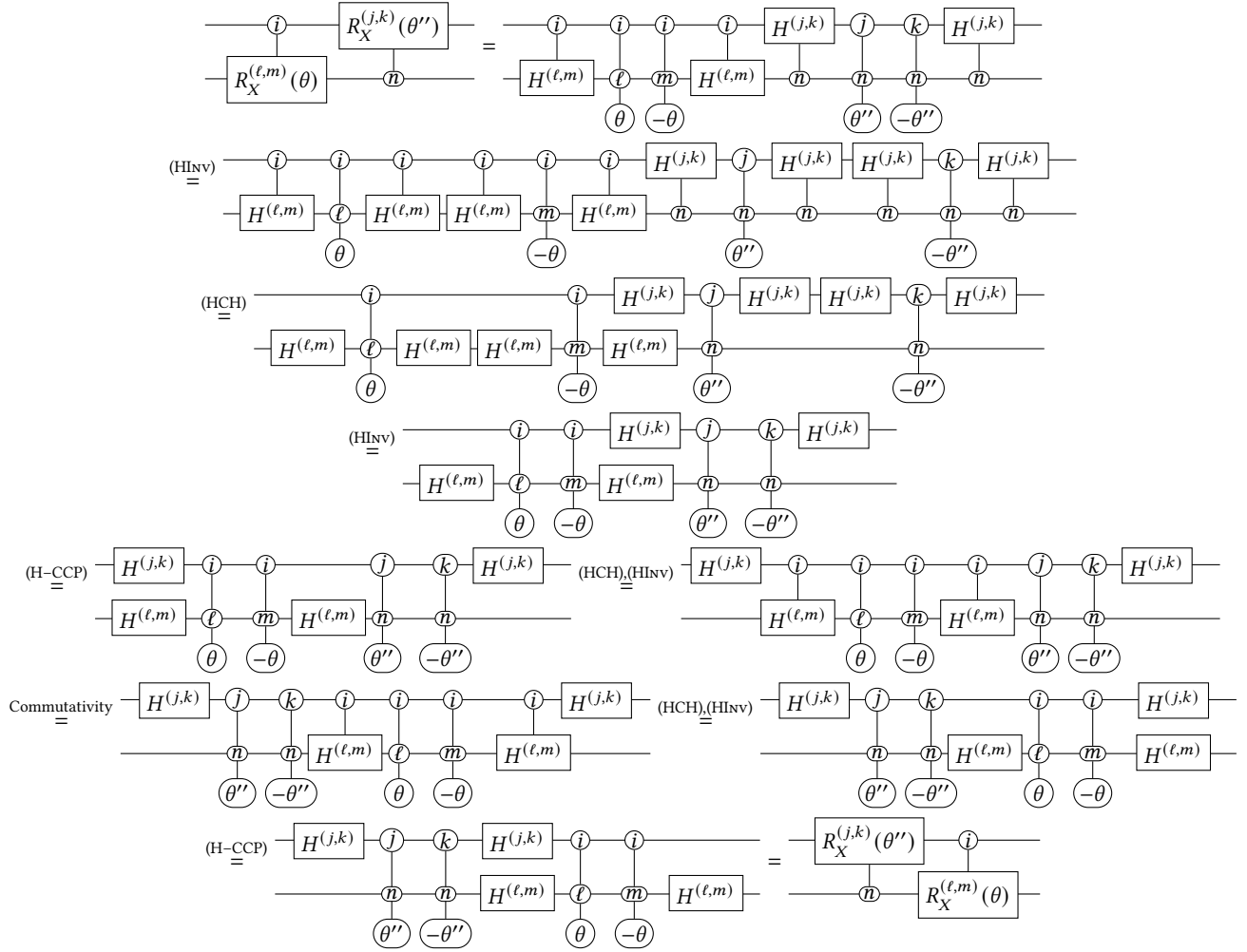
F.2 Proofs of the derived rules

PROOF OF (H-CCP).



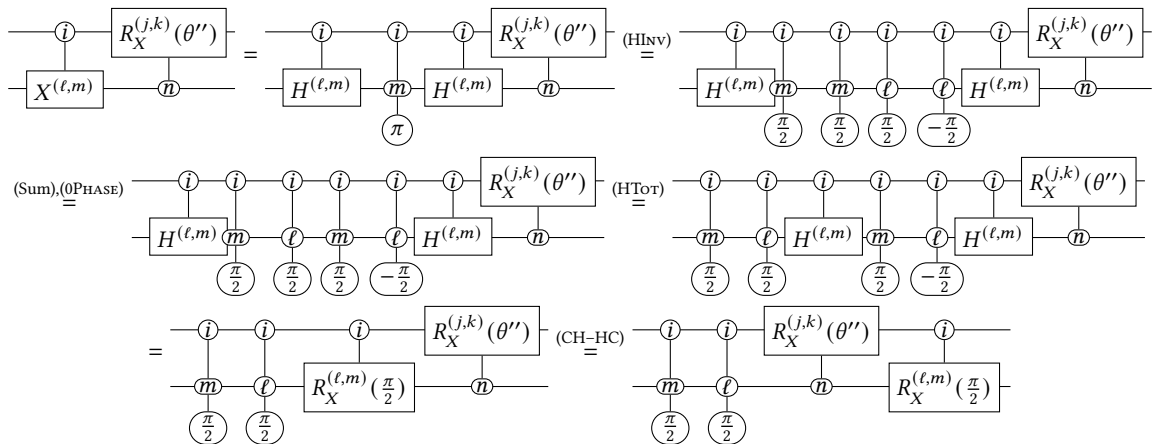
□

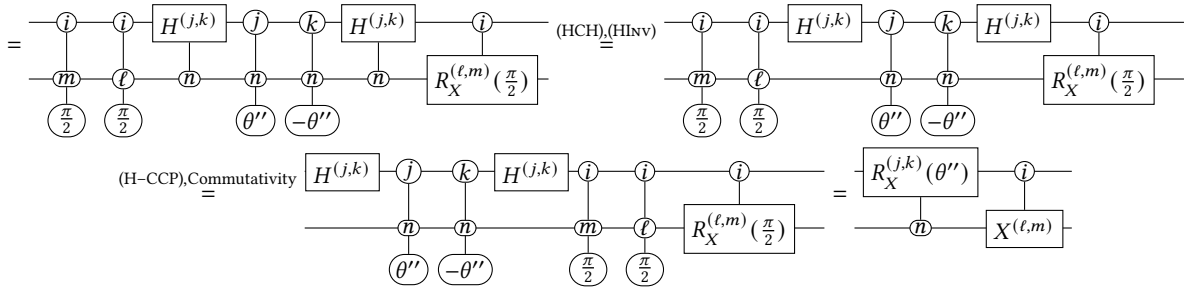
PROOF OF (CH-HC). We treat the case i, j, k pairwise distinct; the case with ℓ, m, n is analogous.



□

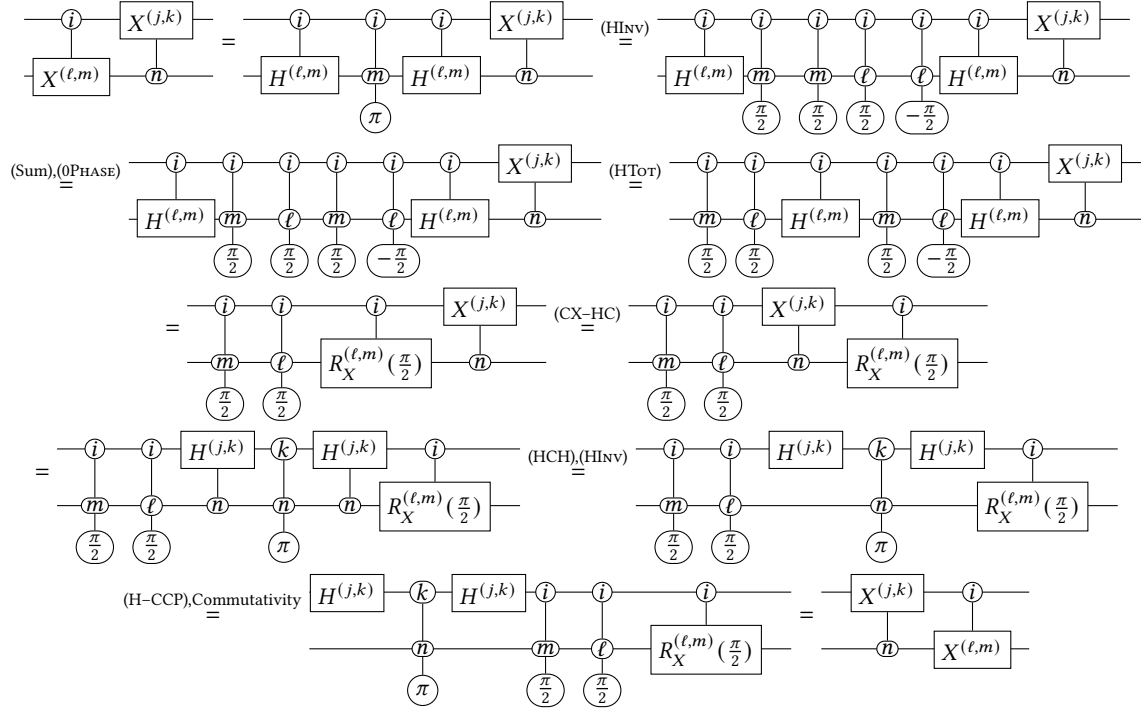
PROOF OF (CX-HC). Again consider i, j, k pairwise distinct; the case with ℓ, m, n is analogous.





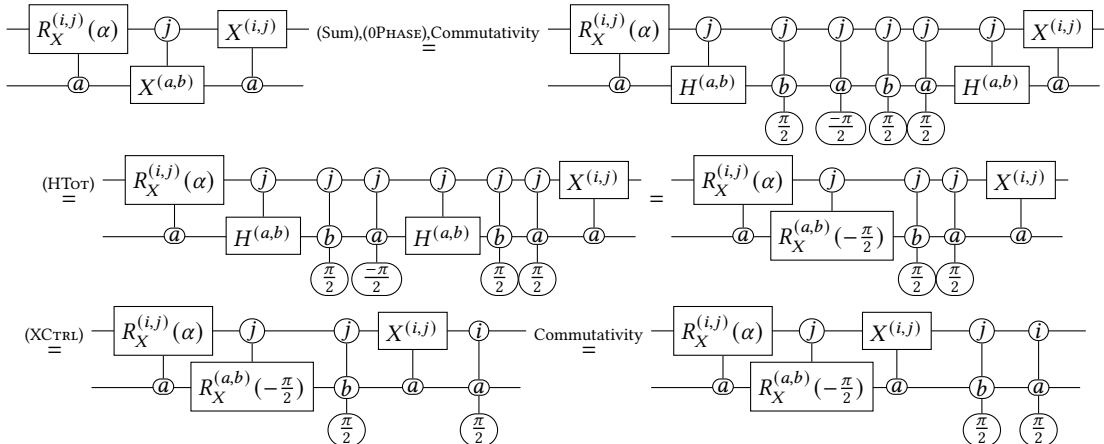
□

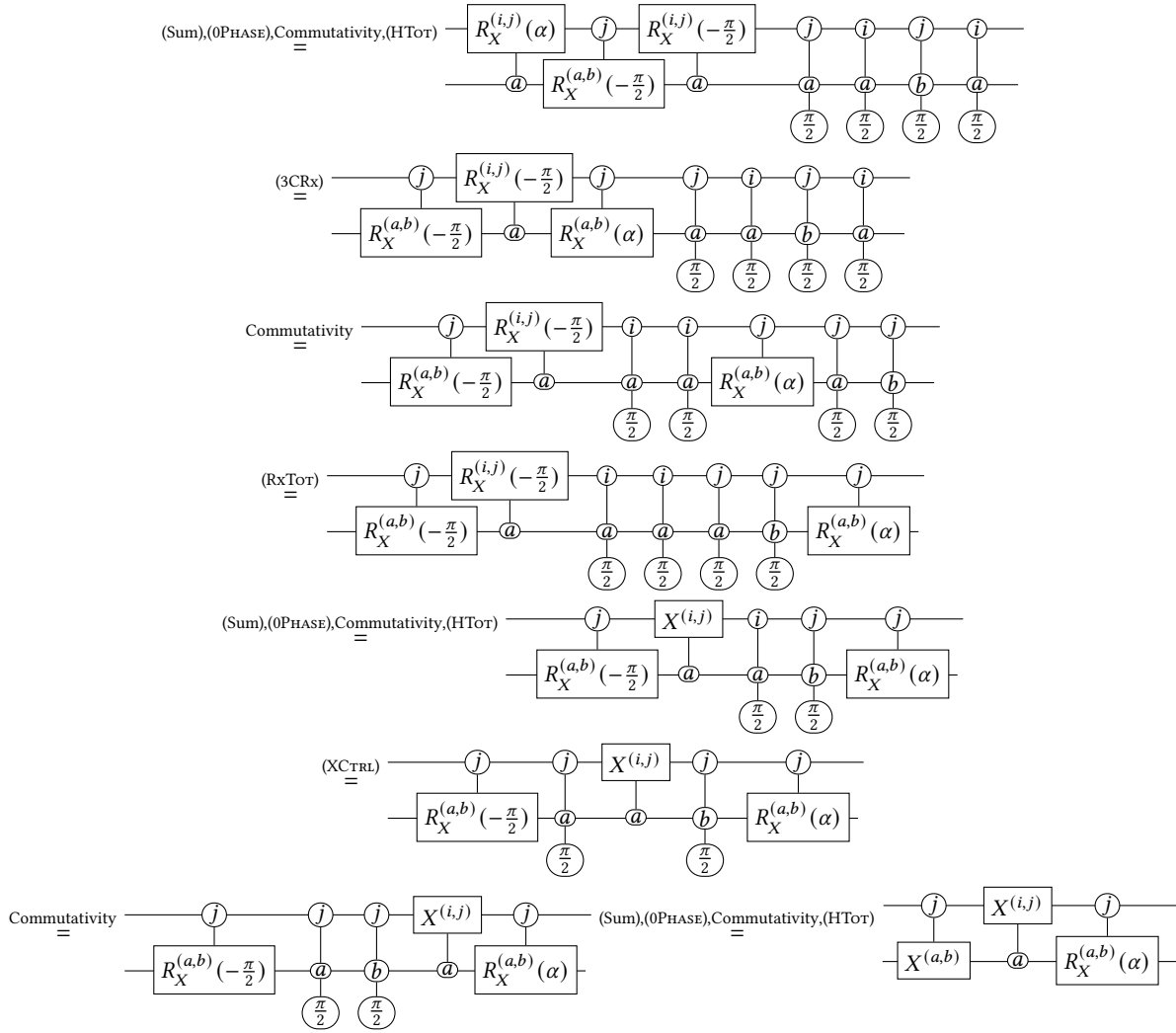
PROOF OF (CX-XC). For i, j, k pairwise distinct (the case with ℓ, m, n is similar),



□

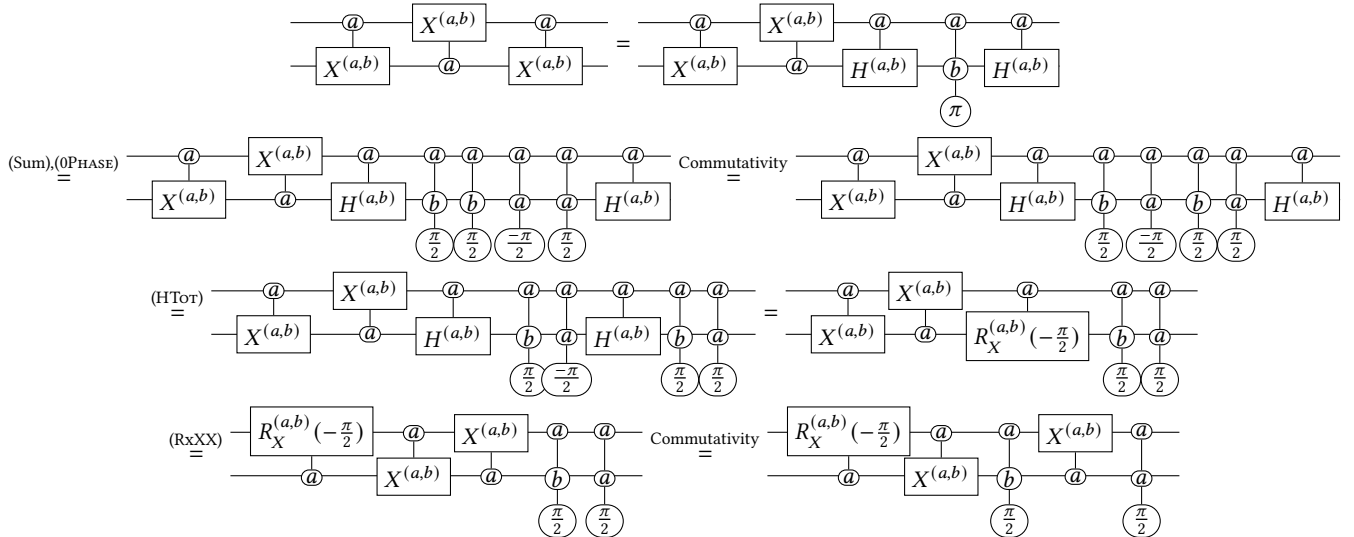
PROOF OF (RxXX).

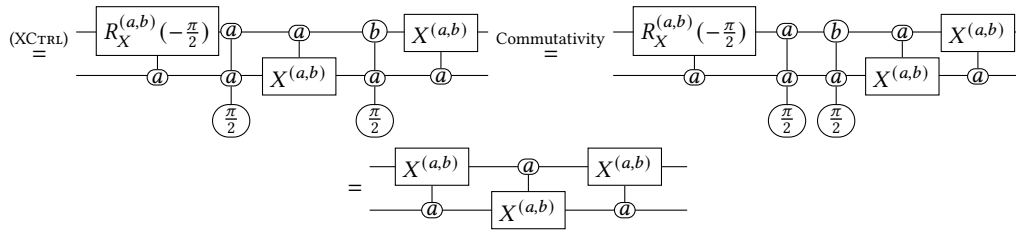




□

PROOF OF (CX-XC-CX).



☐

F.2.1 A π -normalisation lemma (choosing the split parameter x). In the proof of (EULERB) we introduce an auxiliary real parameter x (by splitting a middle rotation/phase into two pieces x and $\alpha_2 - x$). This produces a *family* of circuit identities indexed by x , in which the angles $\beta_i(x)$ and $\gamma_i(x)$ produced by the Euler-extraction depend on x (while the input angles $\alpha_1, \alpha_2, \alpha_3$ are fixed).

To continue the purely diagrammatic part of the derivation, we need one additional property: we can choose a value x_0 such that the *bridge phase* $\beta_3(x_0) + \gamma_0(x_0)$ is a multiple of π . Since angles are always understood modulo 2π , this means that $\beta_3(x_0) + \gamma_0(x_0)$ is either 0 or π modulo 2π , which is exactly what justifies the two cases treated in the proof of (EULERB).

Lemma F.1 (Existence of a π -normalised split [32]). *Fix $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ and consider the continuous function $N : [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow \mathbb{R}$ defined by*

$$N(x) := \sin(\alpha_1) \cos(x) \cos(\alpha_3) + \cos(\alpha_1) \sin(\alpha_3) \cos(\alpha_2 - x).$$

Then there exists $x_0 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $N(x_0) = 0$.

Moreover, in the Euler-extraction step used in the proof of (EULERB), the corresponding bridge phase satisfies

$$\beta_3(x_0) + \gamma_0(x_0) \in \pi\mathbb{Z},$$

hence (modulo 2π) we necessarily have $\beta_3(x_0) + \gamma_0(x_0) \equiv 0$ or $\equiv \pi$.

PROOF. First note that

$$N\left(-\frac{\pi}{2}\right) = \cos(\alpha_1) \sin(\alpha_3) \cos\left(\alpha_2 + \frac{\pi}{2}\right) = -\cos(\alpha_1) \sin(\alpha_3) \sin(\alpha_2),$$

while

$$N\left(\frac{\pi}{2}\right) = \cos(\alpha_1) \sin(\alpha_3) \cos\left(\alpha_2 - \frac{\pi}{2}\right) = \cos(\alpha_1) \sin(\alpha_3) \sin(\alpha_2).$$

Hence $N(-\frac{\pi}{2}) = -N(\frac{\pi}{2})$, so $N(-\frac{\pi}{2}) \cdot N(\frac{\pi}{2}) \leq 0$. By continuity of N , the intermediate value theorem yields $x_0 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ with $N(x_0) = 0$.

For the second part, recall how the angles $\beta_3(x)$ and $\gamma_0(x)$ are obtained: they arise from solving the trigonometric matching conditions that identify the two sides of the Euler-extraction step (equivalently: matching the corresponding 2×2 blocks on the relevant two-dimensional subspace). A standard calculation shows that the tangent of the bridge sum $\beta_3(x) + \gamma_0(x)$ can be written as a ratio

$$\tan(\beta_3(x) + \gamma_0(x)) = \frac{N(x)}{D(x)}$$

for some real-valued continuous denominator $D(x)$ (the explicit shape of D is irrelevant here; it is the usual t -addition denominator and is continuous wherever the extraction is defined). At $x = x_0$ we have $N(x_0) = 0$, hence $\tan(\beta_3(x_0) + \gamma_0(x_0)) = 0$, which implies $\beta_3(x_0) + \gamma_0(x_0) \in \pi\mathbb{Z}$. Reducing modulo 2π gives the claimed dichotomy $\equiv 0$ or $\equiv \pi$. $\square$

Remark 4. The lemma only provides *existence* (not uniqueness) of a suitable x_0 , and any such choice is valid for the subsequent diagrammatic derivation. Operationally, this is the only point in the proof where we appeal to a small analytic argument; once x_0 is fixed, the remainder proceeds purely by the equational rules of QC_J .

Figure 10 illustrates the proof of the commutativity of the sum of the HTor and HInv relations. The diagram shows a sequence of 10 string diagrams connected by equals signs, representing the steps of the proof. The diagrams involve boxes labeled $R_X^{(i,j)}$ and $H^{(i,j)}$, and circles labeled with various expressions. The steps are labeled as follows:

- Step 1: Initial expression involving $R_X^{(i,j)}$ and $\alpha_1, \alpha_2, \alpha_3$.
- Step 2: Application of the (Sum) relation, introducing a circle labeled x .
- Step 3: Application of the (Commutativity, (Sum), (HTor)) relation, showing a sequence of $H^{(i,j)}$ boxes and circles.
- Step 4: Application of the (HInv) relation, introducing a circle labeled $-2\alpha_1$.
- Step 5: Application of the (EH) relation, showing a sequence of $H^{(i,j)}$ boxes and circles.
- Step 6: Application of the (Sum) relation, introducing a circle labeled β_0 .
- Step 7: Application of the (Commutativity, (Sum), (HTor)) relation, showing a sequence of $H^{(i,j)}$ boxes and circles.

Case $\beta_3 + \gamma_0 = 0$.

Figure 10 illustrates the simplification of a quantum circuit through a series of transformations:

- Top Diagram:** A sequence of gates J and $H^{(i,j)}$ with inputs β_0 , β' , $-\beta'$, 0 , γ' , $-\gamma'$, ϕ , and $\phi + \gamma_3$.
- Second Diagram:** Labeled $(0\text{Phase}) =$, showing a simplification where some gates are merged.
- Third Diagram:** Labeled $(H\text{Inv}) =$, showing further simplification using the inverse of H .
- Fourth Diagram:** Labeled $\text{Commutativity, (Sum)} =$, showing the combination of terms in the input.
- Bottom Diagram:** The final simplified circuit with a single $R_X^{(i,j)}$ gate acting on β_0 and $\beta' + \gamma'$.

The figure shows a sequence of five quantum circuit diagrams, each representing a step in the simplification of a quantum circuit. The qubits are represented by circles with labels, and the gates are represented by rectangles.

- Diagram 1:** The initial circuit. It starts with a qubit j (labeled β_0) and a qubit i (labeled β'). A gate $H^{(i,j)}$ is applied. This is followed by a qubit j (labeled $-\beta'$) and a qubit i (labeled π). Another gate $H^{(i,j)}$ is applied. This is followed by a qubit j (labeled γ') and a qubit i (labeled $-\gamma'$). A third gate $H^{(i,j)}$ is applied. This is followed by a qubit i (labeled ϕ) and a qubit j (labeled $\phi + \gamma_3$).
- Diagram 2:** The circuit after a commutativity step. The gates $H^{(i,j)}$ and $X^{(i,j)}$ are swapped. The qubits β' and $-\beta'$ are now adjacent to the first $H^{(i,j)}$ gate. The qubits γ' and $-\gamma'$ are now adjacent to the $X^{(i,j)}$ gate. The qubits ϕ and $\phi + \gamma_3$ are now adjacent to the third $H^{(i,j)}$ gate.
- Diagram 3:** Labeled $(XCTRL)$. This diagram shows a control qubit j (labeled β_0) and a qubit i (labeled β'). A gate $H^{(i,j)}$ is applied. This is followed by a qubit j (labeled $-\beta'$) and a qubit i (labeled γ'). Another gate $H^{(i,j)}$ is applied. This is followed by a qubit j (labeled $-\gamma'$) and a qubit i (labeled $X^{(i,j)}$). A third gate $H^{(i,j)}$ is applied. This is followed by a qubit i (labeled ϕ) and a qubit j (labeled $\phi + \gamma_3$).
- Diagram 4:** Labeled $Commutativity, (Sum)$. This diagram shows a qubit j (labeled β_0) and a qubit i (labeled $\beta' - \gamma'$). A gate $H^{(i,j)}$ is applied. This is followed by a qubit j (labeled $-\beta' + \gamma'$) and a qubit i (labeled $X^{(i,j)}$). A third gate $H^{(i,j)}$ is applied. This is followed by a qubit i (labeled ϕ) and a qubit j (labeled $\phi + \gamma_3$).
- Diagram 5:** Labeled $(HInv)$. This diagram shows a qubit j (labeled β_0) and a qubit i (labeled $\beta' - \gamma'$). A gate $H^{(i,j)}$ is applied. This is followed by a qubit j (labeled $-\beta' + \gamma'$) and a qubit i (labeled $H^{(i,j)}$). A third gate $H^{(i,j)}$ is applied. This is followed by a qubit i (labeled π) and a qubit j (labeled ϕ). A final gate $H^{(i,j)}$ is applied. This is followed by a qubit i (labeled $\phi + \gamma_3$).

PROOF. Under $\llbracket - \rrbracket_{\text{sp}}$, the phase shifter $P(\phi)$ acts as $[e^{i\phi}]$ on the corresponding mode, so $\llbracket \text{id}_1 \otimes P(\phi) \rrbracket_{\text{sp}} = \text{diag}(1, e^{i\phi})$ on 2 modes. The beam splitter has matrix

$$\llbracket B(\theta) \rrbracket_{\text{sp}} = \begin{pmatrix} \cos(\theta) & i \sin(\theta) \\ i \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

For $\theta = \pi/4$ and $\phi = -\pi/2$ we have $\cos(\pi/4) = \sin(\pi/4) = 1/\sqrt{2}$ and $e^{i\phi} = e^{-i\pi/2} = -i$, so

$$\llbracket H_{\text{BS}} \rrbracket_{\text{sp}} = \text{diag}(1, -i) \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \cdot \text{diag}(1, -i) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H.$$

Since parallel composition in **LOPP** is interpreted as block-diagonal direct sum on disjoint mode blocks, we have

$$\llbracket \text{id}_i \otimes H_{\text{BS}} \otimes \text{id}_{d-i-2} \rrbracket_{\text{sp}} = I_i \oplus H \oplus I_{d-i-2} = H_d^{(i,i+1)}.$$

□

G.2 Alternating mirror symmetry on Gray-coded modes

We now define the GrayLift gadgets used in the encoding, namely the operators $\text{GrayLift}_d^p(-)$ and $j\text{GrayLift}_d^p(-)$ introduced informally in Section 4.3. Their role is easiest to understand by thinking in terms of Gray-coded blocks.

Fix $d \geq 2$ and $n \geq 1$. The reflected d -ary Gray code G_n^d (Definition 4.4) orders the d^n basis words in $[d]^n$ so that consecutive words differ in one digit by ± 1 . This ordering has a crucial structural feature: if we fix a prefix $u \in [d]^{n-1}$ and vary only the last digit, the d words $u0, u1, \dots, u(d-1)$ appear consecutively in Gray order, but they appear either in increasing order or in decreasing order depending on how the reflected recursion traverses that prefix. For example:

- for $d = 2, n = 2$, the reflected Gray order is 00, 01, 11, 10: the block with prefix 0 lists last digits as 0, 1, while the block with prefix 1 lists last digits as 1, 0;
- for $d = 3, n = 2$, the reflected Gray order is 00, 01, 02, 12, 11, 10, 20, 21, 22: prefixes 0 and 2 list last digits as 0, 1, 2, while prefix 1 lists them as 2, 1, 0.

In the encoding, we often want to take a local **LOPP** circuit C acting on a block of d^m consecutive modes (representing the Gray traversal of some suffix register) and apply that same logical action uniformly for every fixed configuration of some surrounding digits. Because the Gray traversal flips direction on certain branches, “the same logical action” sometimes corresponds to applying C and sometimes to applying a mirrored copy of C , where the order of the modes inside that block is reversed. The GrayLift operators implement exactly this systematic replication-with-optional-mirroring.

Definition G.2. For each $r \geq 0$, let $\rho_r : 1^{\otimes d^r} \rightarrow 1^{\otimes d^r}$ be the **LOPP** permutation circuit that reverses the order of the d^r modes: on the single-photon basis $\{|0\rangle, \dots, |d^r - 1\rangle\}$ it acts by $\rho_r |k\rangle = |d^r - 1 - k\rangle$.

For any raw **LOPP** circuit $C : 1^{\otimes d^r} \rightarrow 1^{\otimes d^r}$, define its mirror by $\text{Rev}(C) := \rho_r \circ C \circ \rho_r^{-1}$.

Since ρ_r is an involutive permutation, $\rho_r^{-1} = \rho_r$. Equivalently, at the matrix level, $\text{Rev}(C)$ conjugates $\llbracket C \rrbracket_{\text{sp}}$ by the reversal permutation matrix.

For $q \in [d]$, set

$$\text{Rev}^q(C) := \begin{cases} C & \text{if } q \text{ is even,} \\ \text{Rev}(C) & \text{if } q \text{ is odd.} \end{cases}$$

This parity test matches the reflected recursion in the Gray code: odd leading digits reverse the traversal of the suffix.

Definition G.3. Fix $d \geq 2$, let $m \geq 0$, and let $C : 1^{\otimes d^m} \rightarrow 1^{\otimes d^m}$ be a raw **LOPP** circuit. For each $p \geq 0$ define a raw **LOPP** circuit

$$\text{GrayLift}_d^p(C) : 1^{\otimes d^{m+p}} \rightarrow 1^{\otimes d^{m+p}}$$

recursively by

$$\text{GrayLift}_d^0(C) := C, \quad \text{GrayLift}_d^{p+1}(C) := \bigotimes_{q=0}^{d-1} \text{Rev}^q(\text{GrayLift}_d^p(C)).$$

Fix also $j \in [d]$. Define the controlled GrayLift $j\text{GrayLift}_d^p(C) : 1^{\otimes d^{m+1+p}} \rightarrow 1^{\otimes d^{m+1+p}}$ recursively by

$$j\text{GrayLift}_d^0(C) := \bigotimes_{q=0}^{d-1} \text{Rev}^q(C^{(q)}), \quad C^{(q)} := \begin{cases} C & \text{if } q = j, \\ \text{id}_{d^m} & \text{if } q \neq j, \end{cases}$$

and

$$j\text{GrayLift}_d^{p+1}(C) := \bigotimes_{q=0}^{d-1} \text{Rev}^q(j\text{GrayLift}_d^p(C)).$$

The recursion is best read as “one more surrounding digit”. At level $p + 1$, we split the d^{m+p+1} mode into d consecutive blocks of size d^{m+p} , indexed by the new leading digit $q \in [d]$. On each block, we place a copy of the previous lift $\text{GrayLift}_d^p(C)$, but mirrored when q is odd.

Two small expansions illustrate the pattern. For $p = 1$ we have

$$\text{GrayLift}_d^1(C) = \text{Rev}^0(C) \otimes \text{Rev}^1(C) \otimes \cdots \otimes \text{Rev}^{d-1}(C),$$

so for $d = 2$ this is $C \otimes \text{Rev}(C)$, and for $d = 3$ this is $C \otimes \text{Rev}(C) \otimes C$.

For $p = 0$, the controlled lift is already informative:

$$j\text{GrayLift}_d^0(C) = \text{Rev}^0(C^{(0)}) \otimes \text{Rev}^1(C^{(1)}) \otimes \cdots \otimes \text{Rev}^{d-1}(C^{(d-1)}),$$

so it places a single (possibly mirrored) copy of C on the j -th block and identities elsewhere. For example, for $d = 3$ and $j = 1$ we get $j\text{GrayLift}_3^0(C) = \text{id} \otimes \text{Rev}(C) \otimes \text{id}$, reflecting the fact that the $q = 1$ Gray branch is traversed in reverse order.

The next lemma isolates the combinatorial feature of Gray order that makes GrayLift correct: fixing a prefix produces a contiguous block of indices, and within that block the last digit runs either forward or backward.

Lemma G.4. *Fix $d \geq 2$ and $n \geq 0$. For each word $u \in [d]^n$, they're exist an index $s(u)$ with $0 \leq s(u) \leq d^{n+1} - d$ and a sign $\varepsilon(u) \in \{+1, -1\}$ such that*

$$\{t \in \{0, \dots, d^{n+1} - 1\} \mid G_{n+1}^d(t) \text{ has prefix } u\} = \{s(u), s(u) + 1, \dots, s(u) + d - 1\},$$

and for every $r \in \{0, \dots, d - 1\}$ we have

$$G_{n+1}^d(s(u) + r) = \begin{cases} ur & \text{if } \varepsilon(u) = +1, \\ u(d - 1 - r) & \text{if } \varepsilon(u) = -1, \end{cases}$$

where ur denotes concatenation of the word u with the digit r .

Moreover, one may take ε to be the unique function $\varepsilon : [d]^* \rightarrow \{\pm 1\}$ defined by $\varepsilon(\epsilon) = +1$ and

$$\varepsilon(qu) = \begin{cases} \varepsilon(u) & \text{if } q \text{ is even,} \\ -\varepsilon(u) & \text{if } q \text{ is odd.} \end{cases}$$

In particular, $\varepsilon(u)$ flips sign exactly once for each odd digit encountered when reading u from left to right.

PROOF. We prove the block statement and the recursion for ε simultaneously by induction on n .

Base case $n = 0$: we have $u = \epsilon$ and $G_1^d(t) = (t)$ for $0 \leq t < d$, so we may take $s(\epsilon) = 0$ and $\varepsilon(\epsilon) = +1$.

Inductive step: let $n \geq 1$ and write $u = qv$ with $q \in [d]$ and $v \in [d]^{n-1}$. Any $t \in \{0, \dots, d^{n+1} - 1\}$ can be written uniquely as $t = q \cdot d^n + r$ with $0 \leq r < d^n$.

If q is even, then by Theorem 4.4 we have $G_{n+1}^d(qd^n + r) = q \cdot G_n^d(r)$. Thus $G_{n+1}^d(qd^n + r)$ has prefix qv iff $G_n^d(r)$ has prefix v . By the induction hypothesis, the set of such r is an interval of length d , hence so is the set of such t , with the same last-digit direction. This corresponds to setting $\varepsilon(qv) = \varepsilon(v)$.

If q is odd, then $G_{n+1}^d(qd^n + r) = q \cdot G_n^d(d^n - 1 - r)$. Put $r' := d^n - 1 - r$. By the induction hypothesis, the set of r' for which $G_n^d(r')$ has prefix v is an interval $[a, a + d - 1]$. Its image under $r = d^n - 1 - r'$ is the interval $[d^n - d - a, d^n - 1 - a]$, again of length d . Since the map $r' \mapsto d^n - 1 - r'$ reverse order, the last digit is traversed in the opposite direction on this block; this corresponds to setting $\varepsilon(qv) = -\varepsilon(v)$. $\square$

The takeaway of Theorem G.4 is that each fixed prefix u determines a single contiguous mode block of length d in which only the last digit varies, and it varies either as $0, 1, \dots, d - 1$ or as $d - 1, d - 2, \dots, 0$. This is exactly the reason the GrayLift recursion toggles mirroring when a leading digit is odd: it ensures that, once we identify modes with basis words in reflected Gray order, every lifted copy of a local subcircuit acts on its intended suffix register with the same logical orientation.

H Encoding and Decoding

In this appendix we prove Theorem 4.17 (encoding/decoding retraction). Recall that E_ℓ^k is the contextual encoding (Definition 4.8), and D_n^t is the contextual decoding (Definition 4.11). The theorem states that for every $C : n \rightarrow n$ in $\mathbf{CQC}_d$ and every context size $k, \ell \geq 0$, decoding the encoded circuit recovers C padded by identities:

$$\mathbf{QC}_d \vdash D_{k+n+\ell}^0(E_\ell^k(C)) = \text{id}_k \otimes C \otimes \text{id}_\ell.$$

It is helpful to keep in mind what this statement is saying operationally. The encoding E_ℓ^k turns an n -qudit circuit into a $d^{k+n+\ell}$ -mode single-photon optical circuit by using the reflected d -ary Gray ordering of computational basis states. The decoding D_n^t goes the other way: it reads a local optical generator acting on one mode (a phase shifter) or two consecutive modes (a swap or a beam splitter) as a qudit operation supported on one two-dimensional subspace of $\mathcal{H}_n$, controlled on the surrounding Gray digits. Because $\otimes$ means tensor product on qudits but direct sum on optics, both translations are forced to be context-sensitive and to serialise tensor products. The retraction theorem is the formal statement that these design choices are coherent: once the surrounding context is fixed, encoding and decoding cancel each other exactly (up to the expected identity padding on context wires).

The proof proceeds in three steps:

- (1) we analyse how D_n^t behaves on a contiguous Gray-code block corresponding to a fixed value of the most significant qudit;
- (2) we apply this to the alternating mirror-symmetry networks GrayLift_d and $j\text{GrayLift}_d$, which are the basic optical “lifting gadgets” used in the encoding;
- (3) we conclude by a structural induction on circuits in $\mathbf{CQC}_d$.

H.1 Decoding a fixed Gray-code block

The key combinatorial fact is that, in reflected d -ary Gray order, mode indices are grouped into blocks according to the first Gray digit. Fix $a \geq 0$. The d^{a+1} mode labels split into d consecutive blocks of length d^a :

$$\{0, \dots, d^a - 1\}, \{d^a, \dots, 2d^a - 1\}, \dots, \{(d-1)d^a, \dots, d^{a+1} - 1\},$$

and these blocks correspond to leading Gray digit $j = 0, 1, \dots, d-1$. Inside the block for a fixed j , every Gray word has prefix j , so one expects every decoded generator to acquire an outer C_j . The only subtlety is that the reflected Gray code traverses odd blocks in reverse order. For example, when $d = 3$ and $a = 1$ (so we are looking at words of length 2), the reflected Gray order is

$$00, 01, 02, \quad 12, 11, 10, \quad 20, 21, 22,$$

so the $j = 0$ block is forward, the $j = 1$ block is reversed, and the $j = 2$ block is forward again. The next two lemmas formalise exactly how decoding behaves in the forward and reversed cases.

Lemma H.1. *Let C be a **LOPP** circuit acting on ℓ consecutive modes, and fix $a \geq 0$ and s with $s + \ell \leq d^a$. For every even $j \in \{0, \dots, d-1\}$ we have*

$$D_{a+1}^{d^a j + s}(C) = C_j(D_a^s(C)).$$

PROOF. We argue by structural induction on C . The intuition is that, for even j , the Gray code factors on that block as

$$G_{a+1}^d(d^a j + t) = j \cdot G_a^d(t),$$

so every decoded generator is the usual decoding at the smaller size a , wrapped in a j -control.

Empty and identity. If C is the empty diagram $\begin{array}{c} \text{---} \\ \text{---} \end{array}$ or a single wire --- , then by definition of D we have

$$D_{a+1}^{d^a j + s}(C) = \text{id}_{a+1} = C_j(\text{id}_a) = C_j(D_a^s(C)).$$

Phase shifter. Suppose C is a phase shifter $\begin{array}{c} \boxed{\theta} \\ \text{---} \end{array}$ on the mode with global index $d^a j + s$. Since j is even, the Gray code traverses the block with leading digit j in increasing order, so

$$G_{a+1}^d(d^a j + s) = j \cdot G_a^d(s).$$

Decoding at offset $d^a j + s$ therefore yields

$$D_{a+1}^{d^a j + s} \left(\begin{array}{c} \boxed{\theta} \\ \text{---} \end{array} \right) = C_j(C_{G_a^d(s)}(\boxed{\theta})) = C_j(D_a^s(\begin{array}{c} \boxed{\theta} \\ \text{---} \end{array})).$$

As a concrete instance, for $d = 3$, $a = 1$, $j = 0$ and $s = 1$ we have $G_2^3(1) = 01$ and the decoded phase shifter is $C_0(C_1(\boxed{\theta}))$, i.e. the phase is applied only on the computational basis vector $|01\rangle$.

Beamsplitter. Let $C = \begin{array}{c} \theta \\ \text{---} \end{array}$ act on two consecutive modes $d^a j + s$ and $d^a j + s + 1$. Since $s + 1 < d^a$ and j is even, we have

$$G_{a+1}^d(d^a j + s) = j \cdot G_a^d(s), \quad G_{a+1}^d(d^a j + s + 1) = j \cdot G_a^d(s + 1).$$

Write $G_a^d(s) = u'vw$ and $G_a^d(s + 1) = u'xw$, so that

$$G_{a+1}^d(d^a j + s) = ju'vw, \quad G_{a+1}^d(d^a j + s + 1) = ju'xw.$$

In words, the two consecutive modes differ only in the *suffix* digit (the unique position where Gray neighbours differ); the leading digit stays equal to j . Decoding $\begin{array}{c} \theta \\ \text{---} \end{array}$ at offset $d^a j + s$ therefore produces exactly the same two-level gate as in the a -qudit decoding, but additionally controlled on the leading digit j :

$$D_{a+1}^{d^a j + s} \left(\begin{array}{c} \theta \\ \text{---} \end{array} \right) = C_j(D_a^s(\begin{array}{c} \theta \\ \text{---} \end{array})).$$

Swap. The argument for the **LOPP** swap $\begin{array}{c} \text{---} \\ \text{---} \end{array}$ is identical to the beamsplitter case, using the decoding clause of Definition 4.11 for swaps instead of beamsplitters.

Sequential composition. If $C = C_1 \circ C_2$ (so the two subcircuits act on the same mode interval), then

$$D_{a+1}^{d^a j + s}(C) = D_{a+1}^{d^a j + s}(C_1) \circ D_{a+1}^{d^a j + s}(C_2)$$

by definition. By the induction hypothesis,

$$D_{a+1}^{d^a j + s}(C_t) = C_j(D_a^s(C_t)) \quad (t = 1, 2),$$

so

$$D_{a+1}^{d^a j+s}(C) = C_j(D_a^s(C_1)) \circ C_j(D_a^s(C_2)) = C_j(D_a^s(C)),$$

using functoriality of C_j .

Parallel composition. If $C = C_1 \otimes C_2$ with C_1 on ℓ_1 modes and C_2 on ℓ_2 modes, then D serialises the tensor by decoding the left block first and the right block afterwards. Concretely,

$$D_{a+1}^{d^a j+s}(C) = D_{a+1}^{d^a j+s+\ell_1}(C_2) \circ D_{a+1}^{d^a j+s}(C_1).$$

By the induction hypothesis applied to C_1 at offset s and to C_2 at offset $s + \ell_1$,

$$D_{a+1}^{d^a j+s}(C) = C_j(D_a^{s+\ell_1}(C_2)) \circ C_j(D_a^s(C_1)) = C_j(D_a^s(C)),$$

again by functoriality of C_j . $\square$

Lemma H.2. *Let C be a circuit of **LOPP** acting on ℓ consecutive modes, and fix $a \geq 0$ and s with $s + \ell \leq d^a$. For every odd $j \in \{0, \dots, d-1\}$ we have*

$$D_{a+1}^{d^a j+s}(C) = C_j(D_a^{d^a-s-\ell}(\text{Rev}(C))),$$

where $\text{Rev}(C)$ denotes the circuit obtained from C by reversing the order of the d^a modes in the relevant block.

PROOF. This is the same induction as in Lemma H.1, but with one extra bookkeeping step reflecting the fact that odd Gray blocks are traversed in reverse order. When j is odd, the block with leading digit j is read backwards by the Gray code:

$$G_{a+1}^d(d^a j + t) = j \cdot G_a^d(d^a - 1 - t).$$

For example, with $d = 3$ and $a = 1$, the $j = 1$ block corresponds to the three consecutive mode indices 3, 4, 5, but their Gray labels are 12, 11, 10 rather than 10, 11, 12. Thus, a subcircuit C acting on modes starting at offset s in that block corresponds to the same pattern as a forward-traversed circuit, but located at the mirrored starting position $d^a - s - \ell$.

The role of $\text{Rev}(C)$ is precisely to compensate for this reversal at the syntactic level: it renames mode t within the block to mode $(d^a - 1 - t)$. Once Rev is inserted, the decoding of each generator reduces to the even case with the mirrored offset, and then acquires the outer C_j because the leading Gray digit is j throughout the block.

The inductive steps for sequential and parallel composition are identical to those in Lemma H.1, using the respective induction hypotheses and functoriality of C_j . $\square$

H.2 Decoding GrayLifts

The operators GrayLift_d^m and $j\text{GrayLift}_d^m$ are the optical gadgets used in the encoding to “lift” a local d^n -mode circuit through m additional Gray digits. Intuitively, $\text{GrayLift}_d^m(C)$ is a block circuit consisting of d^m copies of C spread across the Gray blocks determined by the extra digits, mirroring the copy whenever the reflected Gray traversal reverses the local order. Decoding should therefore produce m untouched qudits (identities), together with the same decoded action as C on the remaining n qudits. Similarly, $j\text{GrayLift}_d^m(C)$ performs the same lift but only on the branch where a designated digit equals j ; decoding should therefore add a j -control, again padded by m identities for the extra context digits.

We first record a simple arity property: each GrayLift layer multiplies the number of modes by d , corresponding to adding one more Gray digit.

Lemma H.3. *Let C be a **LOPP** circuit acting on d^n modes. Then for every $k \geq 0$ the circuit $\text{GrayLift}_d^k(C)$ acts on d^{n+k} modes.*

PROOF. We proceed by induction on k . For $k = 0$ we have $\text{GrayLift}_d^0(C) = C$ by definition, so $\text{GrayLift}_d^0(C)$ acts on d^n modes.

Assume the statement holds for some $k \geq 0$, i.e. that $\text{GrayLift}_d^k(C)$ acts on d^{n+k} modes. By definition,

$$\text{GrayLift}_d^{k+1}(C) = \bigotimes_{q=0}^{d-1} \text{Rev}^q(\text{GrayLift}_d^k(C)).$$

The mode-reversal $\text{Rev}^q(\cdot)$ is implemented by a permutation of modes and therefore does not change the number of modes, so each tensor factor acts on d^{n+k} modes. The tensor product of d such factors therefore acts on $d \cdot d^{n+k} = d^{n+k+1}$ modes, completing the induction. $\square$

Lemma H.4. *Let C be a **LOPP** circuit acting on d^n modes, and let $m \geq 0$. Then*

$$D_{n+m}^0(\text{GrayLift}_d^m(C)) = \text{id}_m \otimes D_n^0(C)$$

as circuits in CQC_d .

PROOF. We argue by induction on m . It may help to first consider the case $m = 1$: $\text{GrayLift}_d^1(C)$ is a tensor of d blocks, one per leading digit j , and decoding each block yields a j -controlled copy of $D_n^0(C)$. Composing these d controlled copies and using exhaustivity collapses the whole product to $\text{id}_1 \otimes D_n^0(C)$. The general case iterates this argument one Gray digit at a time.

Base case ($m = 0$). By definition $\text{GrayLift}_d^0(C) = C$, so

$$D_n^0(\text{GrayLift}_d^0(C)) = D_n^0(C) = \text{id}_0 \otimes D_n^0(C),$$

since id_0 is the monoidal unit.

Inductive step. Assume the lemma holds for some $m \geq 0$, i.e.

$$D_{n+m}^0(\text{GrayLift}_d^m(C)) = \text{id}_m \otimes D_n^0(C).$$

Set $F := \text{GrayLift}_d^m(C)$, so by Lemma H.3 F acts on d^{n+m} modes. By definition,

$$\text{GrayLift}_d^{m+1}(C) = \bigotimes_{j=0}^{d-1} \text{Rev}^j(F),$$

a tensor of d blocks, each occupying a contiguous mode block of size d^{n+m} .

Decoding in an $(n + m + 1)$ -qudit context serialises the tensor product:

$$D_{n+m+1}^0(\text{GrayLift}_d^{m+1}(C)) = D_{n+m+1}^{d^{n+m}(d-1)}(\text{Rev}^{d-1}(F)) \circ \dots \circ D_{n+m+1}^{d^{n+m}}(\text{Rev}^1(F)) \circ D_{n+m+1}^0(\text{Rev}^0(F)).$$

For each j , the block $\text{Rev}^j(F)$ lives inside the Gray-code block with leading digit j . Applying Lemma H.1 when j is even and Lemma H.2 when j is odd (always with $s = 0$ and $\ell = d^{n+m}$) yields, in either case,

$$D_{n+m+1}^{d^{n+m}j}(\text{Rev}^j(F)) = C_j(D_{n+m}^0(F)).$$

Substituting back, we obtain

$$D_{n+m+1}^0(\text{GrayLift}_d^{m+1}(C)) = \prod_{j=0}^{d-1} C_j(D_{n+m}^0(F)),$$

where the product is sequential composition in some fixed order. By commutativity of distinct controls (Theorem 3.3) the order does not matter, and by exhaustivity (Theorem 3.4) the product collapses to

$$\prod_{j=0}^{d-1} C_j(D_{n+m}^0(F)) = \text{id}_1 \otimes D_{n+m}^0(F).$$

Using the induction hypothesis $D_{n+m}^0(F) = \text{id}_m \otimes D_n^0(C)$, we conclude

$$D_{n+m+1}^0(\text{GrayLift}_d^{m+1}(C)) = \text{id}_1 \otimes (\text{id}_m \otimes D_n^0(C)) = \text{id}_{m+1} \otimes D_n^0(C).$$

□

Lemma H.5. *Let C be a LOPP circuit acting on d^n modes, and let $m \geq 0$. Then*

$$D_{n+1+m}^0(j\text{GrayLift}_d^m(C)) = \text{id}_m \otimes C_j(D_n^0(C))$$

as circuits in CQC_d .

PROOF. We proceed by induction on m . The base case is the branch-selection behaviour: only the block with leading digit j contains C , and decoding turns that into $C_j(D_n^0(C))$. The inductive step then adds one more GrayLift layer, which by exhaustivity contributes exactly one more untouched qudit (an extra identity wire).

Base case ($m = 0$). By definition, $j\text{GrayLift}_d^0(C)$ is a tensor of d blocks on d^n modes each, where all blocks are identities except the j -th block, which contains C (mirrored if needed to match Gray traversal). Decoding serialises this tensor into a product over $q \in \{0, \dots, d-1\}$. For $q \neq j$, the decoded factor is the identity, while for $q = j$ we get, using Lemma H.1 or Lemma H.2 (depending on the parity of j),

$$D_{n+1}^{d^n j}(\text{Rev}^j(C)) = C_j(D_n^0(C)).$$

Hence

$$D_{n+1}^0(j\text{GrayLift}_d^0(C)) = C_j(D_n^0(C)) = \text{id}_0 \otimes C_j(D_n^0(C)).$$

Inductive step. Assume the lemma holds for some $m \geq 0$, i.e.

$$D_{n+1+m}^0(j\text{GrayLift}_d^m(C)) = \text{id}_m \otimes C_j(D_n^0(C)).$$

Set $F := j\text{GrayLift}_d^m(C)$. By the recursive definition of $j\text{GrayLift}$, we have

$$j\text{GrayLift}_d^{m+1}(C) = \bigotimes_{q=0}^{d-1} \text{Rev}^q(F),$$

so decoding in an $(n + 1 + m + 1)$ -qudit context gives

$$D_{n+1+m+1}^0(j\text{GrayLift}_d^{m+1}(C)) = \prod_{q=0}^{d-1} D_{n+1+m+1}^{d^{n+1+m}q}(\text{Rev}^q(F)).$$

Applying Lemma H.1 or Lemma H.2 to each block (with $s = 0$ and $\ell = d^{n+1+m}$) yields

$$D_{n+1+m+1}^{d^{n+1+m}q}(\text{Rev}^q(F)) = C_q(D_{n+1+m}^0(F)).$$

Therefore

$$D_{n+1+m+1}^0(j\text{GrayLift}_d^{m+1}(C)) = \prod_{q=0}^{d-1} C_q(D_{n+1+m}^0(F)).$$

By exhaustivity (Theorem 3.4),

$$\prod_{q=0}^{d-1} C_q(D_{n+1+m}^0(F)) = \text{id}_1 \otimes D_{n+1+m}^0(F).$$

Using the induction hypothesis for F ,

$$\text{id}_1 \otimes D_{n+1+m}^0(F) = \text{id}_1 \otimes (\text{id}_m \otimes C_j(D_n^0(C))) = \text{id}_{m+1} \otimes C_j(D_n^0(C)),$$

which is the desired statement for $m + 1$. $\square$

H.3 Encoding and decoding are inverse

We can now prove that the contextual encoding and decoding are mutually inverse, up to padding with identities on the context wires. The proof is a direct structural induction, but it is worth keeping the intended picture in mind: the encoding of each primitive qudit constructor is built from (i) a small local optical circuit, (ii) GrayLift gadgets (GrayLift or $j\text{GrayLift}$) to place that circuit uniformly across Gray blocks, and (iii) block permutations $\sigma_{-, -, -}^d$ to position the action on the correct qudit wires. The decoding lemmas established above says that (ii) becomes either identity padding or a genuine control, and that (iii) becomes the corresponding qudit-wire symmetry. After these wrappers are removed, what remains is exactly the original qudit generator.

PROOF. We prove Theorem 4.17 by structural induction on f .

Typing convention. Fix $k, \ell \geq 0$ and a circuit $f : n \rightarrow n$ in $\mathbf{CQC}_d$. Then $E_\ell^k(f)$ is a $d^{k+n+\ell}$ -mode **LOPP** circuit, and the decoding appearing in the theorem is $D_{k+n+\ell}^0$. For readability, throughout this proof we write

$$D(-) := D_{k+n+\ell}^0(-)$$

when the arity is clear from context.

Empty and identity. If $f = \begin{bmatrix} \text{ } \\ \text{ } \end{bmatrix}$, then by definition

$$E_\ell^k(f) = \text{---}^{\otimes d^{k+\ell}},$$

and hence

$$D(E_\ell^k(f)) = D(\text{---}^{\otimes d^{k+\ell}}) = \text{id}_{k+\ell} = \text{id}_k \otimes \text{id}_0 \otimes \text{id}_\ell.$$

The case $f = \text{---}$ is similar.

Swap. If $f = \text{---} \times \text{---}$, the encoding uses the optical block-permutations $\sigma_{k,n,\ell}^d$:

$$E_\ell^k(\text{---} \times \text{---}) = \sigma_{k,\ell,2}^d \circ \sigma_{k+\ell,1,1}^d \circ \sigma_{k,2,\ell}^d.$$

Intuitively, these permutations implement the qudit-wire swap by reindexing Gray blocks of modes. By Lemma J.11 we have

$$D(\sigma_{k,\ell,2}^d) = \text{id}_k \otimes \sigma_{\ell,2}, \quad D(\sigma_{k+\ell,1,1}^d) = \text{id}_{k+\ell} \otimes \sigma_{1,1}, \quad D(\sigma_{k,2,\ell}^d) = \text{id}_k \otimes \sigma_{2,\ell}.$$

Therefore, using strict PROP coherence,

$$D(E_\ell^k(\text{---} \times \text{---})) = \text{id}_k \otimes \sigma_{1,1} \otimes \text{id}_\ell = \text{id}_k \otimes \text{---} \times \text{---} \otimes \text{id}_\ell.$$

Global phase. If $f = \textcircled{\theta}$ then

$$E_\ell^k(\textcircled{\theta}) = (-\text{---} \boxed{\theta} \text{---})^{\otimes d^{k+\ell}},$$

so decoding produces one basis-controlled phase per Gray basis vector of the context register:

$$D(E_\ell^k(\textcircled{\theta})) = \prod_{u \in [d]^{k+\ell}} C_u(\textcircled{\theta}).$$

For instance, when $k + \ell = 1$ this products is $C_0((\theta)) \circ \dots \circ C_{d-1}((\theta))$, i.e. apply the same scalar phase on every basis value of the single context qudit. By exhaustivity of control (Theorem 3.4), the product over all words u yields

$$D(E_\ell^k((\theta))) = \text{id}_{k+\ell} \otimes (\theta) = \text{id}_k \otimes (\theta) \otimes \text{id}_\ell.$$

Two-level Hadamard. Let $f = \boxed{H^{(r,r+1)}}$. By Definition 4.8 we have

$$E_\ell^k(\boxed{H^{(r,r+1)}}) = \sigma_{k,1,\ell}^d \circ \text{GrayLift}_d^{k+\ell}(B_d^{(i,i+1)}) \circ \sigma_{k,\ell,1}^d.$$

Applying D and functoriality for sequential composition gives

$$D(E_\ell^k(\boxed{H^{(r,r+1)}})) = D(\sigma_{k,1,\ell}^d) \circ D(\text{GrayLift}_d^{k+\ell}(B_d^{(i,i+1)})) \circ D(\sigma_{k,\ell,1}^d).$$

By Lemma J.11,

$$D(\sigma_{k,1,\ell}^d) = \text{id}_k \otimes \sigma_{1,\ell}, \quad D(\sigma_{k,\ell,1}^d) = \text{id}_k \otimes \sigma_{\ell,1}.$$

By Lemma H.4, instantiated with $C = B_d^{(i,i+1)}$ and $m = k + \ell$,

$$D(\text{GrayLift}_d^{k+\ell}(B_d^{(i,i+1)})) = \text{id}_{k+\ell} \otimes D_1^0(B_d^{(i,i+1)}).$$

Finally, by the choice of $B_d^{(i,i+1)}$ (Lemma G.1) and the decoding correctness for this fixed single-qudit network,

$$D_1^0(B_d^{(i,i+1)}) = \boxed{H^{(r,r+1)}}.$$

Combining these equalities and simplifying using PROP coherence yields

$$D(E_\ell^k(\boxed{H^{(r,r+1)}})) = \text{id}_k \otimes \boxed{H^{(r,r+1)}} \otimes \text{id}_\ell.$$

Sequential composition. If $f = g \circ h$, then by Definition 4.8,

$$E_\ell^k(g \circ h) = E_\ell^k(g) \circ E_\ell^k(h),$$

so

$$D(E_\ell^k(g \circ h)) = D(E_\ell^k(g)) \circ D(E_\ell^k(h)).$$

The induction hypothesis gives $D(E_\ell^k(g)) = \text{id}_k \otimes g \otimes \text{id}_\ell$ and $D(E_\ell^k(h)) = \text{id}_k \otimes h \otimes \text{id}_\ell$, hence

$$D(E_\ell^k(g \circ h)) = \text{id}_k \otimes (g \circ h) \otimes \text{id}_\ell.$$

Parallel composition. If $f = g \otimes h$ with $g : n_g \rightarrow n_g$ and $h : n_h \rightarrow n_h$, then Definition 4.8 serialises the tensor product:

$$E_\ell^k(g \otimes h) = E_\ell^{k+n_g}(h) \circ E_{\ell+n_h}^k(g).$$

Decoding preserves $\circ$, so

$$D(E_\ell^k(g \otimes h)) = D(E_\ell^{k+n_g}(h)) \circ D(E_{\ell+n_h}^k(g)).$$

Applying the induction hypotheses to h at context $(k + n_g, \ell)$ and to g at context $(k, \ell + n_h)$ yields

$$D(E_\ell^k(g \otimes h)) = \text{id}_k \otimes (g \otimes h) \otimes \text{id}_\ell.$$

Control. Finally, let $f = C_j(g)$ with $g : m \rightarrow m$. By Definition 4.8,

$$E_\ell^k(C_j(g)) = \sigma_{k,m+1,\ell}^d \circ j\text{GrayLift}_d^{k+\ell}(E_0^0(g)) \circ \sigma_{k,\ell,m+1}^d.$$

Decoding and using Lemma J.11 reduces the outer permutations to the corresponding qudit-wire symmetries. For the middle term we use Lemma H.5 to obtain

$$D(j\text{GrayLift}_d^{k+\ell}(E_0^0(g))) = \text{id}_{k+\ell} \otimes C_j(D_m^0(E_0^0(g))).$$

By the induction hypothesis applied to g with $k = \ell = 0$, $D_m^0(E_0^0(g)) = g$, hence the middle factor is $\text{id}_{k+\ell} \otimes C_j(g)$, and the surrounding symmetries cancel by PROP coherence. Therefore

$$D(E_\ell^k(C_j(g))) = \text{id}_k \otimes C_j(g) \otimes \text{id}_\ell.$$

This exhausts the constructors of $\mathbf{CQC}_d$, hence $\mathbf{QC}_d \vdash D_{k+n+\ell}^0(E_\ell^k(f)) = \text{id}_k \otimes f \otimes \text{id}_\ell$ for all f , as required. $\square$

I Mimicking rules

This appendix proves Theorem 4.16 from Section 4: every equational step in the LOPP calculus can be *mimicked* inside our qudit theory via decoding. Concretely, for any $n \geq 0$ and any d^n -mode LOPP circuits C_1, C_2 , if $\text{LOPP} \vdash C_1 = C_2$ then their decodings satisfy $\text{QC}_d \vdash D(C_1) = D(C_2)$.

The proof has two independent checks:

- (1) decoding is compatible with the strict-PROP congruence on raw LOPP terms (i.e. the strict symmetric monoidal/PROP equations);
- (2) decoding maps each non-structural axiom of the LOPP calculus (Figure 9) to a derivable equation in QC_d .

Since both equational theories are congruences closed under arbitrary contexts, these two checks suffice to mimic arbitrary LOPP derivations.

We work with *raw* circuits, i.e. string diagrams freely generated from the LOPP (or qudit) generators using $\circ$ and $\otimes$, before quotienting by any equations. We use $\equiv$ for the strict-PROP congruence on raw LOPP circuits as in Definition 4.1. For convenience we reproduce the generating equations in Figure 8.

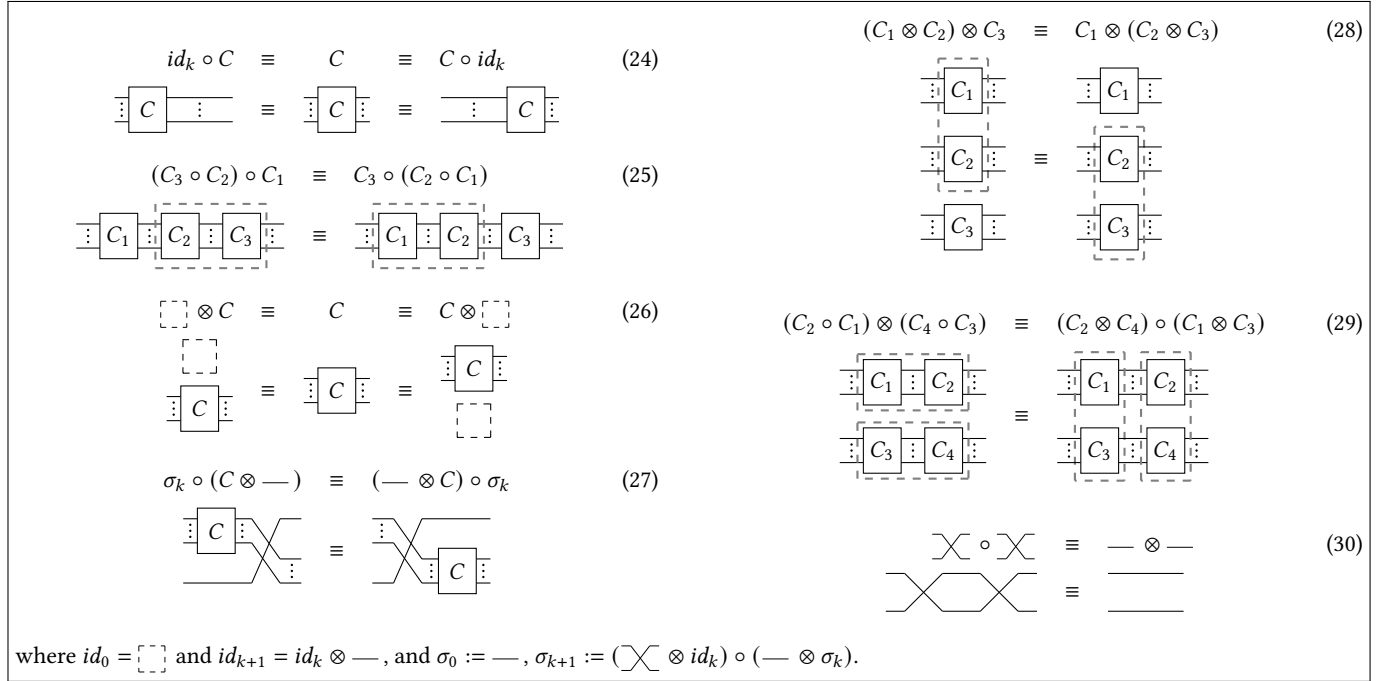


Figure 8: Structural axioms generating the congruence $\equiv$ on raw circuits (either raw quantum circuits or raw optical circuits).

Commutation of decoded circuits on disjoint blocks

We first record a basic commuting property of decoding when two optical subcircuits act on disjoint mode intervals.

Lemma I.1. *Let $C_1 : \ell_1 \rightarrow \ell_1$ and $C_2 : \ell_2 \rightarrow \ell_2$ be raw optical circuits. Let $n \geq 0$ and let $t_1, t_2 \in \mathbb{N}$ satisfy $0 \leq t_1$, $0 \leq t_2$, $t_1 + \ell_1 \leq t_2$, and $t_2 + \ell_2 \leq d^n$. Then*

$$\text{QC}_d \vdash D_n^{t_2}(C_2) \circ D_n^{t_1}(C_1) = D_n^{t_1}(C_1) \circ D_n^{t_2}(C_2).$$

PROOF. We argue by structural induction on the pair (C_1, C_2) .

If either circuit is a sequential composite, the statement follows immediately from the induction hypothesis and the defining clause $D_n^t(C_2 \circ C_1) = D_n^t(C_2) \circ D_n^t(C_1)$.

If either circuit is a tensor composite, we use the serialisation clause $D_n^t(C_1 \otimes C_2) = D_n^{t+\ell_1}(C_2) \circ D_n^t(C_1)$ (Definition 4.11) together with associativity of $\circ$ to reduce to commuting decoded subcircuits whose acted-on mode intervals remain disjoint; the required commutations are then provided by the induction hypothesis.

Thus it suffices to treat the case where both C_1 and C_2 are LOPP generators (phase shifter, beam splitter, or swap). By Lemma 4.12, each decoded generator acts non-trivially on at most one basis vector (phase shifter) or one two-dimensional Gray-neighbour subspace (beam splitter/swap). Since the mode intervals are disjoint, these supports are disjoint in the Gray-ordered computational basis, and the corresponding decoded operations commute in QC_d using the derived distinct-control commutativity principle together with the local commutation schemata of Figure 2. $\square$

Contexts and compatibility with structural congruence

To phrase compatibility with $\equiv$ it is convenient to work with contexts.

Definition I.2. A (one-hole) raw context $\mathbb{C}[\cdot]_i$ is a term over the LOPP signature built from $\circ$ and $\otimes$ with a distinguished placeholder of arity i . If the outer arity of $\mathbb{C}[\cdot]_i$ is N , we say that $\mathbb{C}[\cdot]_i$ is an N -mode context with hole arity i .

Definition I.3. Given a N -mode context $\mathbb{C}[\cdot]_i$ and an i -mode raw circuit C , we write $\mathbb{C}[C]$ for the N -mode circuit obtained by filling the hole in $\mathbb{C}[\cdot]_i$ with C .

Lemma I.4. Let C_1, C_2 be d^n -mode raw optical circuits. If $C_1 \equiv C_2$, then

$$\text{QC}_d \vdash D(C_1) = D(C_2).$$

PROOF. We prove the following offset-and-context strengthening: for any context $\mathbb{C}[\cdot]_i$, any i -mode circuits C_1, C_2 , and any t, n with $t + N \leq d^n$ (where N is the outer arity of $\mathbb{C}[\cdot]_i$),

$$C_1 \equiv C_2 \implies \text{QC}_d \vdash D_n^t(\mathbb{C}[C_1]) = D_n^t(\mathbb{C}[C_2]).$$

The desired statement is the special case $\mathbb{C}[\cdot]_i = [\cdot]_i$ and $t = 0$.

By induction on the context structure it suffices to check preservation of the generating axioms of Figure 8. Identities and tensor units are sent to id_n by Definition 4.11. Associativity of $\circ$ is preserved because D_n^t is defined compositionally on $\circ$. Associativity of $\otimes$ is preserved because D_n^t serialises $\otimes$ into a fixed sequential composite, and the two bracketings decode to the same composite up to associativity of $\circ$. For the mixed-product/interchange axiom (29), expanding both sides via the serialisation clause yields composites of decoded subcircuits acting on disjoint mode blocks; Lemma I.1 provides the required commutations. Naturality and involutivity of swap follow from the fact that decoding of permutation circuits agrees with the corresponding qudit-wire symmetries (see Appendix J). $\square$

Mimicking the LOPP equational theory

Recall the non-structural axioms of LOPP (Figure 9).

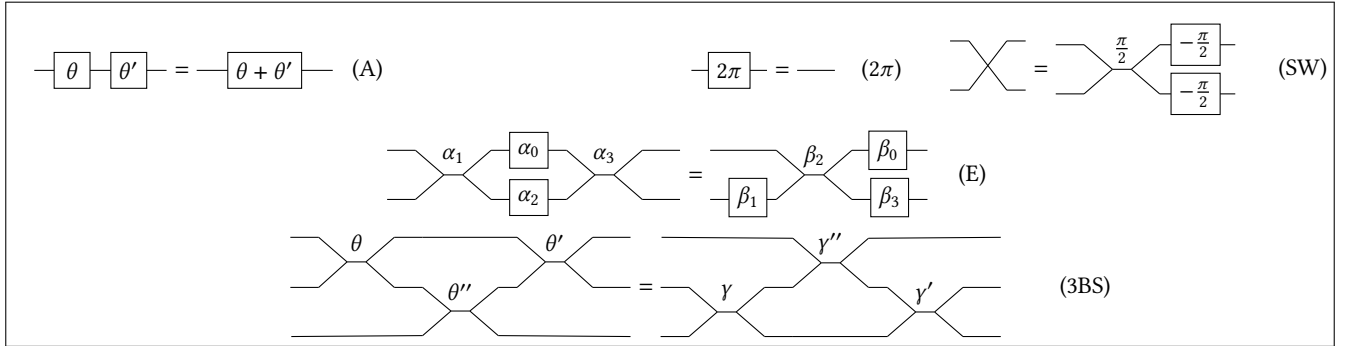


Figure 9: Non-structural axioms of the LOPP calculus [18]: phase addition, 2π -periodicity, swap decomposition, and Euler/three-beamsplitter identities.

Theorem I.5. For any d^n -mode raw optical circuits C_1, C_2 , if $\text{LOPP} \vdash C_1 = C_2$, then

$$\text{QC}_d \vdash D(C_1) = D(C_2).$$

PROOF. The LOPP derivability relation is the congruence generated by the strict-PROP axioms (Figure 8) and the non-structural axioms (A)–(3BS). By Lemma I.4, decoding respects the strict-PROP congruence, so it suffices to verify that each non-structural axiom is preserved by decoding at an arbitrary offset t (closure under contexts is handled exactly as in the proof of Lemma I.4).

Let t, n be such that the relevant local LOPP circuit sits inside d^n modes.

- **Phase addition (A).** Decoding a phase shifter on mode t yields $C_{G_n^d(t)}(\theta)$ (Definition 4.11). The decoded equation is therefore an instance of the QC_d phase arithmetic axiom (Sum), closed under iterated control.
- **2π -periodicity (2π).** Similarly, decoding $\boxed{2\pi}$ yields a multi-controlled 2π -phase, which is equal to id_n by the QC_d axiom (2π) (again closed under iterated control). The right-hand side decodes to id_n .
- **Swap decomposition (SW).** This is a two-mode local identity. Decoding at offset t yields a (multi-)controlled two-level swap on the unique Gray-neighbour subspace determined by modes $t, t + 1$ (Definition 4.11). The right-hand side decodes to the same controlled two-level swap by a direct diagrammatic calculation in QC_d , using the axioms (TP) and (TH) together with the core two-level gate identities of Figure 1.

- *Euler identity* (E). Decoding sends each beam splitter on the consecutive modes $t, t + 1$ to a (multi-)controlled two-level rotation $\boxed{R_X^{(v, v+\epsilon)}(\theta)}$ on the corresponding Gray-neighbour computational subspace. The resulting decoded equation is an instance of the QC_d Euler schema (EH) (closed under control and conjugation by the surrounding wire permutations used in $\Lambda_w^u(-)$).
 - *Three-beamsplitter identity* (3BS). Decoding of the three-mode local pattern depends only on which Gray digit changes between $t, t + 1, t + 2$ (Lemma 4.5). If all three consecutive modes lie in a single adjacent-level chain for one digit, the decoded equation is an instance of (3Rx). Otherwise, one of the rotation is additionally controlled by a fixed value of another digit, and the decoded equation is an instance of (3CRx).
- Thus decoding preserves each generator of the LOPP equational theory and is stable under contexts; hence $LOPP \vdash C_1 = C_2$ implies $QC_d \vdash D(C_1) = D(C_2)$. $\square$

J Explicit encoding and decoding of swaps

In this section we spell out the encoding and decoding of wire-swaps in the LOPP calculus, and show that the optical block-swaps used in the encoding indeed decode to the intended qudit-level swaps in CQC_d .

J.1 Gray-code indices for words

We first fix a convenient notation for addressing modes in Gray-code order. Recall that a mode in the d -ary Gray code for n qudits is labelled by a word $w \in [d]^n$. We now define the inverse map, which returns the position of a given word in the Gray ordering.

Definition J.1. For a base- d word $w \in [d]^n$ we write $G_d^{-1}(w)$ for its index in the d -ary Gray ordering. It is defined recursively as follows.

- For the empty word ϵ we set $G_d^{-1}(\epsilon) = 0$.
- For a non-empty word $w = h \cdot t$ of length $n + 1$, with $h \in \{0, \dots, d - 1\}$ the first digit and $t \in [d]^n$ the suffix, we put

$$G_d^{-1}(h \cdot t) = \begin{cases} h d^n + G_d^{-1}(t) & \text{if } h \text{ is even,} \\ (h + 1) d^n - 1 - G_d^{-1}(t) & \text{if } h \text{ is odd.} \end{cases}$$

Intuitively, when the leading digit h is even, the block $\{h\} \times [d]^n$ appears in the Gray ordering in increasing order of t ; when h is odd, the block is traversed in reverse order. This mirrors the usual “reflect-and-prefix” construction of Gray codes.

J.2 Elementary LOPP swap gadgets

We now define an optical gadget that swaps two basis states.

Definition J.2. Let $w_1, w_2 \in [d]^n$ be two distinct words of the same length. We write

$$C_{d, w_1, w_2} = \begin{array}{c} \text{---} \\ \vdots \\ \text{---} \\ \text{---} \\ \vdots \\ \text{---} \\ \vdots \\ \text{---} \\ \vdots \\ \text{---} \end{array} \begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \begin{array}{c} G_d^{-1}(w_1) \\ G_d^{-1}(w_2) \end{array}$$

for the LOPP circuit that swaps the two modes labelled by w_1 and w_2 and acts as the identity on all other modes.

In what follows we use this notation also in more structured instances such as $C_{d, \beta \cdot q_1 \cdot \alpha, \beta \cdot p_1 \cdot \alpha}$, where β and α denote fixed prefixes/suffixes and the middle digit changes from q_1 to p_1 .

At the level of decoding, the basic behaviour is as expected:

- Decoding $C_{d, \beta \cdot q_1 \cdot \alpha, \beta \cdot p_1 \cdot \alpha}$ yields a two-level $X^{(q_1, p_1)}$ on the corresponding qudit, with all other qudits providing (multi-)controls determined by β and α .
- Decoding $C_{d, \beta \cdot q_1 \cdot q_2 \cdot \alpha, \beta \cdot p_1 \cdot p_2 \cdot \alpha}$ yields a pattern of three such controlled X gates, i.e. three CNOT-like gates acting on the appropriate two-level subspace.

To remove the additional controls arising from β and α , we introduce a “saturation” of these gadgets over all possible prefix/suffix values.

Definition J.3. For integers $k, \ell \geq 0$ and digits $q_1, q_2, p_1, p_2 \in \{0, \dots, d - 1\}$ we define

$$C_{d, k, \ell, [q_1, q_2], [p_1, p_2]}^* = \prod_{\beta=0}^{d^{k-1}-1} \prod_{\alpha=0}^{d^{\ell}-1} C_{d, \beta \cdot q_1 \cdot q_2 \cdot \alpha, \beta \cdot p_1 \cdot p_2 \cdot \alpha}.$$

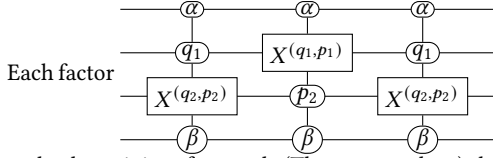
Intuitively, C^* applies the same three-CNOT pattern for every choice of prefix β and suffix α . By exhaustivity and commuting of controls (Thms. 3.3 and 3.4), these extra controls cancel when decoding:

Lemma J.4. *We have*

$$D\left(C_{d,k,\ell,[q_1,q_2],[p_1,p_2]}^\star\right) = id_k \otimes \begin{array}{c} \text{---} \textcircled{q_1} \text{---} \boxed{X^{(q_1,p_1)}} \text{---} \textcircled{q_1} \text{---} \\ \boxed{X^{(q_2,p_2)}} \text{---} \textcircled{p_2} \text{---} \boxed{X^{(q_2,p_2)}} \end{array} \otimes id_\ell.$$

PROOF. By definition,

$$\begin{aligned} D\left(C_{d,k,\ell,[q_1,q_2],[p_1,p_2]}^\star\right) &= D\left(\prod_{\beta,\alpha} C_{d,\beta \cdot q_1 \cdot q_2 \cdot \alpha, \beta \cdot p_1 \cdot p_2 \cdot \alpha}\right) \\ &= \prod_{\beta,\alpha} D\left(C_{d,\beta \cdot q_1 \cdot q_2 \cdot \alpha, \beta \cdot p_1 \cdot p_2 \cdot \alpha}\right) \\ &= \prod_{\beta,\alpha} \begin{array}{c} \text{---} \textcircled{\alpha} \text{---} \textcircled{\alpha} \text{---} \textcircled{\alpha} \text{---} \\ \textcircled{q_1} \text{---} \boxed{X^{(q_1,p_1)}} \text{---} \textcircled{q_1} \text{---} \\ \boxed{X^{(q_2,p_2)}} \text{---} \textcircled{p_2} \text{---} \boxed{X^{(q_2,p_2)}} \\ \textcircled{\beta} \text{---} \textcircled{\beta} \text{---} \textcircled{\beta} \end{array}. \end{aligned}$$



acts on the same pair of target levels and differs only by its outer controls; by commutation

and exhaustivity of controls (Thms. 3.3 and 3.4) the product simplifies to a single three-CNOT pattern with no remaining controls, tensored with identities on the untouched wires. This yields the claimed form. $\square$

J.3 Recursive construction of word-swaps

To describe the optical implementation C_{d,w_1,w_2} more concretely, it is convenient to use a parity based notation on words.

Definition J.5. For a word $w = h_1 \cdots h_m \in [d]^m$ define its (binary) parity by

$$\text{Parity}(w) = (h_1 \bmod 2) \oplus \cdots \oplus (h_m \bmod 2) \in \{0, 1\},$$

where $\oplus$ is addition modulo 2.

Given a word u and an integer $n \geq 0$, the *parity tail* $\text{ParityTail}_{u,n} \in [d]^n$ is

$$\text{ParityTail}_{u,n} = \begin{cases} 0^n & \text{if } \text{Parity}(u) = 0, \\ (d-1)^n & \text{if } \text{Parity}(u) = 1. \end{cases}$$

The following inductive clauses specify C_{d,w_1,w_2} for arbitrary $w_1, w_2 \in [d]^m$.

(1) **Base case (adjacent Gray neighbours).** Suppose

$$w_1 = \text{pref} \cdot p \cdot \text{suf}, \quad w_2 = \text{pref} \cdot q \cdot \text{suf},$$

with $|q - p| = 1$ and $\text{suf} = \text{ParityTail}_{\text{pref} \cdot \max(p,q), |\text{suf}|}$. Then w_1 and w_2 are consecutive in Gray order, and we set

$$C_{d,w_1,w_2} = id_{G_d^{-1}(\text{pref} \cdot \min(p,q) \cdot \text{suf})} \otimes \begin{array}{c} \diagup \quad \diagdown \\ \diagdown \quad \diagup \end{array} \otimes id_{d^{|\text{suf}|} - G_d^{-1}(\text{pref} \cdot \max(p,q) \cdot \text{suf}) - 1}.$$

(2) **Adjusting the suffix.** Suppose again $w_1 = \text{pref} \cdot p \cdot \text{suf}$ and $w_2 = \text{pref} \cdot q \cdot \text{suf}$ with $|q - p| = 1$, but suf is not already a parity tail.

Write $\text{suf} = \alpha_1 \cdot \alpha_i \cdot \alpha_2$ such that $\alpha_2 = \text{ParityTail}_{\text{pref} \cdot \max(p,q) \cdot \alpha_1, |\alpha_2|}$ and $\alpha_i \neq \text{ParityTail}_{\text{pref} \cdot \max(p,q) \cdot \alpha_1, 1}$. Let

$$\alpha'_i = \alpha_i + (-1)^{\text{Parity}(\text{pref} \cdot \alpha_1 \cdot \min(p,q))}$$

and define

$$w'_1 = \text{pref} \cdot p \cdot \alpha_1 \cdot \alpha'_i \cdot \alpha_2, \quad w'_2 = \text{pref} \cdot q \cdot \alpha_1 \cdot \alpha'_i \cdot \alpha_2.$$

Then we set

$$C_{d,w_1,w_2} = C_{d,w_1,w'_1} \circ C_{d,w_2,w'_2} \circ C_{d,w'_1,w'_2} \circ C_{d,w_2,w'_2} \circ C_{d,w_1,w'_1}.$$

(3) **Bridging distant levels.** Suppose $w_1 = \text{pref} \cdot p \cdot \text{suf}$ and $w_2 = \text{pref} \cdot q \cdot \text{suf}$ with $|q - p| > 1$. Let

$$w_{\max} = \text{pref} \cdot \max(p, q) \cdot \text{suf}, \quad w_{\max-1} = \text{pref} \cdot (\max(p, q) - 1) \cdot \text{suf}, \quad w_{\min} = \text{pref} \cdot \min(p, q) \cdot \text{suf}.$$

We then define

$$C_{d,w_1,w_2} = C_{d,w_{\max},w_{\max-1}} \circ C_{d,w_{\min},w_{\max-1}} \circ C_{d,w_{\max},w_{\max-1}}.$$

(4) **Words differing in two positions.** Suppose

$$w_1 = \text{pref} \cdot p_1 \cdot \text{mid} \cdot p_2 \cdot \text{suf}, \quad w_2 = \text{pref}' \cdot q_1 \cdot \text{mid} \cdot q_2 \cdot \text{suf}$$

with $p_1 \neq q_1$ and $p_2 \neq q_2$. Let

$$w'_2 = \text{pref} \cdot q_1 \cdot \text{mid} \cdot q_2 \cdot \text{suf}, \quad w' = \text{pref} \cdot p_1 \cdot \text{mid} \cdot q_2 \cdot \text{suf}.$$

Then we set

$$C_{d, w_1, w_2} = C_{d, w_2, w'_2} \circ C_{d, w_1, w'} \circ C_{d, w', w'_2} \circ C_{d, w_1, w'} \circ C_{d, w_2, w'_2}.$$

The four clauses above recursively express an arbitrary word-swap as a composition of adjacent Gray-neighbour swaps, possibly after “straightening” suffixes and separating differences across positions.

J.4 Decoding of the recursive swap construction

We now record the effect of decoding these gadgets.

Lemma J.6. *For any digits p_1, q_1 and words α, β we have*

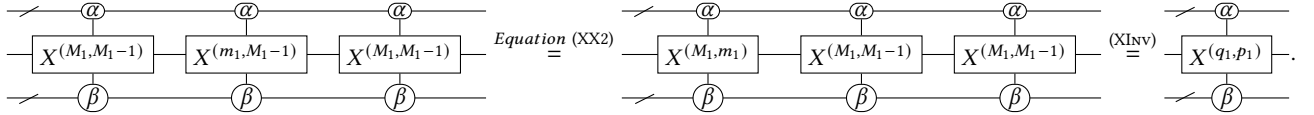
$$D(C_{d, \beta \cdot q_1 \cdot \alpha, \beta \cdot p_1 \cdot \alpha}) = \begin{array}{c} \text{---} \alpha \text{---} \\ | \\ \boxed{X^{(q_1, p_1)}} \\ | \\ \text{---} \beta \text{---} \end{array},$$

i.e. a single two-level $X^{(p_1, q_1)}$ on the appropriate qudit, plus trivial identities on the other wires.

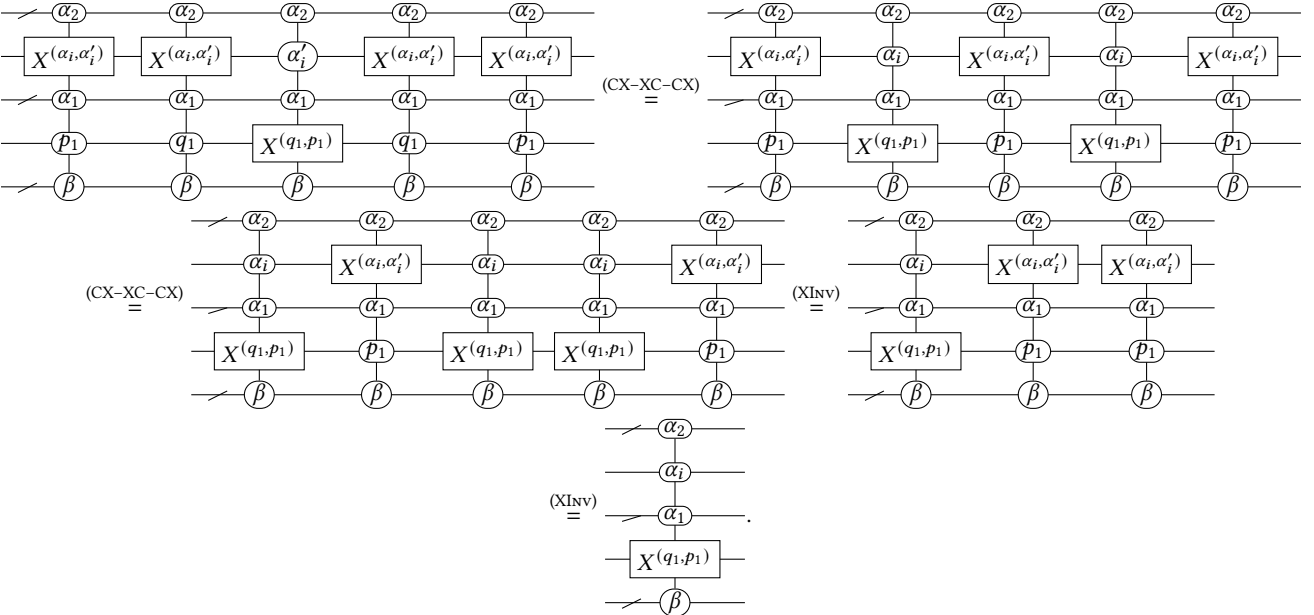
PROOF. The proof proceeds by induction on the definition of C_{d, w_1, w_2} .

In the base case 1 the two words are adjacent in Gray order, and C_{d, w_1, w_2} is literally a single wire-swap between the corresponding modes, so the decoding is immediate.

In case 3, the definition expresses the swap between levels p_1 and q_1 as a composite of swaps involving the intermediate level $M_1 = \max(p_1, q_1)$ and $M_1 - 1$. Using Lemma XX2 together with involutivity (XInv) one obtains the required single $X^{(p_1, q_1)}$ on decoding:



In case 2, the suffix-adjustment circuit is a five-fold composition of swaps. Using repeatedly the three-CX commutation pattern (CX-XC-CX) and symmetry of two-level X gates (By definition), followed again by (XInv), one reduces this pattern to a single $X^{(p_1, q_1)}$:



The remaining case 4 will be handled in the next lemma. This completes the proof for the present statement. $\square$

Lemma J.7. For any digits p_1, p_2, q_1, q_2 and words α, β we have

$$D(C_d, \beta \cdot q_1 \cdot q_2 \cdot \alpha, \beta \cdot p_1 \cdot p_2 \cdot \alpha) =$$

PROOF. In case 4 the two words differ in two positions, and the recursive definition factors the swap into a composition of three single-position swaps, to which Lemma J.6 applies. The result is exactly the depicted three-CNOT pattern. $\square$

Combining these with Lemma J.4 we obtain:

Lemma J.8. For all k, ℓ and digits p_1, p_2, q_1, q_2 we have

$$D(C_{d,k,\ell,[q_1,q_2],[p_1,p_2]}^\star) = id_k \otimes$$

$$\otimes id_\ell.$$

J.5 Decoding block-swaps

We now assemble block-swaps from these saturated gadgets.

Definition J.9. For integers $k, \ell \geq 0$ we define the LOPP block-swap $\sigma_{k,\ell,1}^d$ that exchanges a block of ℓ modes with a single mode to its right by

$$\sigma_{k,\ell,1}^d = \prod_{j=0}^{\ell-1} \prod_{x=0}^{d-2} \prod_{y=x+1}^{d-1} C_{d,k+j,\ell-j-1,[x,y],[y,x]}^\star.$$

Lemma J.10. We have $D(\sigma_{k,\ell,1}^d) = id_k \otimes \sigma_{\ell,1}$, where $\sigma_{\ell,1}$ is the usual PROP symmetry swapping an ℓ -wire block with a single wire.

PROOF. Using Lemma J.8 we obtain

$$\begin{aligned} D(\sigma_{k,\ell,1}^d) &= \prod_{j=0}^{\ell-1} \prod_{x=0}^{d-2} \prod_{y=x+1}^{d-1} D(C_{d,k+j,\ell-j-1,[x,y],[y,x]}^\star) \\ &= \prod_{j=0}^{\ell-1} \prod_{x=0}^{d-2} \prod_{y=x+1}^{d-1} \left(id_{k+j} \otimes \right. \\ &\quad \left. \begin{array}{c} \text{Diagram: A three-CNOT pattern with top wire } X(x,y), \text{ middle wire } X(x,y), \text{ bottom wire } X(x,y). \end{array} \right. \otimes id_{\ell-j-1} \Big) \\ &= \prod_{j=0}^{\ell-1} \left(id_{k+j} \otimes \left(\prod_{x < y} \begin{array}{c} \text{Diagram: A three-CNOT pattern with top wire } X(x,y), \text{ middle wire } X(x,y), \text{ bottom wire } X(x,y). \end{array} \right) \otimes id_{\ell-j-1} \right). \end{aligned}$$

$\underbrace{\quad}_{\stackrel{(S)}{=} \sigma_{1,1}}$

The product over $x < y$ yields exactly the adjacent swap $\sigma_{1,1}$ by the swap decomposition rule (S). Factoring out the leftmost k wires, we obtain

$$\begin{aligned} D(\sigma_{k,\ell,1}^d) &= id_k \otimes \left(\prod_{j=0}^{\ell-1} (id_j \otimes \sigma_{1,1} \otimes id_{\ell-j-1}) \right) \\ &= id_k \otimes \sigma_{\ell,1}, \end{aligned}$$

since the product on the right is precisely the standard factorisation of a block-swap into ℓ adjacent transpositions. $\square$

Finally, we generalise from swapping a single mode with an ℓ -block to swapping two arbitrary blocks.

Lemma J.11. Define inductively $\sigma_{a,b,c}^d := \sigma_{a,b+c-1,1}^d \circ \sigma_{a,b+1,c-1}^d$. Then $D(\sigma_{a,b,c}^d) = id_a \otimes \sigma_{b,c}$.

PROOF. By definition,

$$\begin{aligned} D(\sigma_{a,b,c}^d) &= D(\sigma_{a,b+c-1,1}^d \circ \sigma_{a,b+1,c-1}^d) \\ &= D(\sigma_{a,b+c-1,1}^d) \circ D(\sigma_{a,b+1,c-1}^d) \\ &\stackrel{\text{Theorem J.10}}{=} (id_a \otimes \sigma_{b+c-1,1}) \circ D(\sigma_{a,b+1,c-1}^d). \end{aligned}$$

By the induction hypothesis on c we have $D(\sigma_{a,b+1,c-1}^d) = id_a \otimes \sigma_{b+1,c-1}$, so

$$\begin{aligned} D\left(\sigma_{a,b,c}^d\right) &= id_a \otimes (\sigma_{b+c-1,1} \circ \sigma_{b+1,c-1}) \\ &= id_a \otimes \sigma_{b,c}, \end{aligned}$$

since $\sigma_{b,c}$ is given by first moving the rightmost wire across the block of size $b + c - 1$ and then swapping the resulting $(b + 1)$ -block with the remaining $c - 1$ wires. This is exactly the composition $\sigma_{b+c-1,1} \circ \sigma_{b+1,c-1}$. $\square$